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Solving Class-Agnostic Counting Problem using deep learning

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Abstract

Counting objects in natural images remains a fundamental yet unsolved problem in computer vision progress. Motivated by recent achievements in fundamental computer vision tasks, *e.g.*, Object Detection, Semantic Segmentation, the dissertation focuses on the Class-Agnostic Counting (CAC) problem by exploiting the recent vision models and presents two complementary paradigms: one for Zero-Shot Counting, which dispenses with exemplars altogether, and another one for Few-Shot Counting, which relies on a handful of class-agnostic cues.

For the zero-shot counting task, we propose the Gated Context-Aware Swin-UNet (GCA-SUNet). The architecture employs a Swin Transformer-based encoder to extract rich multi-scale representations. To enhance discriminative feature learning, we incorporate a Gated Context-Aware Modulation (GCAM) block, which dynamically attenuates background noise while amplifying self-similarity tokens indicative of countable object instances. Furthermore, two additional gated modules, *i.e.*, GEFS and GAFU, are introduced within the bottleneck and decoder stages. These components selectively propagate informative features through the upsampling path to rebuild the density map, thereby enabling accurate object count estimation without the need for external exemplar guidance.

For the few-shot counting task, we propose the Cross-modal Semantic-guided Group-attended (CSG) model. This framework jointly leverages visual exemplars and textual prompts to enrich representation learning. A vision–language model generates dual descriptions for both query and exemplar images, wherein textual inputs are encoded using BERT and visual inputs via a Swin Transformer. To facilitate effective feature alignment, two Gated Intra-modal Contrastive Embedding (GICE) blocks, in conjunction with a Cross-modal Semantic Refinement (CSR) module, are employed to capture fine-grained intra- and inter-modal correlations. Finally, a Semantic-Guided Group Attention (SGGA) module aggregates shared semantic attributes, selecting the candidates, thereby improving the accuracy of the final count estimation.

Extensive experiments conducted on challenging benchmark datasets validate the effectiveness of both proposed frameworks. GCA-SUNet and CSG achieve new state-of-the-art performance in their respective settings, reducing the mean absolute error by 0.63 and 2.68 on the primary benchmark. These results highlight the potential of minimally annotated, generalizable paradigms for object counting.

Publications

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Yuzhe Wu, Yipeng Xu, Jialu Zhang, Yiran Li, Jianfeng Ren, and Xudong Jiang. CSG: A Cross-Modal Semantic-Guided Group-Attended Framework for Few-Shot Counting. Under review at *33rd ACM International Conference on Multimedia (CCF-A)*, 2025.

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List of Abbreviations

CAC	Class Agnostic Counting
CSC	Class-Specific Counting
CSG	Cross-modal Semantic-guided Group-attended framework
CSR	Cross-modal Semantic Refinement
EFC	Exemplar Free Counting
FSC	Few Shot Counting
GAFU	Gated Adaptive Fusion Unit
GCA-SUNet	Gated Context-Aware Swin-UNet
GCAM	Gated Context-Aware Modulation
GEFS	Gated Enhanced Feature Selector
GICE	Gated Intra-modal Contrastive Embedding
MAE	Mean Average Error
PBC	Prompt Based Counting
RMSE	Root Mean Squared Error
SAM	Segment Anything Model
SGGA	Semantic-Guided Group-Attention
VLM	Vision-Language Model
ZSC	Zero Shot Counting

Chapter 1

Introduction

This dissertation addresses the Class-Agnostic Counting (CAC) problem using advanced deep learning techniques, particularly emphasizing two challenging sub-problems: Zero-Shot Counting (ZSC) and Few-Shot Counting (FSC). The primary aim is to significantly improve the generalization capabilities of counting models, enabling accurate and robust counting of objects across diverse scenarios without heavy reliance on extensive annotated datasets.

The first part of this research, focusing on the ZSC problem, has been accepted for publication at the **IEEE International Conference on Multimedia & Expo (ICME)**, recognized as a top CCF-B level conference. Subsequently, the second part, addressing the FSC problem, is currently under review for the **33rd ACM International Conference on Multimedia (ACMMM)**, a prestigious CCF-A conference.

In this introductory chapter, we detail the motivations behind this research in Section 1.1. The explicit research objectives are presented in Section 1.2. Section 1.3 succinctly summarizes the key contributions of this dissertation, and Section 1.4 outlines the overall structure of the dissertation.

1.1 Motivation

Object counting is a fundamental task in computer vision aimed at estimating the number of object instances within an image [3]. Traditional methodologies predominantly follow the Class-Specific Counting (CSC) paradigm, concentrating on predefined categories such as people [7], vehicles [8], or biological cells [9], typically relying on extensive annotated datasets. However, the accelerating growth in diversity and volume of real-world visual data has exposed significant limitations of CSC models. These methods inherently require exhaustive, category-specific annotations, significantly inflating data preparation costs and restricting their effectiveness in dynamically changing or novel scenarios [10, 11], where enumerating all potential object classes beforehand is infeasible.

In response to these challenges, contemporary approaches have emerged, aiming to overcome the dependence on extensive annotations and to generalize effectively across new or mixed object categories [3, 12]. This shift parallels human cognitive flexibility in object counting, allowing adaptation even in highly cluttered and visually complex environments [13]. Nevertheless, emulating such flexibility presents substantial challenges for automated systems. These systems must carefully balance that all relevant objects are identified and prevent erroneous activations caused by implicit features inadvertently representing irrelevant objects or background

noise [14]. Additionally, scenarios characterized by clutter, occlusion, and varying object scales further complicate accurate detection and localization, posing significant barriers to reliability.

ZSC methods specifically address the human-annotated cost bottleneck by counting objects without explicit input annotations. Typically, ZSC methods identify dominant or prevalent object categories using implicit mechanisms, including attention modules [2] or majority voting strategies over region proposals [15]. Although these approaches enhance model generality, they also introduce mislocalization vulnerabilities, particularly in highly heterogeneous scenes where non-target objects might be erroneously counted. Thus, the intricate balance between broad generalization capabilities and precise discrimination remains a critical challenge driving continued exploration in ZSC methods.

FSC, as a representative and foundational paradigm of CAC, seeks to alleviate the dependency on extensive annotations by leveraging only a limited number of annotated visual exemplars [3]. In traditional FSC methods, object representations are derived from the exemplar features and subsequently correlated with features from the query image to generate a density map [3, 16, 17]. The final object count is obtained by integrating the predicted density map. Although FSC approaches generally achieve high counting accuracy, they often overlook fine-grained spatial and scale information of the counted objects. This limitation is particularly problematic in domains such as biomedical imaging [9], where accurate spatial localization and size estimation are critical. Recently, detection-based approaches [18–20] have emerged as strong contenders in the pursuit of state-of-the-art performance. However, these methods still face challenges related to semantic misalignment between guidance and the visual context, leading to inconsistencies in countable object identification and localization.

In summary, the inherent limitations of conventional CSC methods have motivated the exploration of more generalizable paradigms, *i.e.*, CAC. Within the problem, ZSC and FSC have emerged as two prominent sub-tasks. While these paradigms reduce dependence on exhaustive annotations and extend applicability to novel object categories, they still face significant challenges. Chief among these are the complexities of interpreting cluttered visual scenes, mitigating false activations, and preserving fine-grained spatial and semantic details in the counting outputs. These persistent obstacles underscore the need for continued innovation in the field of object counting.

1.2 Objectives

1.2.1 Research Questions

Considering the high cost and scalability limitations associated with human-annotated exemplars in CAC, the dissertation addresses the following overarching research question:

How can deep learning paradigms be advanced to eliminate the dependency on human-annotated inputs, while simultaneously enhancing the robustness and generalization ability for CAC tasks?

To address this central challenge, we formulate the following sub-research questions (subRQs), each targeting a distinct yet complementary dimension of the broader CAC problem:

- **subRQ1: How can deep learning models autonomously extract and utilize intrinsic visual cues to accurately count objects in ZSC scope?**

This question aims to investigate architectural designs and learning strategies that empower deep neural networks to perform ZSC. Specifically, it explores how models can uncover latent object-level semantics, spatial self-similarity patterns, and contextual dependencies directly from the input image. Solving this sub-question is critical for reducing annotation overhead and advancing scalable counting frameworks that generalize effectively across unseen object categories and open-world visual environments.

- **subRQ2: How can interactions between automatically generated textual exemplars and visual features from both query and exemplar images be effectively modeled to enable fine-grained visual-linguistic alignment in the FSC task?**

In the context of FSC, where only a limited number of visual exemplars are available, CountGD [20] highlights the benefit of incorporating both textual and visual exemplars. However, its reliance on human-annotated textual inputs limits scalability. To overcome this, we explore the use of Vision-Language Models (VLMs) to automatically generate textual exemplars, removing the need for manually labeled textual input. This sub-question focuses on modeling cross-modal interactions between these semantic representations and visual features from query and exemplar images. By capturing fine-grained visual-linguistic alignments, the model can more accurately identify semantically relevant regions and enhance object reasoning in previously unseen categories, ultimately improving generalization in complex counting scenarios.

1.2.2 Research Methods

Aligned with the identified research questions, we propose two methodological directions aimed at enhancing robustness, generalizability, and annotation efficiency in the CAC setting. These methods aim to either eliminate the reliance on exemplar inputs entirely or enhance their semantic effectiveness through multimodal interactions, under the premise of minimizing the requirement for input exemplars.

Chapter 3 addresses *subRQ1* by proposing a Gated Context-Aware Swin-UNet (GCA-SUNet) for the ZSC task. The core of this framework is built upon the Swin-UNet [21], enhanced by three novel gated modules. The resulting end-to-end architecture reduces annotation requirements and enhances scalability. Empirical evaluations show that the proposed framework achieves strong generalization across diverse datasets, demonstrating its robustness and effectiveness in complex, real-world counting scenarios.

Chapter 4 addresses *subRQ2* by introducing a Cross-modal Semantic-guided Group-attended (CSG) framework for the FSC task. The framework begins by leveraging a VLM to automatically generate rich textual descriptions. These multimodal representations are then processed through three newly proposed modules that exploit deeply coupled interaction between four sets of features. Experimental results demonstrate significant performance improvements, particularly under low-annotation conditions, underscoring the effectiveness of the proposed approach.

1.3 Contributions

The primary contributions of this dissertation are outlined as follows:

- To tackle the challenge of exemplar-free object counting in ZSC scenarios, we propose the Gated Context-Aware Swin-UNet (GCA-SUNet). This framework is built upon the Swin-UNet [21] backbone, which enables hierarchical and context-rich feature extraction through shifted window-based attention. To enhance the model’s capacity to localize and quantify repetitive objects, we introduce a Gated Context-Aware Modulation (GCAM) module that leverages self-similarity matrices to selectively amplify features corresponding to countable instances. Furthermore, the three different modules with the gating mechanism are integrated throughout the encoder, bottleneck, and decoder to consistently suppress background noise and emphasize salient object-level cues. Extensive experiments on the ZSC benchmarks demonstrate that GCA-SUNet achieves state-of-the-art performance among existing ZSC approaches, effectively mitigating the sample bias prevalent in exemplar-guided models [15].
- To investigate how visual-linguistic interactions among exemplar images, query images, and countable objects influence counting performance, particularly after the automatic generation of textual exemplars by a VLM, we propose the Cross-modal Semantic-guided Group-attended (CSG) framework for the FSC task. The CSG framework leverages the LLaVA model to generate detailed textual descriptions for both query and exemplar images, producing four distinct sets of multimodal features without the need for handcrafted prompts. To achieve fine-grained alignment across modalities, CSG incorporates dedicated cross-modal interaction modules that systematically refine and integrate these features. Extensive evaluations of the FSC benchmarks confirm that CSG consistently outperforms existing approaches, demonstrating its effectiveness.

1.4 Dissertation Structure

As shown in Fig. 1.1, the remainder of this dissertation is structured as follows:

1. **Chapter 2** presents a comprehensive literature review that traces the progression from traditional CSC methods to more recent advancements in CAC. Particular emphasis is placed on developments in ZSC and FSC, outlining key techniques, prevailing challenges, and representative architectural two dominant paradigms within the CAC tasks.
2. **Chapter 3** introduces the proposed GCA-SUNet framework, detailing its novel architecture to achieve exemplar-free counting. This chapter elucidates the model’s methodological innovations, implementation details, and presents extensive experimental analyses demonstrating its effectiveness and robustness in the ZSC task.
3. **Chapter 4** presents the novel CSG framework, designed to tackle the FSC task. It describes the integration of multimodal interactions involving visual and linguistic features. The chapter elaborates on the model’s structural components and provides exhaustive empirical evaluations that underscore its superior performance against contemporary benchmarks.

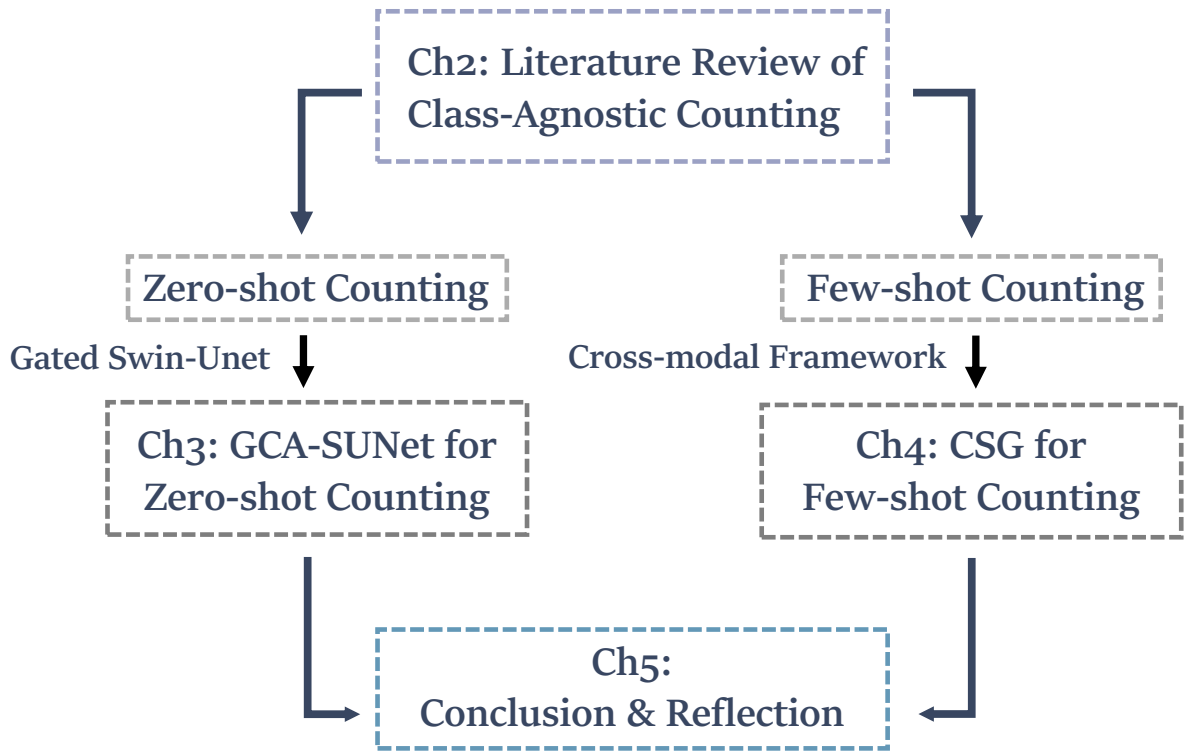


Figure 1.1: Overview of the dissertation.

4. **Chapter 5** concludes the dissertation by summarizing the primary research contributions and achievements, reflecting on the overall progression and insights gained throughout the research. It also outlines potential limitations and proposes several promising directions for future research to further advance the field of CAC.

Chapter 2

Literature Review

2.1 Background

Object counting has been extensively studied in computer vision, with methods evolving to address challenges such as annotation overhead, adaptability, and generalization. In the literature, object-counting approaches can be broadly divided into CSC and CAC.

Class-Specific Counting (CSC). Early efforts in object counting relied on class-specific detectors trained on predefined object categories [6, 22–24]. While these methods offer precise localization and counting, their dependence on category-specific annotations limits their ability to generalize to unseen classes or adapt to diverse real-world scenarios.

Class-Agnostic Counting (CAC). To overcome the limitations of CSC, Class-Agnostic Counting approaches have been developed that do not rely on fixed class labels. These methods typically utilize external guidance, either in the form of visual exemplars or textual prompts, to learn a universal mapping from image features to density maps. As illustrated in Fig. 2.1, it could be further categorized into FSC, PBC, and ZSC.

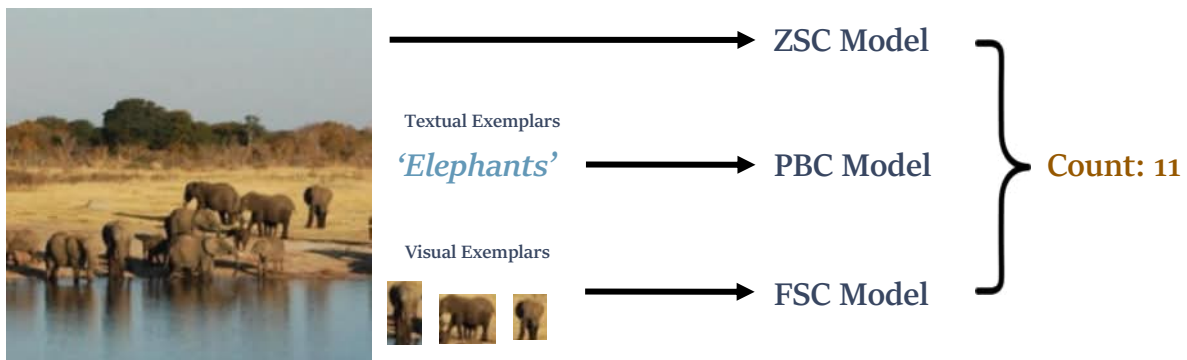


Figure 2.1: Demonstration of the different settings in CAC.

Few-Shot Counting (FSC). Few-Shot Counting, introduced as a foundational method for CAC, aims to reduce dependency on extensive labeled data by leveraging a limited number of visual exemplars (*e.g.*, bounding boxes or exemplar patches) [12]. The advent of the FSC147 dataset [3] has significantly advanced research in this field. Despite the recent advancements, FSC still faces considerable challenges, particularly due to limited annotations, sensitivity to exemplar selection, and significant intra-class variability in object appearance, scale, and occlusion.

Prompt-Based Counting (PBC). Prompt-Based Counting minimizes annotation effort by

substituting visual exemplars with textual prompts. Utilizing VLMs, these methods align textual instructions with visual features to pinpoint countable regions.

Zero-Shot Counting (ZSC). Zero-Shot Counting, also known as Exemplar-Free Counting (EFC), eliminates the requirement for external guidance such as exemplars or textual prompts, relying exclusively on intrinsic visual features to autonomously detect and quantify countable objects [1, 2, 15, 25]. This paradigm significantly reduces annotation overhead and minimizes biases introduced through external inputs. However, the absence of explicit guidance introduces substantial difficulties in reliably identifying and localizing countable objects, as well as achieving robust generalization to diverse, unseen real-world scenarios.

2.2 Related Work

Recent advances in CAC have led to the emergence of several distinct methodological paradigms. In this section, we provide a detailed review of representative approaches across three major categories: FSC, PBC, ZSC, as well as CSC.

Class-Specific Counting. Early object counting efforts are tightly coupled with class-specific detectors trained on predefined categories [6, 23, 24]. These frameworks offer accurate localization and counting, but their reliance on class-specific training data hinders their ability to generalize to unseen object classes. As a result, scaling these approaches to more diverse scenarios remains challenging and computationally expensive.

Few-Shot Counting. Class-agnostic counting was first introduced in the context of Few-Shot Counting [12] to address the limitations of class-specific counting. Many methods have been designed to map from image features to density maps, guided by visual exemplars [3]. This exemplar-based strategy reduces the reliance on class-specific annotations, enabling broader generalization across diverse scenarios. Prominent examples include Generic Matching Network (GMN) [12], which infers counts through similarity comparisons between query images and exemplar shots. For example, FamNet [3] refines query features by leveraging feature correlations and enhances count estimation accuracy through test-time adaptation. CountTR [1] employs cross-attention mechanisms to strengthen interactions between exemplars' features and query features, thereby refining these features. CACViT [17] employs decoupled attention in Vision Transformer [26] to maximize the interactions between the query and exemplars. DAVE [14] integrates a detector to generate a high-recall object set and filters outliers using cosine similarity. Recently, GeCo [27] integrates Segment Anything Model [28] into an encoder-decoder model and dynamically generates candidate regions.

Prompt-Based Counting. PBC aims to minimize the effort of annotating visual exemplars by utilizing text prompts instead, *i.e.*, utilizing the capabilities of large language models and vision-language pre-training to interpret prompts and align them with visual contents. Patch-Selection [29] utilized text prompts to construct prototypes, effectively identifying class-relevant patches as visual exemplars [29]. CLIP-Count [30] utilizes the CLIP text encoder [31] to obtain text descriptions and adopts a contrastive loss and a hierarchical patch-text interaction module to adapt VLMs for visual counting tasks. VLCounter [32] combines semantic-aware prompt tokens with visual tokens, highlighting target-specific regions to enhance the features for zero-shot counting. CountGD [20] utilizes BERT [33] to extract the text descriptions and

a set of self-attentions and cross-attentions to facilitate the multi-modal interaction between visual tokens and textual tokens. Despite these advancements, their performance highly depends on the quality and informativeness of the text guidance.

Zero-Shot Counting. ZSC eliminates all forms of external guidance, such as exemplars or text prompts. It aims to autonomously identify and focus on regions containing countable objects. ZSC significantly reduces annotation overhead and avoids biases induced by external inputs, making it suitable for diverse and unpredictable real-world scenarios. For example, pre-trained vision transformers with weakly-supervised training have been utilized to extract visual patterns without explicit exemplars [2]. Repetitive Region Proposal Network (RepRPN) [15] is developed for identifying exemplars from repetitive objects and estimating their approximate count [15]. Indeed, the absence of annotated exemplars presents significant challenges, particularly in reliably discovering and locating countable objects.

2.2.1 Regression-Based vs. Detection-Based CAC methods

As shown in Fig. 2.2, CAC methods can be broadly categorized into two major paradigms: detection-based and regression-based approaches.

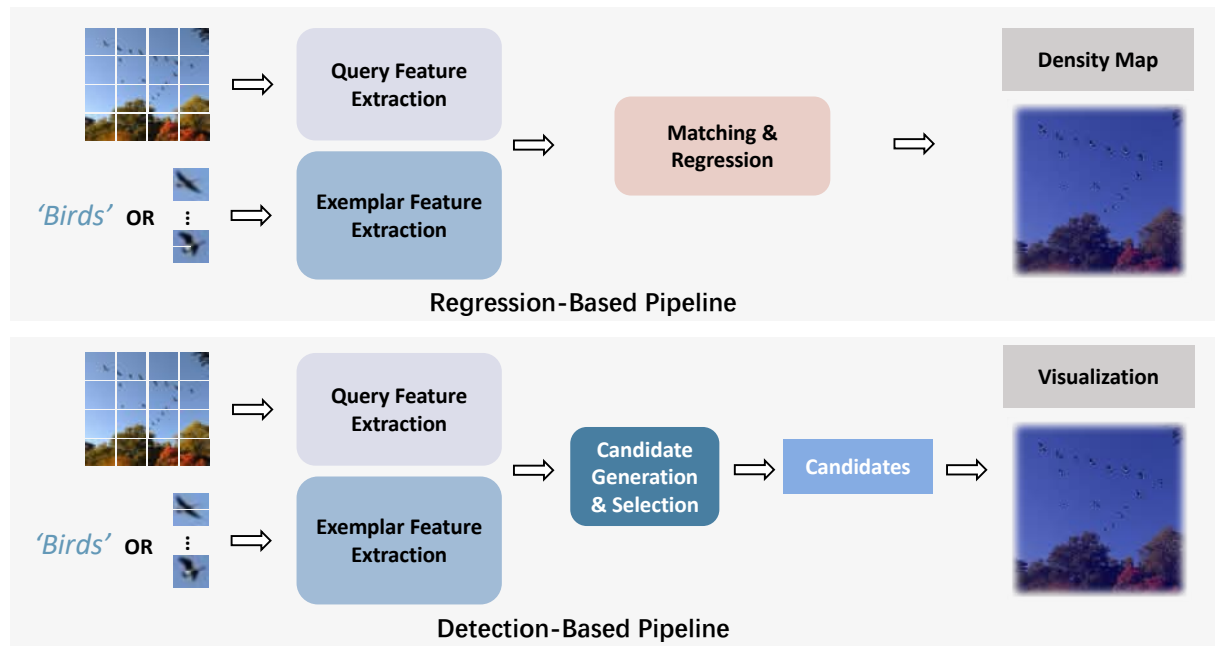


Figure 2.2: High-level ideas between the two paradigms for CAC.

Detection-based methods focus on identifying and localizing individual object instances within an image. Early CSC works in this domain relied heavily on object detectors [4, 22, 34], which require substantial class-specific supervision and a large volume of annotated data. While these methods offer precise spatial localization, they often struggle in scenarios with severe object overlap or when annotated exemplars are limited. Recent advancements have started to alleviate these limitations through the emergence of *open-world* methods that integrate VLMs, such as CLIP [31] and Grounding-DINO [18]. These models enrich the semantic understanding of visual content, enabling detection-based pipelines to operate effectively in open-world settings. Contemporary approaches [14, 19, 20, 32] typically perform candidate object detection

followed by semantic selection guided by exemplar prompts or textual cues, achieving state-of-the-art performance in CAC tasks.

Regression-based methods, by contrast, bypass the need for instance-level localization by learning a direct mapping from image features to object counts. A simple yet effective strategy involves regressing a global count value [2, 13], while more advanced techniques employ density map estimation [1, 3, 16, 17, 35, 36], wherein each pixel indicates the estimated local density of objects. This formulation is particularly well-suited for scenes with occlusions and overlapping instances, as it captures spatial distribution implicitly and reduces reliance on precise bounding box annotations [1, 17]. Recent progress in this line of work includes the integration of multi-scale feature aggregation and hierarchical architectures [36], enabling robust extraction of contextual and structural cues across varying image resolutions. These advancements have significantly improved generalization to complex visual scenes, making regression-based approaches highly competitive in the ZSC setting. Moreover, the end-to-end differentiability of these models facilitates the learning of discriminative features that are well-aligned with counting objectives, even under challenging visual conditions.

While prior approaches have progressively transitioned from the highly supervised CSC paradigm to more flexible CAC scenarios, substantial challenges remain in both ZSC and FSC settings. To address the sample bias of exemplars as well as reduce the reliance on the exemplars, we propose GCA-SUNet, a framework that directly estimates object density maps without relying on external exemplars. The architecture incorporates gating mechanisms strategically embedded across the encoder, bottleneck, and decoder stages, significantly enhancing robustness and scalability for deployment in diverse and unconstrained visual environments. To tackle the challenge that appears after integrating the linguistic feature in FSC, we introduce the CSG framework, which utilizes dual automatically generated textual descriptions for both query and exemplar images. These multimodal representations are refined and fused through three dedicated modules, enabling fine-grained cross-modal alignment and improved object reasoning under minimal supervision.

Extensive evaluations on FSC-147 [3], CARPK [4], and CountBench [5] demonstrate the superior accuracy and robustness of the proposed frameworks, setting new performance benchmarks in both ZSC and FSC tasks. Together, GCA-SUNet and CSG advance the state-of-the-art in ZSC and FSC tasks, offering scalable solutions for real-world applications with constrained annotation resources.

Chapter 3

GCA-SUNet: A Gated Context-Aware Swin-UNet for Zero-Shot Counting

Zero-Shot Counting aims to count objects of interest without intensive annotations of objects or exemplars. To achieve this, we propose a Gated Context-Aware Swin-UNet (GCA-SUNet) to directly map an input image to the density map of countable objects. Specifically, a set of Swin-Transformers forms an encoder to derive a robust feature representation, and a Gated Context-Aware Modulation block is designed to suppress irrelevant objects or background through a gate mechanism and exploit the attentive support of objects of interest through a self-similarity matrix. The gate strategy is also incorporated into the bottleneck network and the decoder of the Swin-UNet to highlight the features most relevant to objects of interest. By explicitly exploiting the attentive support among countable objects and eliminating irrelevant features through the gate mechanisms, the proposed GCA-SUNet focuses on and counts objects of interest without relying on predefined categories or exemplars. Experimental results on the real-world datasets such as FSC-147 and CARPK demonstrate that GCA-SUNet significantly and consistently outperforms state-of-the-art methods.

3.1 Introduction

Object counting determines the number of instances of a specific object class in an image, *e.g.*, vehicles [37, 38], crowd [7], and cells [9]. It can be broadly categorized into the following. 1) CSC, counting specific categories like fruits [39] and animals [40]; 2) CAC, counting objects based on visual exemplars [1, 3, 35] or text prompts [29, 41]; 3) ZSC, a specialized sub-task of CAC, aims to estimate object counts without relying on any exemplar inputs, posing substantial challenges in identifying countable objects and accurately inferring their repetitions within diverse and unconstrained visual scenes [1, 2, 15].

ZSC shows promise for automated systems such as wildlife monitoring [42], healthcare [43], and anomaly detection [44]. Hobley and Prisacariu directly regressed the image-level features learned by attention modules into a density map [2]. CounTR [1] and LOCA [16] are originally designed for another setting in CAC tasks, but can be adapted to ZSC tasks by using trainable components to simulate exemplars. RepRPN-Counter identifies exemplars from region proposals by majority voting [15], and DAVE selects valuable objects using a strategy similar to majority vote [14].

Existing models [1, 14, 16] often explicitly require exemplars to count similar objects. ZSC methods such as RepRPN-Counter do not require exemplars but generate them through region

proposal [15]. Either explicit or implicit exemplars may induce sample bias as exemplars can't cover the sample distribution. To address the challenge, we propose Gated Context-Aware Swin-UNet (GCA-SUNet), which directly maps an input image to the density map of countable objects, without any exemplars. Specifically, the encoder consists of a set of Swin-Transformers to extract features, and a set of Gated Context-Aware Modulation (GCAM) blocks to exploit the attentive supports of countable objects. The bottleneck network includes a Gated Enhanced Feature Selector (GEFS) to emphasize the encoded features that are relevant to countable objects. The decoder includes a set of Swin-Transformers for generating the density map, with the help of Gated Adaptive Fusion Units (GAFUs) to selectively weigh features based on their relevance to countable objects. Finally, a regression head is utilized to derive the density map from the aggregated features.

One key challenge in ZSC is to effectively differentiate countable objects from other objects. The GCAM blocks tackle this challenge by first evaluating the feature quality by computing the feature score for each token, and then prioritizing those with informative content. To uncover the informative features, GCAM computes pairwise similarities between tokens through a self-similarity matrix, exploiting the support of repeating objects in the same scene. Lastly, a gate mechanism is incorporated to highlight the most relevant features while suppressing irrelevant ones.

Another challenge is that foreground objects often share similar low-level features with background content. The skip connections directly fuse low-level features in the encoder with high-level semantics in the decoder, potentially impeding counting performance as the background could disturb the foreground objects. To tackle this issue, gate mechanisms are incorporated into both GEFS and GAFU to suppress irrelevant low-level features while preserving as much information on objects of interest as possible. The former selectively enhances the condensed features at the bottleneck, and the latter filters the irrelevant features in the decoder.

Our contributions can be summarized as follows. 1) The proposed GCA-SUNet achieves zsc through a UNet-like architecture that utilizes Swin-Transformer blocks for feature encoding and decoding, avoiding the sample bias of exemplar-based methods [15]. 2) The proposed GCAM focuses on countable objects by exploiting attentive support of repetitive objects through the self-similarity matrix. 3) The gate mechanism is integrated into various modules, *e.g.*, GCAM, GEFS, and GAFU, which suppresses the features of irrelevant objects or background while highlighting the most relevant features of countable objects. 4) Our GCA-SUNet is evaluated on the FSC-147 and CARPK datasets. It outperforms state-of-the-art methods for ZSC.

3.2 Proposed Method

3.2.1 Overview of Proposed Method

The proposed GCA-SUNet is built upon a Swin-UNet architecture [21], with three new types of building blocks, GCAM, GEFS, and GAFU, to exploit attentive support of countable objects and suppress irrelevant tokens or features, as outlined in Fig. 4.2. The primary motivation to choose Swin-UNet is that it has demonstrated superior performance in many tasks [45]. It is also well suited for our task, in which we need to generate object density maps of the same size as the input image. Specifically, the proposed GCA-SUNet begins with patched image feature

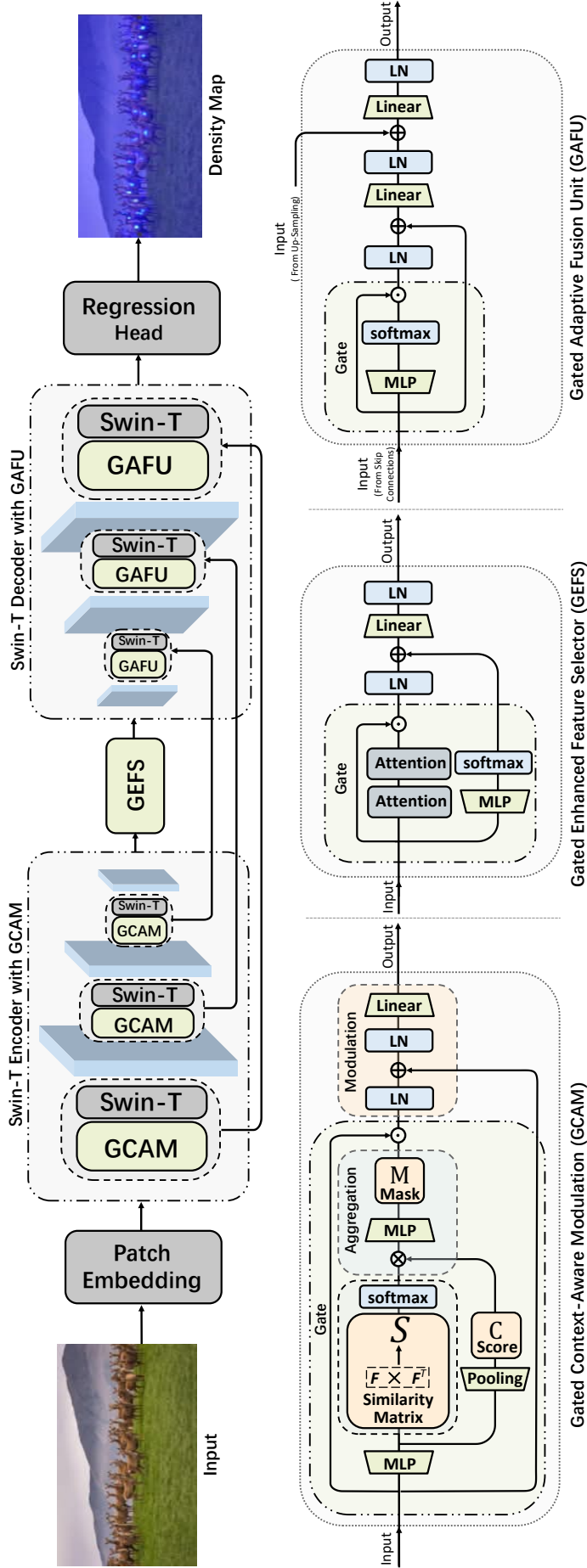


Figure 3.1: Overview of proposed GCA-SUNet. It consists of an encoder, a bottleneck, and a decoder. The encoder consists of a set of GCAM blocks to highlight the features relevant to countable objects while suppressing others, and a set of Swin-Transformers to extract features. The GEFS in the bottleneck and the GAFU in the decoder enhance the features of objects of interest. Finally, a regression head generates a density map for estimating the number of objects.

F , following by feature encoding,

$$\mathbf{F}_i^E = \mathcal{F}_i^{\text{Down}}(\mathcal{F}_i^{\text{Swin-T}}(\mathcal{F}_i^{\text{GCAM}}(\mathbf{F}_{i-1}^E))), \quad (3.1)$$

where $\mathcal{F}_i^{\text{Down}}$, $\mathcal{F}_i^{\text{Swin-T}}$, $\mathcal{F}_i^{\text{GCAM}}$ denote down-sampling, GCAM, and Swin-T processing, and $\mathbf{F}_{i-1}^E$ and $\mathbf{F}_i^E$ are the input and output features at the i -th stage, respectively. GCAM enhances the token for countable objects and suppresses others.

At the bottleneck, the features are enhanced through GEFS, *i.e.*, $\mathbf{F}^{\text{BN}} = \mathcal{F}^{\text{GEFS}}(\mathbf{F}_K^E)$, where $\mathcal{F}^{\text{GEFS}}(\cdot)$ denotes the operation of GEFS, and $\mathbf{F}_K^E$ denotes the output features at the K -th stage of the encoder. GEFS selects the features corresponding to the countable object using a gate mechanism.

Subsequently, a set of Swin-Transformer blocks is utilized as the decoder to derive the density map. Specifically, the features at the j -th stage of the decoder are derived as,

$$\mathbf{F}_j^D = \mathcal{F}_j^{\text{Up}}(\mathcal{F}_j^{\text{Swin-T}}(\mathcal{F}_j^{\text{GAFU}}(\mathbf{F}_{j-1}^D, \mathbf{F}_{K+1-j}^E))), \quad (3.2)$$

where $\mathcal{F}_j^{\text{Up}}$, $\mathcal{F}_j^{\text{Swin-T}}$, and $\mathcal{F}_j^{\text{GAFU}}$ denote the operation of up-sampling, Swin-Transformer, and GAFU block, respectively. The GAFU enhances features through a gate mechanism, prioritizing crucial information with a dynamic assigned weight.

Finally, features are processed through a regression head,

$$\mathbf{F}^{\text{head}} = \mathcal{F}^{\text{Head}}(\mathbf{F}_K^D), \quad (3.3)$$

where $\mathcal{F}^{\text{Head}}$ denotes the regression head consisting of a series of convolutional blocks. The output is a density map that accurately represents the object count.

3.2.2 Swin-T Encoder with GCAM

A set of Swin-Transformer blocks forms the encoder to extract features related to countable objects. The GCAM adapts a dynamic token modulation to simultaneously exploit the attentive support of tokens relevant to countable objects and suppress features of irrelevant objects [46–48]. This process facilitates self-probing among objects and precise capture of objects of the same category for ZSC.

Specifically, we first compress token features $\mathbf{F}_i^E$ using an MLP, $\mathbf{F}_i^{\text{proj}} = \mathcal{F}^{\text{MLP}}(\mathbf{F}_i^E)$. To identify the objects of interest, we resort to two key observations: 1) The objects should be salient enough to step out from the background; 2) Similar objects could support each other to boost the saliency. The former is exploited by computing the average feature score C_i for each token through average pooling $\mathcal{F}^{\text{AVG}}$ as,

$$C_i = \mathcal{F}^{\text{AVG}}(\mathbf{F}_i^{\text{proj}}). \quad (3.4)$$

The score reflects the importance of tokens, prioritizing those with rich content. Tokens that frequently appear in similar contexts are more likely to be related to the target object of interest.

To identify them, we employ a similarity matrix,

$$\mathbf{S}_i = \sigma \left(\mathbf{F}_i^{\text{proj}} \times \mathbf{F}_i^{\text{proj}^T} \right), \quad (3.5)$$

where σ is a softmax function to normalize similarities across rows. $\mathbf{S}_i$ captures the semantic similarity of tokens in a spatial context to emphasize tokens that repeatedly share similar features, thereby emphasizing potential countable objects. A mask $\mathbf{M}_i$ is derived by aggregating $\mathbf{S}_i$ and $\mathbf{C}_i$ as,

$$\mathbf{M}_i = \sigma \left(\mathcal{F}^{\text{MLP}} (\mathbf{S}_i \times \mathbf{C}_i) \right). \quad (3.6)$$

$\mathbf{C}_i$ encodes the token importance when considering the token alone, while $\mathbf{S}_i$ encodes the token importance after interacting with other tokens. The tokens are then filtered by the mask as,

$$\mathbf{F}_i^{\text{GCAM}} = \mathcal{F}^{\text{Linear}}(\mathcal{F}^{\text{LN}}(\mathcal{F}^{\text{LN}}(\mathbf{F}_i^{\text{E}} \odot \mathbf{M}_i) + \mathbf{F}_i^{\text{proj}}))), \quad (3.7)$$

where $\odot$, $\mathcal{F}^{\text{LN}}$, and $\mathcal{F}^{\text{Linear}}$ denote element-wise product, layer normalization, and linear layer, respectively. The GCAM applies the mask $\mathbf{M}_i$ to $\mathbf{F}_i^{\text{E}}$, filtering out less relevant features and reinforcing those critical ones for countable objects.

The proposed GCAM selectively amplifies the importance of tokens related to significant object features through pairwise similarities. It is significantly different from LOCA [16] and DAVE [14], which depend on predefined prototypes to predict object densities. Instead, our GCAM leverages the self-similarity matrix for more dynamic and precise modulation of features. It is also different from RCC [2] that relies on global feature comparisons, and CounTR [1] that utilizes attention-driven similarity matrices. The GCAM emphasizes a clear distinction between relevant and irrelevant tokens.

Table 3.1: Comparison with other methods on the FSC-147 dataset [3], with best results highlighted in bold.

Category	Method	Backbone	Resolution	Test Set		Val Set	
				MAE	RMSE	MAE	RMSE
CAC	GMN [12] ACCV'18	ResNet-50	255	37.86	141.39	39.02	106.06
	FamNet [3] CVPR'21	ResNet-50	384	32.27	131.46	32.15	98.75
	LOCA [16] ICCV'23	ResNet-50	512	16.22	103.96	17.43	54.96
	DAVE [14] CVPR'24	ResNet-50	512	15.14	103.49	15.54	52.67
	CounTR [1] BMVC'22	ViT-B & CNN	384	14.71	106.87	18.07	71.84
ZSC	RepRPN-C [15] ACCV'22	ResNet-50	[384,1504]	26.66	129.11	29.24	98.11
	RCC [2] CVPRW'23	ViT-B	224	17.12	104.53	17.49	58.81
	Proposed GCA-SUNet	SwinT-B	384	14.08	84.77	17.02	55.37

3.2.3 Bottleneck with GEFS

The proposed GEFS selectively filters out features in the bottleneck that are semantically irrelevant to the objects of interest, while allowing critical condensed features to pass. It is implemented by first deriving the local token weights as $\mathbf{W}^{\text{GEFS}} = \sigma(\mathcal{F}^{\text{MLP}}(\mathbf{F}_K^{\text{E}}))$, and applying them

to features, as,

$$\mathbf{F}_0^D = \mathbf{F}_K^E + (\mathbf{W}^{\text{GEFS}} \odot (\mathcal{F}^{\text{ATTN.}}(\mathcal{F}^{\text{ATTN.}}(\mathbf{F}_K^E)))). \quad (3.8)$$

The GEFS is positioned at the bottleneck where features transit from the down-sampling pathway to the up-sampling pathway. As a vital bottleneck, the GEFS compresses and filters essential object-related features, ensuring that only the most relevant information of countable objects is advanced into the up-sampling pathway. Specifically, the attention blocks allow the model to capture a more comprehensive range of relationships and refine its ability to extract high-level semantic representations, resulting in more accurate and semantically enriched feature embeddings. Furthermore, a gate mechanism is incorporated into GEFS to selectively prioritize specific aspects of this condensed representation, effectively filtering out less relevant semantics. This process not only refines features by strengthening inter-dependencies between relevant features, but also lays a solid foundation for the comprehensive reconstruction of the up-sampling pathway.

3.2.4 Swin-T Decoder with GAFU

The decoder contains a set of Swin-T blocks to articulate the density map, and a set of Gated Adaptive Fusion Units (GAFUs) to integrate low-level features from the encoder with abstract features from the up-sampling pathway. In each GAFU, the token weights are determined by a gate mechanism as, $\mathbf{W}^{\text{GAFU}} = \sigma(\mathcal{F}^{\text{MLP}}(\mathbf{F}_i^E))$, and modulate the features as,

$$\mathbf{F}_i^G = \mathbf{F}_i^E + (\mathbf{W}^{\text{GAFU}} \odot \mathbf{F}_i^E). \quad (3.9)$$

Subsequently, these features are fused with the decoder features as $\mathbf{F}_j^{\text{GAFU}} = \mathcal{F}^{\text{Linear}}([\mathbf{F}_j^D, \mathbf{F}_i^G])$. By weighing the features during the fusion process, the GAFU effectively concentrates on semantic information pertinent to countable objects, minimizing the interference from irrelevant details.

3.2.5 Discussion of Gate Mechanisms

Gate mechanisms are incorporated into the three proposed modules, GCAM, GEFS, and GAFU. They share a similar underlying mechanism, but they are tailored to address distinct challenges. 1) Gates in GCAM tackle the challenges of differentiating countable objects from others by emphasizing the relevant tokens through a similarity-based gate. It effectively probes countable objects within the same category through a similarity matrix. 2) Gates in GEFS or GAFU tackle the challenges of effectively differentiating foreground features from the background context, *i.e.*, to suppress irrelevant low-level features while preserving essential information on objects of interest. By reinforcing relevant inter-dependencies, it prioritizes the semantic information of countable objects, thereby enhancing its ability to distinguish the objects from the background when constructing the density map.

3.3 Experimental Results

3.3.1 Experimental Settings

Two benchmark datasets are utilized for evaluation. Following [1, 14, 16], we employ the Mean Average Error (MAE) and Root Mean Squared Error (RMSE) as evaluation metrics.

FSC-147 [3] is collected from the real world and it consists of 6,135 images of 147 categories with 7 to 3,731 objects per image, mainly composed of foods, animals, kitchen utensils, and vehicles. It is officially split into 3,659, 1,286, and 1,190 images for training, validation, and testing, respectively.

CARPK [6] comprises 1,448 images taken from four parking lots using a bird’s-eye view. It is primarily intended for object counting and vehicle localization tasks, and it is officially split into 989 training images and 459 testing images.

The Swin-T blocks are pre-trained on ImageNet-22k [49], and other modules are randomly initialized. AdamW optimizer is employed, with an initial learning rate of 0.003, a decay rate of 0.95, and a batch size of 16. The model is trained with a warm-up period of 50 epochs and 300 epochs in total. The input image size is 384×384 . Data augmentation [1] is employed to facilitate efficient training. All experiments are conducted using two NVIDIA RTX A5000 GPUs.

3.3.2 Comparison Results on FSC-147 Dataset

Comparison experiments are conducted on the FSC-147 dataset. The results are summarized in Table 3.1. Following the practice in [1, 2], we report the errors on the test and validation sets. We have the following observations. 1) The proposed GCA-SUNet outperforms all the compared methods regarding test errors, while performing slightly poorer than DAVE [14] regarding validation errors. The superior results demonstrate its effectiveness for ZSC tasks. It outperforms not only the dedicated methods for solving ZSC tasks, such as RepRPN-C [15] and RCC [2], but also state-of-the-art models for CAC tasks. Compared to the second-best method, CounTR [1], the performance gain is 0.63 for MAE and 16.47 for RMSE. 2) Although the GCA-SUNet performs slightly poorer than DAVE [14] in terms of validation errors, it significantly outperforms DAVE in terms of test errors, *i.e.*, a performance gain of 1.06 for MAE and 18.72 for RMSE. DAVE performs well on the validation set and generalizes poorly on the test set. In contrast, the GCA-SUNet generalizes well on the novel test set with minimal errors.

We visually compare the density maps with CounTR [1] and RCC [2] on the FSC-147 dataset. As shown in Fig. 3.2, our method can capture fine-grained details of objects. In contrast, CounTR sometimes generates density maps that do not accurately distinguish between individual objects in the map, *e.g.*, in the fourth column, CounTR can’t identify the far-away small fruit, while our method can.

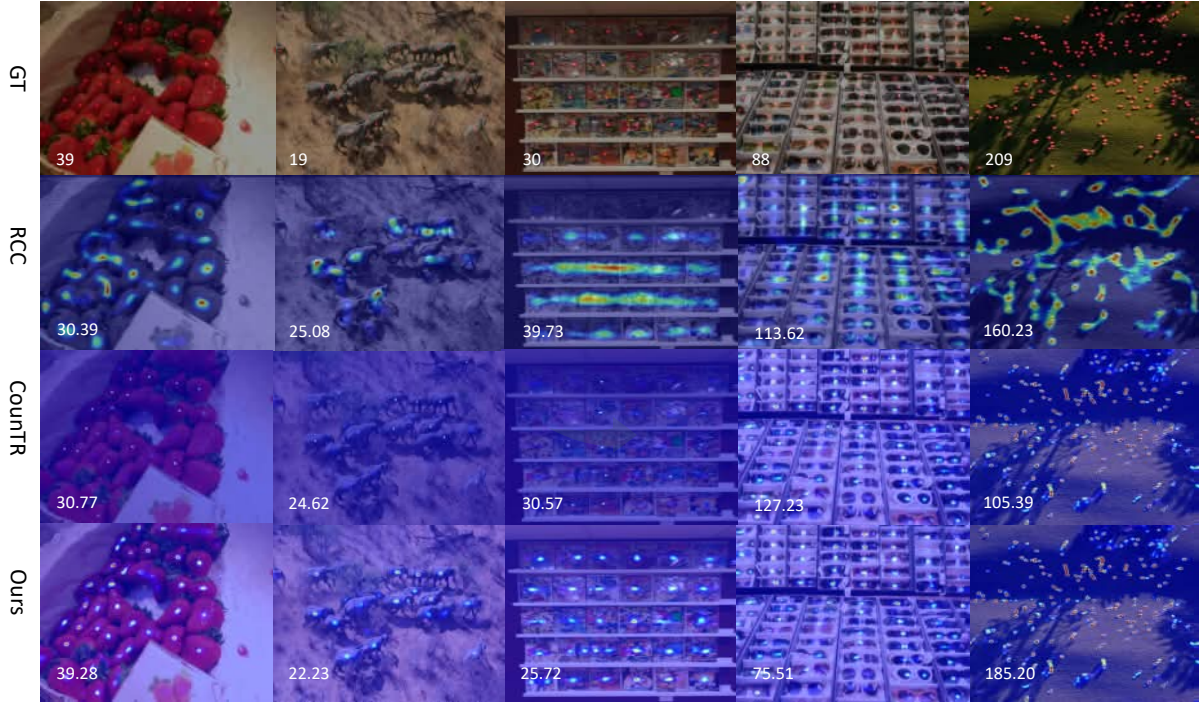


Figure 3.2: Visual comparisons of the generated density maps with CounTR [1] and RCC [2] on the FSC-147 dataset.

3.3.3 Cross-Domain Evaluation

Following [17], we conduct a cross-domain evaluation, *i.e.*, training the model on the FSC-147 dataset [3] while directly evaluating on the CARPK dataset [6]. The results are summarized in Table 4.3. The results for all the compared methods are obtained from [1, 2, 16] under the same settings. As shown in Table 4.3, our model shows superior cross-domain performance compared with other methods, achieving a performance gain of 0.56 on MAE and 0.61 on RMSE compared to the previous best-performing method, CounTR. Compared to the earlier ZSC model [2], the gains are even more significant, highlighting GCA-SUNet’s superior generalization over the compared methods.

Table 3.2: Training on the FSC-147 dataset [3] while testing on the CARPK dataset [6].

Category	Method	Test Set	
		MAE	RMSE
CAC	LOCA [16] ICCV’23	16.84	19.72
	CounTR [1] BMVC’22	11.52	14.56
ZSC	RCC [2] CVPR’23	21.38	26.61
	Proposed GCA-SUNet	10.96	13.95

3.3.4 Ablation Study of Major Modules

We ablate the three major modules of the proposed method on the FSC-147 dataset [3]. Swin-UNet [21] serves as the baseline model. The results are summarized in Table 3.3. The GCAM

Table 3.3: Ablation study of each component on the FSC-147 dataset [3].

$\mathcal{F}^{\text{GCAM}}$	$\mathcal{F}^{\text{GEFS}}$	$\mathcal{F}^{\text{GAFU}}$	Test Set		Val Set	
			MAE	RMSE	MAE	RMSE
✗	✗	✗	18.86	89.00	22.85	85.16
✓	✗	✗	16.54	85.20	21.13	81.30
✗	✓	✗	16.63	80.33	20.13	62.98
✗	✗	✓	16.87	84.36	21.60	79.37
✓	✗	✓	15.27	84.81	20.08	81.58
✗	✓	✓	15.07	78.79	18.58	59.17
✓	✓	✗	15.21	85.71	17.97	55.67
✓	✓	✓	14.08	84.77	17.02	55.37

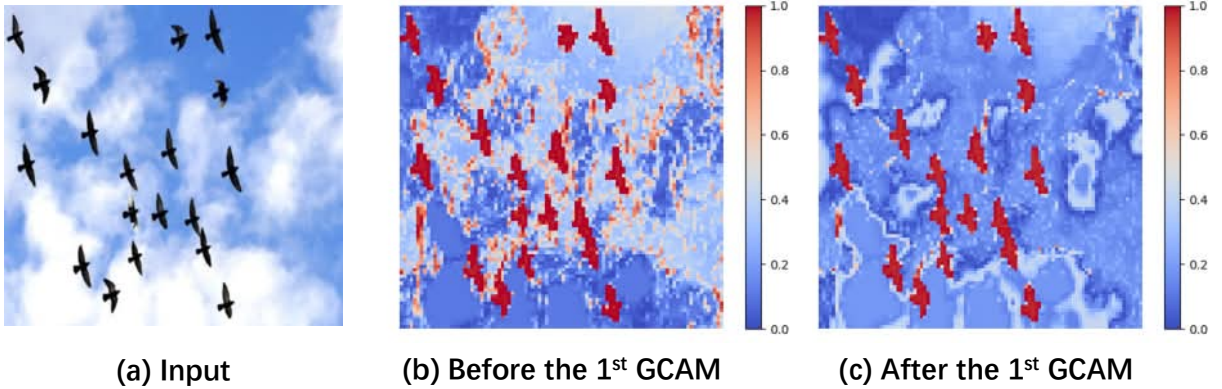


Figure 3.3: Feature maps before/after the 1st GCAM in the Swin-T encoder.

module alone significantly reduces the MAE by 2.32 on the test set and by 1.72 on the validation set compared with the baseline, underscoring its capability to enhance feature selectivity that is essential for complex scenes. Similarly, utilizing GEFS or GAFU individually also greatly reduces errors on both test and validation sets, showcasing the proposed gate mechanism’s ability to highlight relevant features while suppressing irrelevant ones. The combination of these modules also achieves significant performance gains. Specifically, the overall framework with all three components achieves the most substantial improvement compared with the baseline, reducing the MAE by 4.78 on the test set and by 5.83 on the validation set. The results show the effectiveness of the synergistic interaction among different modules and emphasize the individual contribution of each major component.

3.3.5 Analysis of GCAM

As shown in Table 4.5, GCAM contributes most significantly to the overall performance gain. We further visualize its effects as the density maps in Fig. 3.3. Following the practice in [2], sub-figures (b) and (c) are obtained by projecting the density maps into two-dimensional spaces. Clearly, by applying GCAM, the noisy estimation in the background areas (sky) is significantly reduced, with the foreground objects (birds) becoming more prominent. The results show that the proposed GCAM effectively enhances the representation of foreground tokens while sup-

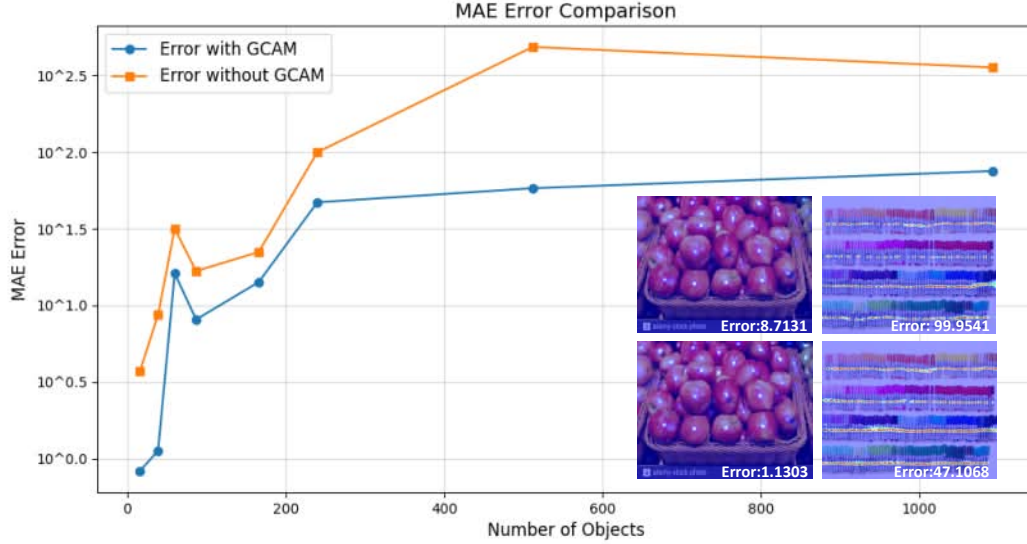


Figure 3.4: Evaluation of GCAM in various object density scenarios.

pressing irrelevant ones in the background.

We further compare the results with/without the GCAM across scenarios of different object densities in Fig. 3.4. For different object densities, the GCAM could consistently reduce the MAE, signifying its importance. On the other hand, Fig. 3.4 reveals a clear trend of increasing error as the number of objects grows. Indeed, as the density of objects increases, they become smaller, and counting tasks become more challenging. Future work will focus on optimizing the proposed approach to improve its performance in such challenging scenes.

3.3.6 Model Complexity Analysis

Tab. 4.6 compares the inference time, model size, FLOPs, and training epochs with the previous best method, CounTR [1]. The proposed model requires inference time comparable to CounTR. It significantly outperforms CounTR in terms of accuracy, at a higher but reasonable computational cost.

Table 3.4: Comparison of inference time, model size, FLOPs, and training epochs.

Method	Inf. time (s)	Parameters (M)	FLOPs (G)	Training (epochs)
CounTR [1]	0.08	100	84	1000
Proposed GCA-SUNet	0.11	192	173	300

3.4 Chapter Summary

The proposed GCA-SUNet effectively tackles the problems of ZSC by using a Swin-UNet architecture to directly map the input image to the density map of countable objects. The proposed GCAM exploits the attention information among the tokens of repetitive objects through the self-similarity matrix, and suppresses the features of irrelevant objects through a gate mechanism. The gate mechanism is also incorporated into the GEFS module and the GAFU module, which highlight the features most relevant to countable objects while suppressing irrelevant ones. Our experiments on the FSC-147 and CARPK datasets demonstrate that GCA-SUNet outperforms state-of-the-art methods, achieving superior performance in both intra-domain and cross-domain scenarios.

Despite its effectiveness, it suffers from inherent limitations. In particular, its performance is generally inferior to that of FSC methods, a consequence attributable to the absence of explicit semantic guidance. While ZSC offers the advantage of circumventing the need for human annotations, which serves to reduce annotation costs, this benefit is offset by a loss in semantic alignment between the model and the visual data. Consequently, the importance of visual exemplars in CAC tasks remains evident. Motivated by these observations, our research further investigates the FSC paradigm to develop models that are both robust and generalizable for CAC applications.

Chapter 4

CSG: A Cross-Modal Semantic-Guided Group-Attended Framework for Few-Shot Counting

CAC aims to accurately count objects of interest using only a few visual exemplars or text prompts without relying on extensive annotations. In this work, we present a Cross-modal Semantic-guided Group-attended (CSG) framework for CAC tasks. Our method leverages a large vision-language model (LLaVA) to generate dual text descriptions for both the query image and exemplars, which are then encoded by BERT. Simultaneously, visual features are extracted from the query and exemplars using a Swin-Transformer, forming four sets of multi-modal inputs. To achieve deep and effective cross-modal interactions, two Gated Intra-modal Contrastive Embedding (GICE) blocks are employed to refine the query features by highlighting tokens that are highly correlated with exemplar features in both the visual and linguistic domains. In addition, a Cross-modal Semantic Refinement (CSR) module is used to extract and refine exemplars’ semantics through cross-modal interactions, and a Semantic-Guided Group-Attention (SGGA) module further aggregates shared characteristics among candidate objects via a set of attention mechanisms. Extensive experiments on FSC-147, CARPK, and Count-Bench datasets demonstrate that CSG significantly outperforms state-of-the-art methods in both accuracy and robustness. The code will be available upon paper acceptance.

4.1 Introduction

Counting objects such as vehicles [50], crowd [51, 52], and cells [9] is crucial in many applications, *e.g.*, traffic monitoring [53], remote sensing [54], and biomarker statistics [55]. Early methods count objects of specific classes, but can’t count objects not covered in training [10, 11]. Recently, Class-agnostic counting has emerged as a promising research area, counting objects during inference without retraining the model [3]. They do not require large-scale annotated samples, but only a few visual exemplars [1, 17] and/or text prompts [29, 32] to specify classes. CAC tasks are more challenging but potentially have wider applications than counting specific classes of objects.

Existing CAC models can be broadly grouped as follows. 1) FSC [1], which counts objects using limited visual exemplars extracted from the query image. Recent models focus on estimating object density maps using exemplar patches [17, 36], or detecting candidate objects and selecting relevant ones through exemplar comparison [14, 20]. 2) PBC [29, 30], which utilizes textual prompts to quantify objects, leveraging Vision-language models to establish

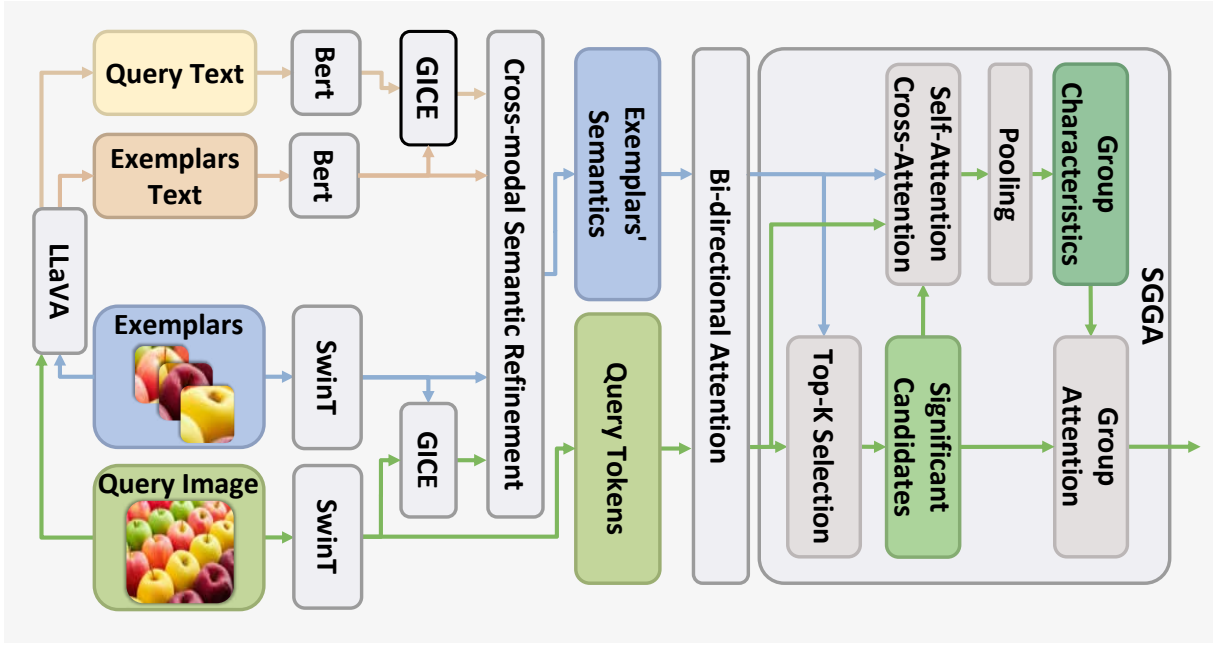


Figure 4.1: The proposed CSG leverages LLaVA to generate text descriptions of both query image and exemplars, forming four sets of features. The query and exemplars perform intra-domain interactions through GICEs and cross-domain interactions through CSR modules. Significant candidates are extracted from query features, and group common characteristics are extracted in SGGA, which in turn highlight the significant candidates, further enhancing the counting accuracy.

cross-modal alignment between textual prompts and visual query features, and counting objects without explicit visual exemplars. 3) ZSC [2, 25], which autonomously identifies countable regions by self-probing potential objects through pre-trained vision transformers [2] or synthesizing pseudo-exemplars [15], without explicit external guidance.

Despite these advancements, existing CAC models still face challenges. 1) Visual exemplars often overemphasize visual details, but limited exemplars may not provide sufficient semantic guidance [3]. In contrast, textual cues provide a comprehensive understanding of the target object but lack a specific visual context assigned to that object [32]. Only very recently, CountGD [20] integrated both visual exemplars and text prompts to exploit visual clues and textual semantics of countable objects [20], but they heavily relied on expensive prompt annotations. 2) Four sets of visual and linguistic features can be extracted from the query and exemplars, posing significant challenges for deeply interacting with them, while most methods utilize either visual clues only [27], or text prompts only [30]. 3) Lastly, the shared characteristics of countable objects in the query images are crucial clues but only very few methods partially exploit them, *e.g.*, Patch-Selection [29] aggregated high-quality exemplars from class-relevant patches [29], and CountGD [20] aggregated a unified semantics of countable objects through the token similarity [20].

To leverage multi-modal features while avoiding extensive annotations of prompts, we propose a **Cross-modal Semantic-guided Group-attended (CSG)** framework for few-shot counting. As shown in Fig. 4.1, we first leverage LLaVA [56] to automatically generate two distinct descriptions with high-level semantics by using two specifically designed prompts. Trained on a large number of vision-language pairs, LLaVA incorporates extensive knowledge about

the target objects that limited exemplars alone cannot provide. These descriptions eliminate the need for labor-intensive annotations while enabling multi-modal object counting. We then utilize BERT [33] to extract two sets of linguistic features from the descriptions, and Swin-Transformer [49] to extract two sets of visual features from the query and exemplars, respectively, forming four sets of features.

Then, we propose two modules to deeply interact with these four sets of features. 1) **Gated Intra-modal Contrastive Embedding (GICE)**. Two GICE blocks are developed to interact with the query and exemplars in the visual and linguistic domains, separately. Specifically, we leverage the contrastive embeddings [18] to derive the similarity matrix between the query and exemplars, highlighting the query tokens that are most similar to the exemplars. Then, salient tokens are emphasized by using saliency scores that are derived by average pooling over the query tokens. These two mechanisms help the model dynamically adapt to multimodal inputs and selectively enhance salient and relevant query tokens. 2) **Cross-modal Semantic Refinement (CSR)**, which extracts and refines the exemplars’ semantics through cross-modal interaction among the four sets of features, and mutually enhances the query tokens and exemplars’ semantics through bi-directional attention. Following [33], we embed the query’s semantics in the $[CLS]$ token by interacting with their visual and linguistic features through a self-attention mechanism, and similarly embed the exemplars’ semantics. The two semantics are then concatenated and compressed by a Multi-Layer Perceptron (MLP), and further interacted with exemplars’ visual tokens and linguistic tokens through two self-attention blocks, respectively, to derive the exemplified semantics. Then, visual query tokens and exemplified semantics are mutually enhanced through bi-directional attention [18], achieving both query-to-semantic and semantic-to-query cross-attention enhancements.

Lastly, to exploit the common characteristics of significant objects, we propose a **Semantic-Guided Group-Attention (SGGA)** module. We first perform Top- K selection to select the most relevant object candidates from the query, guided by the exemplars’ semantics. The selected candidates are further boosted by two cross-attention operations, *i.e.*, the first integrates the visual-linguistic clues from the query image, and the second incorporates visual-linguistic semantics from the exemplars. Then, we obtain the common characteristics of candidates through average pooling and utilize them to highlight the relevant candidates through a set of attention mechanisms. The proposed SGGA significantly improves the distinctiveness and granularity of candidate features by deeply exploiting the common characteristics embedded in the group of candidates.

Our contributions can be summarized as follows. 1) The proposed CSG leverages LLaVA to generate text descriptions, extracting and exploiting four sets of visual and linguistic features from the query image and exemplars, avoiding extensive annotations of prompts. 2) The two GICE blocks deeply interact with the four distinct sets of features in visual and linguistic domains separately, and the CSR module extracts the visual-linguistic semantics of exemplars and mutually enhances the query tokens and exemplars’ semantics through bi-directional attention. 3) The SGGA further highlights countable objects under the guidance of exemplars’ semantics, and exploits the group characteristics of objects to further enhance their features, effectively differentiating salient objects from background and other obfuscations. 4) The proposed model is comprehensively evaluated on the FSC-147 [3], CARPK [4] and CountBench datasets [5], outperforming state-of-the-art methods for few-shot counting.

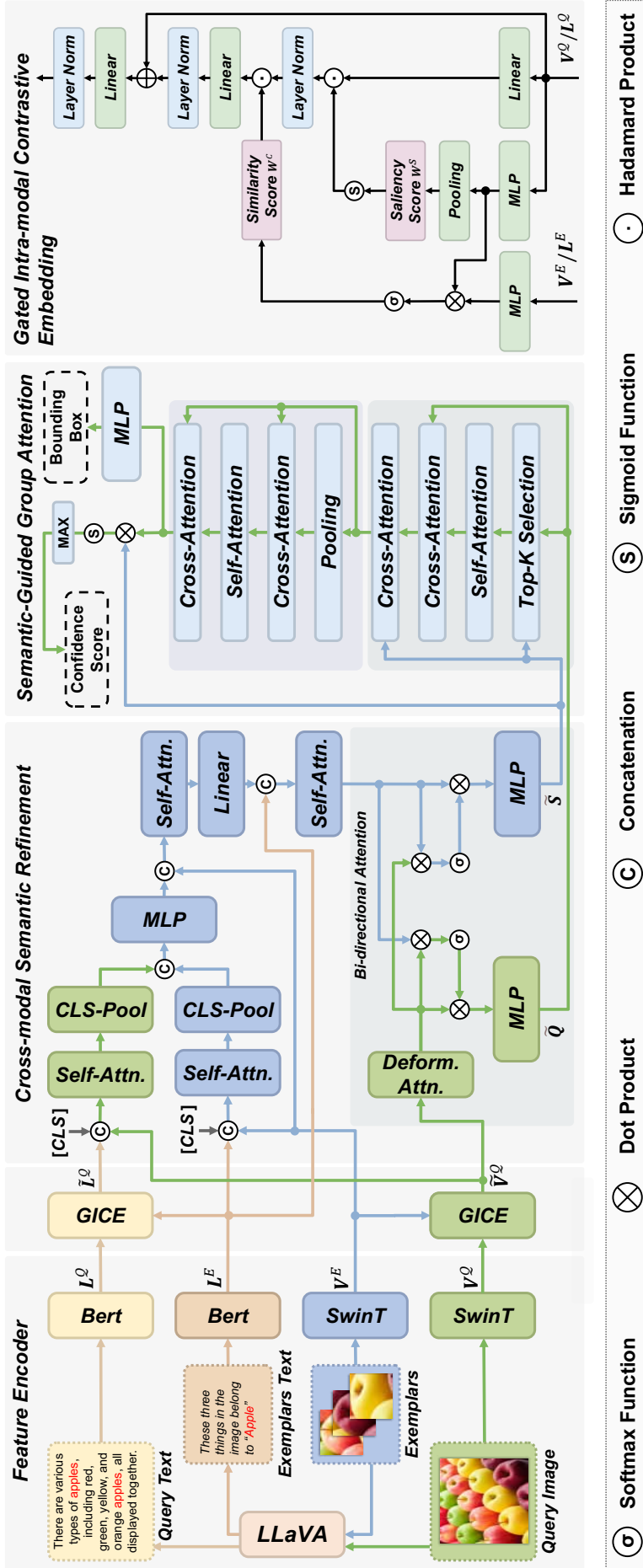


Figure 4.2: The proposed CSG contains four building blocks. 1) Feature Learning: LLaVA is utilized to generate text descriptions for the query image and exemplars, forming four sets of features by using BERT and SwinT. 2) Gated Intra-modal Contrastive Embedding: The query and exemplars interact in visual and linguistic domains separately. 3) Cross-modal Semantic Refinement: Exemplar’s semantics are extracted and interact with the query tokens through a bi-directional attention. 4) Semantic-Guided Group Attention: The common features are extracted from object candidates and interact with candidates through group attention.

4.2 Proposed CSG

4.2.1 Overview of Proposed Method

To tackle the challenges of few-shot counting, we propose the Cross-modal Semantic-guided Group-attended (CSG) framework. Fig. 4.1 illustrates the block diagram. Formally, given a query image $\mathbf{I}^Q \in \mathbb{R}^{H \times W \times 3}$ and a few visual exemplars $\{\mathbf{I}_i^E \in \mathbb{R}^{H^E \times W^E \times 3}\}_{i=1}^K$, the proposed CSG counts the number of objects similar to the exemplars, where H, W, H^E and W^E are the dimensions for the query image and exemplars, respectively, and K is the number of exemplars.

Firstly, a pre-trained vision-language model, LLaVA [56] Φ^{LLaVA} , is utilized to generate text descriptions $\mathbf{t}^Q \in \mathbb{R}^{T^Q \times 1}$ and $\mathbf{t}^E \in \mathbb{R}^{T^E \times 1}$ for the query and exemplars with two specifically designed prompts respectively, *i.e.*, $\mathbf{P}^Q = \text{"What kinds of objects are there?"}$ and $\mathbf{P}^E = \text{"What do these three things in the image belong to?"}$,

$$\mathbf{t}^Q = \Phi^{\text{LLaVA}}(\mathbf{I}^Q, \mathbf{P}^Q), \quad \mathbf{t}^E = \Phi^{\text{LLaVA}}(\{\mathbf{I}_i^E\}_{i=1}^K, \mathbf{P}^E), \quad (4.1)$$

where T^Q and T^E denote the lengths of the respective text descriptions. We then utilize a Swin-Transformer Φ^{SWINT} to generate the visual features $\mathbf{V}^Q = \Phi^{\text{SWINT}}(\mathbf{I}^Q) \in \mathbb{R}^{V^Q \times D}$ and $\mathbf{V}^E = \Phi^{\text{SWINT}}(\{\mathbf{I}_i^E\}_{i=1}^K) \in \mathbb{R}^{K \times D}$ for the query and exemplars, where D is the channel dimension. Note that one token is generated for each visual exemplar. In parallel, a pre-trained BERT Φ^{BERT} is utilized to obtain the corresponding linguistic embeddings $\mathbf{L}^Q = \Phi^{\text{BERT}}(\mathbf{t}^Q) \in \mathbb{R}^{L^Q \times D}$ and $\mathbf{L}^E = \Phi^{\text{BERT}}(\mathbf{t}^E) \in \mathbb{R}^{L^E \times D}$, forming four sets of visual and linguistic features, $\{\mathbf{V}^Q, \mathbf{V}^E, \mathbf{L}^Q, \mathbf{L}^E\}$, for object counting.

Secondly, two **G**ated **I**ntra-modal **C**ontrastive **I**nteraction (**GICE**) blocks interact with the query and exemplars visually and linguistically,

$$\tilde{\mathbf{V}}^Q = \Phi^{\text{GICE}}(\mathbf{V}^Q, \mathbf{V}^E) \in \mathbb{R}^{V^Q \times D}, \quad (4.2)$$

$$\tilde{\mathbf{L}}^Q = \Phi^{\text{GICE}}(\mathbf{L}^Q, \mathbf{L}^E) \in \mathbb{R}^{L^Q \times D}. \quad (4.3)$$

Φ^{GICE} highlights the significant query tokens relevant to exemplars.

Thirdly, the **C**ross-modal **S**emantic **R**efinement (**CSR**) block Φ^{CSR} extracts the exemplars' semantics, progressively refines them by exemplars' visual tokens and linguistic tokens to generate aligned visual-linguistic semantics, and mutually boosts the query tokens through a bi-directional attention [18, 57]. Formally,

$$\{\tilde{\mathbf{Q}}, \tilde{\mathbf{S}}\} = \Phi^{\text{CSR}}(\tilde{\mathbf{V}}^Q, \tilde{\mathbf{L}}^Q, \mathbf{V}^E, \mathbf{L}^E), \quad (4.4)$$

where $\tilde{\mathbf{Q}} \in \mathbb{R}^{V^Q \times D}$ denotes the enhanced query and $\tilde{\mathbf{S}} \in \mathbb{R}^{(K+L^E) \times D}$ denotes the exemplar's semantics in the visual-linguistic domain.

Lastly, the proposed **S**emantic-**G**uided **G**roup-**A**ttention (**SGGA**) module Φ^{SGGA} exploits the attentive supports of the group of object candidates. The SGGA first selects the top-K representative object candidates, and then boosts the candidates' features through one self-attention, and two cross-attentions with the query tokens and exemplars' semantics. The common characteristics among these candidates are obtained through an average pooling, and refined through a self-attention to enforce intra-class coherence and two cross-attentions with candidates' fea-

tures to highlight relevant features. The refined common characteristics are then applied to candidates' features to derive countable objects. Formally,

$$\{\mathbf{B}, \gamma\} = \Phi^{\text{SGGA}}(\tilde{\mathbf{Q}}, \tilde{\mathbf{S}}), \quad (4.5)$$

where $\mathbf{B} \in \mathbb{R}^{Z \times 4}$ denote the set of bounding boxes with associated confidence scores $\gamma \in \mathbb{R}^{Z \times 1}$, and Z is the number of objects.

4.2.2 Gated Intra-modal Contrastive Embedding

As shown in Fig. 4.2, the proposed GICE blocks facilitate effective intra-domain interactions between the query and exemplars, one in the visual domain and the other in the linguistic domain. They highlight the query tokens relevant to the exemplars. For illustration, we introduce the GICE block in the visual domain here. The linguistic GICE block follows the same process.

Token Saliency

Salient tokens are emphasized by their saliency. Specifically, we first align the query features $\mathbf{V}^Q$ and exemplars' features $\mathbf{V}^E$ using an MLP as $\mathbf{F}^Q = \mathcal{F}^{\text{MLP}}(\mathbf{V}^Q) \in \mathbb{R}^{V^Q \times D}$ and $\mathbf{F}^E = \mathcal{F}^{\text{MLP}}(\mathbf{V}^E) \in \mathbb{R}^{K \times D}$. The token saliency is derived by an average-pooling function $\mathcal{F}^{\text{AVG}}(\cdot)$ over the query tokens $\mathbf{F}^Q$, followed by a sigmoid activation function $\mathcal{F}^{\text{SIGMOID}}(\cdot)$,

$$\mathbf{w}^S = \mathcal{F}^{\text{SIGMOID}}(\mathcal{F}^{\text{AVG}}(\mathbf{F}^Q)), \quad (4.6)$$

where $\mathbf{w}^S \in \mathbb{R}^{V^Q \times 1}$ denotes the saliency scores. The average pooling aggregates the query tokens across the channel dimension, summarizing the overall response of each token and indicating their saliency, *i.e.*, tokens that consistently exhibit large responses are more important. We then replicate $\mathbf{w}^S$ D times to form the weighting matrix $\mathbf{W}^S \in \mathbb{R}^{V^Q \times D}$, and apply it on $\mathbf{F}^Q$ as,

$$\mathbf{F}^G = \mathcal{F}^{\text{LN}}(\mathbf{W}^S \odot \mathcal{F}^{\text{LINEAR}}(\mathbf{F}^Q)), \quad (4.7)$$

where $\mathbf{F}^G$, $\mathcal{F}^{\text{LN}}(\cdot)$, $\odot$, and $\mathcal{F}^{\text{LINEAR}}(\cdot)$ denote the resulting salient tokens, layer normalization, Hadamard product, and linear transformation, respectively. The linear layer projects $\mathbf{F}^Q$ into a new representation space for enhancing the model's dynamics, while the layer normalization mitigates channel-wise scale variations.

Different from uniformly processing all tokens [17], this adaptive weighting strategy enables the model to selectively emphasize informative tokens, thus enhancing the robustness and fine-grained representations of countable objects.

Contrastive Embedding

Many mechanisms have been developed to facilitate interactions between two sets of features, *e.g.*, attention-based fusion [58], cross-attention [59], gated multimodal unit [60], and feature-

wise linear modulation [61]. But these methods fail to establish an effective and tightly coupled feature interaction [62]. In particular, attention-based fusion [58] requires extensive calculations, leading to increased computational overhead. Contrastive embedding [18] has been utilized as a loss for cross-modal matching. We adapt it to emphasize query tokens that are similar to exemplars' tokens with a relatively low computational cost, highlighting strongly correlated query-exemplar token pairs.

Specifically, we obtain the contrastive embedding between the query and exemplars as $\mathbf{E} = \sigma(\mathbf{F}^Q \times \mathbf{F}^{E^\top}) \in \mathbb{R}^{V^Q \times K}$, where $\sigma(\cdot)$ denotes the softmax operation. $\mathbf{E}$ emphasizes the query tokens that are similar to the exemplars. Then, we derive the maximum similarity scores for query tokens as $\mathbf{w}^C = \mathcal{F}^{\text{MAX}}(\mathbf{E}) \in \mathbb{R}^{V^Q \times 1}$, where $\mathcal{F}^{\text{MAX}}(\cdot)$ denotes max-pooling, highlighting the strongest correspondences between the query's and exemplars' tokens. We then replicate $\mathbf{w}^C$ D times to form the weighting matrix $\mathbf{W}^C \in \mathbb{R}^{V^Q \times D}$, and apply it on $\mathbf{F}^G$ to produce the enhanced tokens $\mathbf{F}^P \in \mathbb{R}^{V^Q \times D}$ as,

$$\mathbf{F}^P = \mathcal{F}^{\text{LN}}(\mathcal{F}^{\text{LINEAR}}(\mathbf{W}^C \odot \mathbf{F}^G)). \quad (4.8)$$

Again, the linear layer enhances the model's dynamics, and the layer normalization mitigates scale variations. Compared to the traditional attention mechanism [26], the contrastive embedding requires much less computational power. Additionally, by explicitly modeling the pairwise similarities between query tokens and exemplar tokens, it highlights the query tokens relevant to exemplars. It is also more robust to deal with background noise and other irrelevant objects than cross-attention [59] by filtering out irrelevant or noisy tokens.

Finally, the enhanced tokens are integrated back into the query tokens to produce $\tilde{\mathbf{V}}^Q$, the output of the GICE block as,

$$\tilde{\mathbf{V}}^Q = \mathcal{F}^{\text{LN}}(\mathcal{F}^{\text{LINEAR}}(\mathbf{F}^P + \mathbf{F}^Q)). \quad (4.9)$$

The residual connection preserves the original query tokens.

The proposed GICE integrates two complementary gating mechanisms to enhance intra-domain interactions: 1) Saliency gating assigns token importance based on their informativeness; 2) Contrastive embedding computes query-exemplar similarity in a shared space to emphasize tokens relevant to exemplars. Together, these mechanisms highlight salient and relevant query tokens.

4.2.3 Cross-Modal Semantic Refinement

The CSR module consists of the following four steps: 1) It first extracts semantic features from the query and exemplars in the visual-linguistic domain through self-attention. 2) The semantics are further enhanced by visual and linguistic features of exemplars through self-attention. 3) In parallel, visual query tokens are enhanced through a deformable attention [63]. 4) A bi-directional attention [57] is utilized to mutually enhance both the query tokens and exemplars' semantics. As a result, we achieve cross-modal interactions between the query and exemplars in the aligned visual-linguistic domain.

Semantics Extraction

Following [33], we first aggregate two respective semantic tokens, $\mathbf{s}^Q \in \mathbb{R}^{1 \times D}$ and $\mathbf{s}^E \in \mathbb{R}^{1 \times D}$ for the query and exemplars respectively as,

$$\mathbf{s}^Q = \mathcal{F}^{\text{CLS-POOL}} \left(\mathcal{F}^{\text{ATTN}} \left(\{[\mathbf{CLS}], \tilde{\mathbf{V}}^Q, \tilde{\mathbf{L}}^Q\} \right) \right), \quad (4.10)$$

$$\mathbf{s}^E = \mathcal{F}^{\text{CLS-POOL}} \left(\mathcal{F}^{\text{ATTN}} \left(\{[\mathbf{CLS}], \mathbf{V}^E, \mathbf{L}^E\} \right) \right), \quad (4.11)$$

where $\mathcal{F}^{\text{CLS-POOL}}(\cdot)$ represents the CLS-pooling [33], $\mathcal{F}^{\text{ATTN}}(\cdot)$ represents the self-attention [59], and $[\mathbf{CLS}]$ denotes the token encapsulating learnable semantics. $\mathbf{s}^Q$ and $\mathbf{s}^E$ are then concatenated and passed through an MLP to generate a unified semantic token $\mathbf{s}^C = \mathcal{F}^{\text{MLP}}(\{\mathbf{s}^Q, \mathbf{s}^E\}) \in \mathbb{R}^{1 \times D}$, aggregating the multimodal information into a compact and informative semantic token. It greatly saves the subsequent computations compared to a traditional attention mechanism that utilizes the whole set of self-attended tokens [20].

Semantics Enhancement

The semantic token $\mathbf{s}^C$ is compact but lacks exemplars' visual and linguistic details. We hence enhance it through two self-attentions with exemplars' visual and linguistic tokens. Firstly, the self-attention followed by a linear projection is applied to $\mathbf{s}^C$ and exemplar's visual token $\mathbf{V}^E$ to obtain a visually-enhanced semantics $\mathbf{S}^V \in \mathbb{R}^{K \times D}$,

$$\mathbf{S}^V = \mathcal{F}^{\text{LINEAR}}(\mathcal{F}^{\text{ATTN}}(\{\mathcal{F}^{\text{REP}}(\mathbf{s}^C), \mathbf{V}^E\})), \quad (4.12)$$

where $\mathcal{F}^{\text{REP}}(\cdot)$ replicates the semantic token $\mathbf{s}^C$ K times, $\mathcal{F}^{\text{ATTN}}(\cdot)$ exploits the interactions between $\mathbf{s}^C$ and $\mathbf{V}^E$, and $\mathcal{F}^{\text{LINEAR}}(\cdot)$ compresses the $2K$ tokens back to K tokens. This step integrates high-level semantics $\mathbf{s}^C$ with exemplars' fine-grained visual tokens $\mathbf{V}^E$.

Secondly, $\mathbf{S}^V$ is further refined by interacting with exemplars' linguistic cues $\mathbf{L}^E$ via another self-attention, to derive the linguistically-enhanced semantic tokens $\mathbf{S}^A = \mathcal{F}^{\text{ATTN}}(\{\mathbf{S}^V, \mathbf{L}^E\}) \in \mathbb{R}^{(K+L^E) \times D}$. As a result, the semantic coherence is enhanced for exemplars' visual and linguistic features, ensuring consistent and complementary integration of both visual and linguistic modalities.

Different from cross-attention [19], our method aligns visual-linguistic semantics within a joint query-exemplar space, with special emphasis on exemplars' tokens. This is particularly critical for FSC tasks, where achieving the semantic consistency often proves more beneficial than directly augmenting query features [30].

Bi-Directional Attention

Bridging the gaps between low-level visual features and high-level semantics remains a fundamental challenge in vision-language understanding. To tackle this challenge, we leverage a Bi-Directional Attention [18, 57] to facilitate mutual refinement of query features and exemplars' semantics. Specifically, visual query tokens $\mathbf{V}^Q$ are first refined via a deformable

self-attention [63] as $\hat{\mathbf{V}}^Q = \mathcal{F}^{\text{DSA}}(\mathbf{V}^Q) \in \mathbb{R}^{V^Q \times D}$. As shown in [63], the deformable self-attention could dynamically warp sampling locations based on both image content and spatial structure. Then, a bi-direction attention mechanism [18, 57] is utilized to interact $\hat{\mathbf{V}}^Q$ with $\mathbf{S}^A$, capturing cross-modal relevance and generating mutually enhanced query features and exemplars' semantics as,

$$\tilde{\mathbf{Q}} = \mathcal{F}^{\text{MLP}}(\sigma(\hat{\mathbf{V}}^Q \times \mathbf{S}^{A^\top}) \times \mathbf{S}^A), \quad (4.13)$$

$$\tilde{\mathbf{S}} = \mathcal{F}^{\text{MLP}}(\sigma(\mathbf{S}^A \times \hat{\mathbf{V}}^{Q^\top}) \times \hat{\mathbf{V}}^Q). \quad (4.14)$$

A soft attention map is produced based on the similarity between $\hat{\mathbf{V}}^Q$ and $\mathbf{S}^A$, which guides the feature refinement, yielding semantically-enhanced visual query features $\tilde{\mathbf{Q}}$ and contextually-informed semantics $\tilde{\mathbf{S}}$. Unlike unidirectional attentions, the bi-directional attention enables a symmetric and continuous exchange of complementary information between the two sets of features, leading to more coherent feature integration, more effectively mitigating noise and ambiguity in the inputs, and yielding a richer joint representation.

4.2.4 Semantic-Guided Group-Attention Module

The SGGA module first extracts a set of object candidates from the query features $\tilde{\mathbf{Q}}$ by ranking $\tilde{\mathbf{Q}}$ based on their similarity with the semantics $\tilde{\mathbf{S}}$, and selects the top-K candidates. Then, a sequence of self-attention and cross-attention is adopted to highlight significant candidates by exploiting their similarities. Lastly, the common characteristics are extracted from these candidates, and progressively interact with candidates' features through group attention.

Specifically, the significant candidates $\mathbf{C}^S \in \mathbb{R}^{Z \times D}$ are generated by selecting K most significant candidates as $\mathbf{C}^S = \mathcal{F}^{\text{TOPK}}(\tilde{\mathbf{Q}})$, where $\mathcal{F}^{\text{TOPK}}(\cdot)$ denotes the Top-K Selection [64]. Subsequently, $\mathbf{C}^S$ is refined by a self-attention and two cross-attentions with the query tokens $\tilde{\mathbf{Q}}$ and the semantic guidance $\tilde{\mathbf{S}}$ as,

$$\mathbf{C} = \mathcal{F}^{\text{CA}} \left(\mathcal{F}^{\text{CA}} \left(\mathcal{F}^{\text{ATTN}}(\mathbf{C}^S), \tilde{\mathbf{Q}} \right), \tilde{\mathbf{S}} \right), \quad (4.15)$$

where $\mathcal{F}^{\text{CA}}(\cdot)$ represents the cross-attention, and $\mathbf{C} \in \mathbb{R}^{Z \times D}$ denotes the refined candidates' features. The first self-attention aggregates the context among the candidates to emphasize the common characteristics, while the cross-attention incorporates complementary cues from both the query features and semantic guidance, further refining the candidates' features. Compared to directly utilizing convolutional matching between features for candidate generation [14], our design leverages semantic similarity to choose highly confident object candidates and further refines them through cross-attention, resulting in more precise and effective candidates' representations.

The absence of sufficient exemplars often results in biased and ambiguous semantic representations. To tackle this issue, we exploit the common characteristics of candidates through group attention, *i.e.*, we first extract the common characteristics of candidates' features $\mathbf{C}$ through average-pooling $\mathcal{F}^{\text{AVG}}(\cdot)$ to generate group features, and then utilize the group features to refine

the candidates' features through a sequence of cross-attentions and self-attentions,

$$\tilde{C} = \mathcal{F}^{\text{CA}}(C, \mathcal{F}^{\text{ATTN}}(\mathcal{F}^{\text{CA}}(\mathcal{F}^{\text{AVG}}(C), C))), \quad (4.16)$$

where the first cross-attention integrates the contexts from candidates' features C into the group features, thus enriching their semantics. The self-attention is applied to further boost the description power, resulting in more coherent and robust group features. Subsequently, the candidates' features are enhanced via the cross-attention with the group features. The entire process enables contextualized refinement of C through the interaction with the group features.

Finally, the enhanced candidates' features $\tilde{C} \in \mathbb{R}^{Z \times D}$ are utilized to generate the bounding box $B \in \mathbb{R}^{Z \times 4}$ via an MLP head as $B = \mathcal{F}^{\text{MLP}}(\tilde{C})$. The confidence scores $\gamma \in \mathbb{R}^{Z \times 1}$ are computed as $\gamma = \mathcal{F}^{\text{MAX}}(\mathcal{F}^{\text{SIGMOID}}(\tilde{C} \times \tilde{S}^\top))$, *i.e.*, candidates that are most similar to exemplars' semantics are counted as objects.

The proposed SGGA leverages semantic similarity to extract significant candidates and then refines them through a set of attentions. Then, it extracts the group features and progressively refines them by interacting with significant candidates, effectively enhancing the semantic coherence among objects and mitigating ambiguity.

4.2.5 Loss Function

The loss comprises two terms: a localization loss and a classification loss, jointly encouraging spatial precision and robust categorization.

$$\mathcal{L} = \lambda_{\text{LOC}} \|\mathbf{c}^P - \mathbf{c}^T\|_1 + \lambda_{\text{FL}} \mathcal{F}^{\text{FL}}(\gamma, \gamma^T), \quad (4.17)$$

where $\mathbf{c}^P$ denotes the predicted object centers, derived from the predicted bounding box B . $\mathbf{c}^T$ denotes the ground-truth object centers. γ^T denotes the one-hot encoding of ground-truth labels. $\mathcal{F}^{\text{FL}}(\cdot, \cdot)$ denotes the focal loss [65]. λ_{LOC} and λ_{FL} balance the two terms. The localization loss ensures that the objects are located accurately, while the focal loss ensures that the objects are classified accurately.

4.3 Experimental Results

4.3.1 Experimental Settings

Dataset Description. Experiments are conducted on three datasets. 1) **FSC-147** [3] covers 147 diverse object categories, with significant variations in object density, ranging from 7 objects per image to 3,731 objects per image. The dataset consists of 6,135 images, officially split into 3,659 for training, 1,286 for validation, and 1,190 for testing [3]. The counting categories of the three subsets are mutually exclusive. 2) **CARPK** [4] counts vehicles in aerial imagery. It contains 1,448 bird's-eye view images collected from four parking lots, posing unique challenges of significant scale variations and frequent occlusions. It is officially divided into 989 and 459 samples for training and testing, respectively [4]. Following [20, 25], we employ its test set for cross-domain evaluation, with vehicles ranging from 2 to 188 per image.

3) **CountBench** [5] comprises 540 images, varying from 2 to 10 objects per image. Every sample includes a text description explicitly indicating the object count, facilitating multi-modal research. Following [20], the entire dataset serves as a test set to assess the cross-domain generalization capabilities.

Evaluation Metrics. Following [1, 3, 16, 20], two common evaluation metrics are employed, *i.e.*, Mean Absolute Error (MAE) $\epsilon_M = \frac{1}{N} \sum_{i=1}^N |\mathcal{C}_i^P - \mathcal{C}_i^G|$, which measures the average absolute difference between the predicted object count $\mathcal{C}_i^P$ and the ground truth count $\mathcal{C}_i^G$; and Root Mean Squared Error (RMSE) $\epsilon_R = \sqrt{\frac{1}{N} \sum_{i=1}^N (\mathcal{C}_i^P - \mathcal{C}_i^G)^2}$, which computes the square root of the mean of the squared differences between predictions and ground-truth.

Compared Methods. The compared methods can be categorized into four groups. 1) *Zero-Shot Counting* methods, counting objects solely from the query images without any exemplars or prompts, *e.g.*, RCC [2], and GCA-SUNet [25]. 2) *Prompt-Based Counting* methods, which specify the counted objects using text exemplars, *e.g.*, CounTX [66], CLIP-Count [30], VL-Counter [32], T2ICount [67], and GroudingREC [19]. 3) *Few-Shot Counting* methods, which utilize visual exemplars for object counting, *e.g.*, BMNet+ [35], SPDCN [68], CounTR [1], LOCA [16], SSD [36], CACViT [17], DAVE [14] and GeCo [27]. 4) *Hybrid* methods, utilizing both visual and textual exemplars to specify classes, *e.g.*, CountGD [20].

Implementation Details. The Adam optimizer with a cosine annealing scheduler is used with an initial learning rate of 1×10^{-5} and a weight decay of 10^{-4} . The hyper-parameters λ_{LOC} and λ_{FL} are empirically set to 1 and 5, respectively. Following [20], we utilize data augmentation to improve generalization, test-time normalization to avoid counting one object multiple times, and adaptive crop to handle small objects and high-density scenarios.

Additional Implementation Details

The Adam optimizer with a cosine annealing scheduler is used with an initial learning rate of 1×10^{-5} and a weight decay of 10^{-4} . The hyperparameters λ_{LOC} and λ_{FL} are empirically set to 1 and 5, respectively, to address the class imbalance by emphasizing harder, misclassified examples during training. A confidence threshold of 0.39 is empirically set to filter predictions, ensuring that only those with sufficiently high confidence scores are retained. Following [20], we utilize data augmentation to improve generalization, test-time normalization to avoid counting one object multiple times, and adaptive crop to handle small objects and high-density scenarios. Our model is trained on a single NVIDIA A5000 with 24GB of GPU memory. An early stopping strategy is employed to prevent overfitting during training. A complete training process takes approximately 40 hours.

Data Augmentation. The following data augmentation technique is utilized to generate more training samples. Following [20], the training image is horizontally flipped with a probability of 50%. Subsequently, one of two resizing strategies is applied with a 50% probability. In the first strategy, the minimum side length of the image is resized to one of the following values: {480, 512, 544, 576, 608, 640, 672, 704, 736, 768, 800}, while maintaining the aspect ratio. In the second strategy, the image is randomly cropped to have a minimum side length within the range [384, 600], after which it is resized according to the first strategy. Once processed, the image is normalized and passed into the model.

Test-Time Normalization One of the key challenges for counting models, as noted by [1], is handling self-similarity, where objects are intrinsically repetitive. To address this issue, we adopt the SAM-based test-time normalization strategy in [20]. Specifically, we generate segmentation masks to ascertain whether the model detects multiple instances of the same object in a bounding box. We obtain these masks by using the bounding boxes derived from the visual exemplars as prompts for the Segment Anything Model (SAM) [28]. This procedure confines detection strictly to the object boundaries, thereby reducing the likelihood of over-counting caused by partial or complete overlaps with extraneous regions. By concentrating on object-specific segmentation, our method effectively alleviates overestimation arising from repetitive and overlapping features, thus addressing the issue of self-similarity in counting tasks.

Adaptive Cropping Similar to many other models [1, 17, 20], our model faces a limitation that it can process at most 900 objects in an image. Following [20], we employ adaptive cropping to fully leverage the model’s capacity for counting. Specifically, if the model detects 900 objects in an image (for the hybrid version is 810), the input image is divided into multiple overlapping pieces, and each cropped piece is processed individually by the model. We then aggregate the counting results of all pieces into the final count.

4.3.2 Details of Compared Methods

The proposed method is comprehensively compared with the following state-of-the-art models, categorized into four groups. Note that some methods can be adapted for handling different tasks.

Zero-Shot Counting Methods

ZSC methods achieve object counting solely from the provided images without the use of any exemplars [2, 25].

RCC [2]: The RCC achieves object counting through a reference-less, class-agnostic framework trained with only weak supervision. It uses a pretrained Vision Transformer [64] to extract dense, globally contextualized features from the input image. These features, which capture repetitive patterns and long-range dependencies, are aggregated via global average pooling and passed through a linear regression head to predict the object count.

LOCA [16]: LOCA leverages a pre-trained ResNet-50 network as the encoder to extract multi-scale features from input images. It uses RoI pooling to obtain uniform appearance queries from annotated exemplars, while a light network transforms each exemplar’s width and height into shape queries, which are then iteratively fused through a series of attention modules to create robust object prototypes. The prototypes are matched to the encoded image features using depth-wise correlation, and a decoder subsequently regresses these features into a density map for counting. For zero-shot counting, LOCA adapts its architecture by substituting the annotation-dependent shape and appearance queries with trainable objectness queries to adapt the zero-shot counting.

DAVE [14]: DAVE employs a two-stage detect-and-verify architecture that combines the

CNN and Transformer to achieve low-shot counting. Initially, a ResNet-50 backbone extracts multi-scale image features that are further processed by a Transformer module to predict a location density map, from which high-recall candidate object centers and corresponding bounding boxes are derived using a feature fusion module and a bounding box regression head. In the subsequent verification stage, features from these candidate regions are pooled and refined through a shallow network before being clustered using spectral clustering based on cosine similarity to filter out false positives. For zero-shot counting, DAVE replaces its exemplar-dependent density prediction with a learnable zero-shot variant and modifies its clustering strategy by retaining clusters that exceed a certain size threshold.

CounTR [1]: CounTR employs a Vision Transformer to encode the query image, while a CNN encodes the few exemplars provided by the user. A feature-interaction module performs cross-attention to capture the similarity between the query and the exemplar. The resulting feature map is upsampled by a series of deconvolution layers to generate the density map, which is integrated to obtain the final object count. For zero-shot counting, the exemplars are replaced by a trainable token.

GCA-SUNet [25]: GCA-SUNet is a counting network based on the Swin-UNet architecture that leverages a Swin Transformer to partition and encode the input image into patches. In the encoder, a Gated Context-Aware Modulation (GCAM) module uses a self-similarity matrix to dynamically filter and enhance features related to the counting targets, while suppressing irrelevant background information. At the bottleneck, a Gated Enhanced Feature Selector (GEFS) module further refines key features, and the decoder employs Swin Transformer blocks along with a Gated Adaptive Fusion Unit (GAFU) for feature fusion and upsampling to ultimately produce a density map.

GeCo [27]: GeCo is originally devised for FSC but can be adapted to address ZSC, *i.e.*, GeCo encodes the target category name using a frozen CLIP text encoder to obtain a text-derived semantic prototype, and projects this prototype into the same embedding space as the visual features extracted from the image. In parallel, a dense grid of object queries is generated and used to attend to image features via cross-attention. Each query then computes a similarity score with the text-derived prototype through a dot-product operation in the shared embedding space. Queries with high similarity scores are interpreted as matching regions containing instances of the target object.

Prompt-Based Counting Methods

PBC methods specify the counted objects using text exemplars [18, 19, 29, 30, 32, 66, 67].

Patch-Selection [29]: This method converts the specified class name into a semantic embedding using a pretrained CLIP model, then uses a conditional VAE to generate a class prototype in a discriminative feature space. It samples image patches and selects those closest to the prototype via L2 distance, with an error predictor further refining the selection. The chosen patches are fed into an exemplar-based counting model with ResNet-50 to extract features, compute similarity maps, and regress a density map for the final count.

CounTX [66]: It leverages pretrained CLIP encoders to extract features from the input image and the specified object class name. Visual and textual features are fused through a two-layer

Transformer decoder for cross-attention, producing a feature map that emphasizes regions containing the target object. The fused representation is reshaped and progressively upsampled by a decoder composed of convolutional layers and bilinear interpolation, generating a single-channel density map. The final object count is obtained by summing all pixel values in this map.

CLIP-Count [30]: Given a query image and a text prompt describing the target object, it extracts visual features using a CLIP vision encoder and encodes the prompt with the CLIP text encoder. A patch-text contrastive loss aligns visual and textual features by encouraging spatially informative representations. A hierarchical patch-text interaction module integrates semantic cues across multiple resolutions using cross-attention, where the text embedding serves as the query to attend to visual tokens. A lightweight decoder then generates a density map representing the spatial distribution of the object, and the final count is obtained by summing all pixel values in the map.

VLCounter [32]: It first utilizes a Semantic-conditioned Prompt Tuning within the image encoder to enhance the feature tokens that are related to the target. Then, a Learnable Affine Transformation is designed to extract counting information from the semantic-patch similarity map. Lastly, a Segment-aware Skip Connection is designed to enhance the generalization on unseen object classes.

CountGD [20]: CountGD was originally designed as a hybrid visual-linguistic object counting solution. Under the PBC setting, the model first employs a Swin-Transformer to extract multi-scale spatial features from the queried image. In parallel, a BERT-based encoder processes the textual prompt into semantic embeddings. These visual and linguistic features are then fused through a series of cross-attention and self-attention modules within the Feature Enhancer, enabling effective alignment and interaction across modalities. The enhanced image patch features are subsequently passed through a decoder head to generate the bounding box with a corresponding confidence score.

T2ICount [67]: It integrates input images with category-specific text prompts within a pre-trained text-to-image diffusion model by employing a single denoising step. It refines decoder feature maps through a Hierarchical Semantic Correction Module and supervises intermediate text-image similarity maps using a Representational Regional Coherence Loss.

GDino [18]: It employs a dual-encoder architecture, where visual and textual features are fused via self-attention and cross-attention in a feature enhancer. A language-guided query selection module initializes decoder queries, which are refined in a cross-modality decoder to predict object bounding boxes.

GroundingREC [19]: It takes a referring expression and a query image as inputs, encodes them using GDino [18], and develops a global-local feature fusion module to inject global context into the image embeddings. Subsequently, a contrastive learning module is designed to align the image with the positive text embedding and repel negative ones. Finally, a loss function is defined as the weighted sum of localization loss, classification loss, and contrastive losses.

Few-Shot Counting Methods

FSC methods provide visual exemplars alongside query images for object counting [1, 14, 16–18, 27, 35, 36, 68];

BMNet+ [35]: It extracts features from both the query image and exemplars using the same feature extractor as in FamNet [3], the first four blocks from a pre-trained ResNet-50 backbone, and computes bilinear similarity between the query and exemplars via a Bilinear Matching Network (BMNet). A self-attention mechanism is applied to the query’s feature maps to focus on relevant parts, improving robustness to intra-class variations. The model then matches the query image to the exemplars by comparing the computed bilinear similarities and selects the most relevant matches. The total object count is finally obtained by aggregating these matches.

SPDCN [68]: SPDCN uses a Scale-Prior Deformable Convolution (SPDC) to incorporate scale information from exemplars. It then computes the cosine similarities between the query image and the exemplars to highlight regions in the query that closely match the exemplars. These regions are subsequently processed by a decoder to generate a density map for the final object count.

CounTR [1]: CounTR performs the FSC task by extracting features from both the query image and a set of visual exemplars using a transformer-based architecture. These features are subsequently fed into a cross-attention module to model the relationships between the query and the exemplars. The resulting representations are then processed by a decoder that progressively upsamples the attended features to generate a density map. The final object count is obtained by summing each value in the predicted density map.

LOCA [16]: LOCA was originally designed for FSC, while it can be adapted for ZSC as shown above. Specifically, it decouples the extraction of shape and appearance queries from the exemplars. The refined prototypes are subsequently utilized as convolutional kernels to generate a response map, which is then upsampled to reconstruct a density map for final object counts.

GDino [18]: It was originally designed for object detection by grounding text to visual features. To adapt it to the FSC setting, visual exemplars are provided to capture object-specific visual features. These features are then projected into the linguistic space, allowing them to serve as linguistic tokens. Then, both visual and linguistic features are utilized for object counting.

SSD [36]: It is designed to learn spatial similarity distributions between the exemplar and query images. To achieve this, it first extracts the visual features of both inputs by using ResNet-50 to construct the Query Features Pyramid. The refined features are then passed to a similarity learning module, which employs a 4D convolution operation to capture fine-grained spatial correlations across both spatial and cross-image dimensions. The resulting similarity tensor is subsequently processed by a decoder to reconstruct a density map. The final object count is obtained by summing over all pixel values in the predicted map.

CACViT [17]: It first tokenizes both the query and exemplar images into patches, flattens them into tokens, and concatenates them before feeding them into a shared Vision Transformer encoder. The transformer jointly extracts features and captures cross-image similarities through self-attention. The attention map is then decoupled. Only the output feature of the query image

and the similarity metric from the attention map are used to regress a density map for final object counting.

DAVE [14]: DAVE was originally designed for the FSC task, but it can be adapted for ZSC. Specifically, it achieves counting via a two-stage pipeline: text-guided detection and contrastive verification. In the detection stage, CLIP extracts visual and textual features from the query image and class name. They are fused in a transformer-based density estimator using cross-attention, producing a dense map to indicate candidate object locations. In the verification stage, patches around these candidates are re-encoded by CLIP and compared to the prompt using a contrastive verifier to filter out false positives. The final count is the number of verified candidates.

CountGD [20]: CountGD was originally designed as a hybrid visual-linguistic object counting solution. To perform under the FSC setting, it uses visual exemplars encoded by the visual encoder as input prompts. Through a series of cross-attention, these representations guide the feature extraction process on the query image, enabling the model to attend to visually similar instances. The confidence scores for each object with its bounding boxes are generated, and the final count is obtained by calculating the number of confident objects.

GeCo [27]: GeCo was originally designed as an FSC solution, and it can be adapted to ZSC. Given a few exemplar boxes, it encodes them into object prototypes via feature aggregation. Then, it uses a dense grid of object queries over the input image to search for matches to these prototypes by computing the similarities between the prototype and each dense query. The matched queries are processed by detection and segmentation heads to localize individual object instances through predicted bounding boxes and masks. The final count is obtained by enumerating all matched instances that exceed a confidence threshold.

Visual-Linguistic Hybrid Methods

Hybrid methods simultaneously incorporate both visual and textual exemplars to specify objects alongside the query image [20]. To our best knowledge, CountGD [20] is the only method that belongs to this category in the literature. CountGD adopts a transformer-based architecture with a dual-encoder design: one for the image and one for the prompt. The visual encoder processes the query image to extract dense visual features while the prompt encoder encodes the user-defined target object. These encoded features are fused through a bi-attention mechanism. The decoder uses a query-based transformer head that predicts objects with spatial locations and confidence scores.

4.3.3 Comparison Results on FSC-147 Dataset

Comparisons to State-of-the-Art Models

We conduct four groups of comparison experiments on the FSC-147 dataset, as summarized in Table 4.1. Following [3, 20], we report both test errors and validation errors. The results of all the compared methods except GDino [18] are obtained from their original paper.

Table 4.1: Comparison with state-of-the-art methods on the FSC-147 dataset [3]. V and T represent visual exemplars and text prompts. ϵ_M^T , ϵ_R^T , ϵ_M^V , and ϵ_R^V denote MAE and RMSE for test and validation data, respectively. Best results are in bold.

	Method	ϵ_M^T	ϵ_R^T	ϵ_M^V	ϵ_R^V
ZSC	RCC CVPRW'22 hobley2022learning	17.12	104.53	17.49	58.81
	LOCA ICCV'23 djukic2023loca	16.22	103.96	17.43	54.96
	DAVE CVPR'24 pelhan2024dave	15.14	103.49	15.54	52.67
	CounTR BMVC'22 liu2023countr	14.71	106.87	18.07	71.84
	GCA-SUNet arxiv wu2024gca	14.08	84.77	17.02	55.37
	GeCo NIPS'24 pelhan2024novel	13.30	108.72	14.81	64.95
	Proposed CSG	8.52	83.26	9.40	49.08
PBC	Patch-Selection CVPR'23 xu2023zero	22.09	115.17	26.93	88.63
	CounTX BMVC'23 AminiNaieni23	15.88	106.29	17.10	65.61
	CLIP-Count MM'23 jiang2023clip	17.78	106.62	18.79	61.18
	VLCounter AAAI'24 kang2024vlcounter	17.05	106.16	18.06	65.13
	CountGD NIPS'24 amini2024countgd	12.98	98.35	12.14	47.51
	T2ICount CVPR'25 qian2025t2icount	11.76	97.86	13.78	58.78
	GDino ECCV'24 liu2025grounding	10.82	104.00	10.32	55.54
	GroundingREC CVPR'24 dai2024referring	10.12	107.19	10.06	58.62
	Proposed CSG	8.08	80.53	8.65	41.25
Hybr.	CountGD NIPS'24 amini2024countgd	5.74	24.09	7.10	26.08
	Proposed CSG	4.69	15.23	6.90	28.24
FSC	BMNet+ CVPR'22 shi2022bmnetplus	14.62	91.83	15.74	58.53
	SPDCN BMVC'22 lin2022scale	13.51	96.80	14.59	49.97
	CounTR BMVC'22 liu2023countr	11.95	91.23	13.13	49.83
	LOCA ICCV'23 djukic2023loca	10.79	56.97	10.24	32.56
	GDino ECCV'24 liu2025grounding	10.77	99.51	9.32	51.72
	SSD IJCAI'24 xu2024learning	9.58	64.13	9.73	29.72
	CACViT AAAI'24 wang2024vision	9.13	48.96	10.63	37.95
	DAVE CVPR'24 pelhan2024dave	8.66	32.36	8.91	28.08
	CountGD NIPS'24 amini2024countgd	8.31	91.05	7.46	29.54
	GeCo NIPS'24 pelhan2024novel	7.91	54.28	9.52	43.00
	Proposed CSG	5.23	27.95	7.26	28.77

Few-Shot Counting. We primarily conduct comparison experiments under the FSC setting [3], where three visual exemplars are provided but without text exemplars. We have the following observations. 1) The proposed CSG significantly outperforms all the compared methods in terms of both MAE and RMSE on the test set, showing its effectiveness in FSC scenarios. Compared to the second-best method, GeCo [27], it significantly reduces the MAE from 7.91 to 5.23, and reduces the RMSE from 54.28 to 27.95, highlighting its effectiveness in the FSC setting. 2) CountGD [20] produces small RMSE on validation sets, but it tends to overfit to the validation data and produce larger errors on novel test data. In contrast, CSG demonstrates superior generalization performance, which not only achieves excellent performance on the validation set but also generalizes well on novel test data. 3) CountGD [20] utilizes both text and visual exemplars, but it simply concatenates visual and textual features for multi-modal interac-

tions. Compared with CountGD [20], our model significantly reduces MAE by 3.08 and RMSE by 63.10 on test data, demonstrating the effectiveness of our comprehensive set of cross-modal interactions.

Zero-Shot Counting. Besides FSC, we adapt our method for zero-shot counting, where the visual exemplars are replaced by learnable tokens as in [1], enabling the model to operate without explicitly providing visual exemplars. We can observe the following from Table 4.1. 1) The proposed CSG demonstrates superior performance compared to other ZSC methods, outperforming the second-best method, GeCo [27], with a significant reduction of 4.53 in MAE and 19.57 in RMSE. 2) CSG significantly outperforms dedicated ZSC methods such as RCC [2] and GCA-SUNet [25], as it leverages LLaVA to generate text prompts that carry abstract semantics of objects. In addition, the group’s common characteristics of these object candidates are exploited by SGGA, mitigating the challenges of a lack of visual exemplars or text prompts in the ZSC setting. 3) CSG demonstrates excellent generalization ability, outperforming all the compared methods in terms of both validation errors and test errors.

Prompt-Based Counting. For PBC, besides adopting learnable tokens to represent visual exemplars as in ZSC, the provided text exemplars are integrated with the generated text descriptions to provide additional clues. From Table 4.1, we can observe the following. 1) CSG consistently outperforms all the compared methods on both validation and test sets. Compared to the second-best method, GroundingREC [19], CSG largely reduces the test MAE from 10.12 to 8.08 and the test RMSE from 107.19 to 80.53. The substantial gains are mainly attributed to the customized modality interaction and explicit semantic guidance in the CSR module. 2) Similar performance gains are also confirmed on the validation set, demonstrating CSG’s robustness and ability to mitigate potential overfitting issues.

Hybrid Counting. Only CountGD [20] explicitly makes use of visual and text exemplars for object counting. The proposed CSG achieves substantial improvements over CountGD [20], reducing the MAE from 5.74 to 4.69 and the RMSE from 24.09 to 15.23 on the test set. The performance gains primarily stem from effective intra-domain interaction in GICE blocks, cross-domain interactions in CSR, and group interaction in SGGA.

Discussions on FSC-147 Dataset. Though originally designed for the FSC task, CSG consistently outperforms all competing approaches on the test set across all settings, demonstrating its task adaptability and superior counting capability. Moreover, since the counting categories are disjoint across the training, validation, and test sets, the outstanding performance on the test set verifies its excellent generalization to unseen object categories. Lastly, the performance gain from ZSC to FSC is more substantial than that from ZSC to PBC, indicating that the integration of visual exemplars provides a more significant performance boost, highlighting the importance of exemplar-based guidance for accurate counting.

Density Analysis

The object density ranges from 7 to 3,731 per image on the FSC-147 dataset. Following [17], we divide the 1,190 test images into two groups: a low-density group of 605 images with each containing no more than 37 objects, and a high-density group of 585 images, each with 38 or more objects. Table 4.2 summarizes the results under the FSC setting. CSG outperforms all

Table 4.2: Comparison of counting objects on images of different object densities on the FSC-147 dataset [3].

Method	ϵ_M^{Low}	ϵ_R^{Low}	ϵ_M^{High}	ϵ_R^{High}
BMNet+ CVPR'22 shi2022bmnetplus	6.23	29.96	23.21	127.62
SPDCN BMVC'22 lin2022scale	5.06	31.53	22.25	133.59
CounTR BMVC'22 liu2023countr	5.44	36.96	18.56	124.11
CACViT AAAI'24 wang2024vision	3.14	6.26	15.33	69.54
CountGD NIPS'24 amini2024countgd	1.76	4.97	14.32	120.09
GeCo NIPS'24 pelhan2024novel	1.97	4.34	13.86	81.59
Proposed CSG	1.30	3.33	8.95	39.02

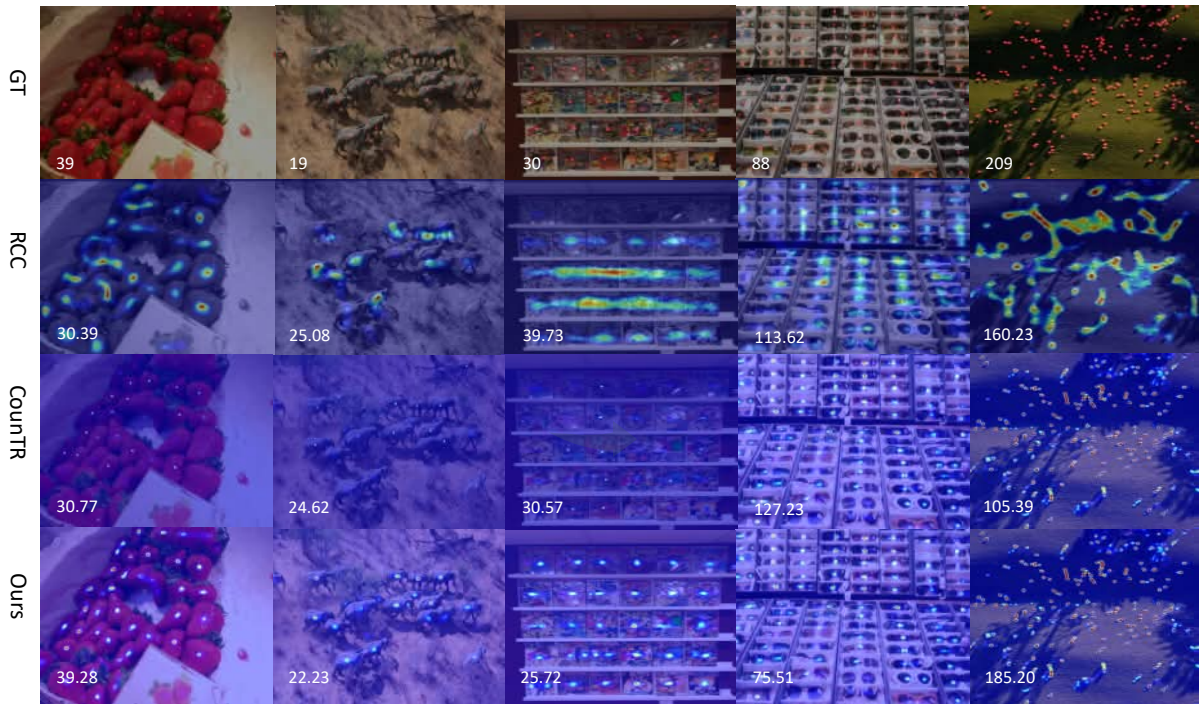


Figure 4.3: Visual comparisons on the FSC-147 dataset [3].

other methods in both low-density and high-density groups. Particularly in the high-density group, compared to the second-best method, GeCo [27], CSG reduces the MSE by 4.91 and RMSE by 42.57, demonstrating its effectiveness in handling high-density images.

Visual Comparisons

CountGD [20], GeCo [27], and CSG are visually compared under the FSC setting. Fig. 4.3 shows five real-world counting scenarios: elephants in the wild, chairs in the canteen, strawberries in the basket, nail polishes on the goods shelf, and flamingos in the sky. CSG performs well in all five scenarios. As the object density increases, all three methods show a decline in accuracy, while CSG experiences the smallest performance degradation. For example, for counting chairs in a canteen, both CountGD [20] and GeCo [27] tend to mistakenly localize tableware instead of chairs, whereas CSG accurately delineates and counts chairs.

Table 4.3: Comparison results of cross-domain evaluation on CARPK dataset [4] and CountBench [5].

		Method	ϵ_M^T	ϵ_R^T
CARPK	ZSC	RCC CVPRW'22 hobley2022learning	21.38	26.61
		GCA-SUNet arxiv [25]	10.96	13.95
		Proposed CSG	4.31	6.28
	PBC	CLIP-Count MM'23 jiang2023clip	11.96	16.61
		CountGD NIPS'24 amini2024countgd	3.83	5.41
		Proposed CSG	3.45	5.20
	FSC	LOCA ICCV'23 djukic2023loca	9.97	12.51
		CACViT AAAI'24 wang2024vision	8.30	11.18
		Proposed CSG	3.07	4.84
CountB	PBC	CounTX BMVC'23 AminiNaieni23	6.64	15.75
		CountGD NIPS'24 amini2024countgd	0.86	3.10
		Proposed CSG	0.77	2.56

4.3.4 Cross-Domain Evaluation

Following [20], we conduct a cross-domain evaluation by training a model on the FSC-147 dataset [3] and evaluating it directly on the CARPK [4] and CountBench [5] datasets without fine-tuning. Following [20], experiments are conducted under ZSC, PBC, and FSC settings for CARPK [4] and PBC for CountBench [5]. The results of the compared methods are taken from their original papers.

CARPK Dataset. As shown in Table 4.3, CSG consistently and significantly outperforms all the compared methods across all three evaluation settings on CARPK [4]. Compared to GCA-SUNet [25] dedicated to ZSC, CSG significantly reduces MAE from 10.96 to 4.31 and RMSE from 13.95 to 6.28. CSG also surpasses the previous best approach, CountGD [20], in both PBC and FSC, reducing MAE from 3.83 to 3.45 for PBC and from 3.68 to 3.07 for FSC. Notably, only CSG adapts to all settings, and it performs best, further highlighting its superior adaptability and generalization capabilities.

CountBench Dataset. The proposed CSG exhibits robust generalization capabilities on CountBench [5], reducing MAE from 0.86 to 0.77 and RMSE from 3.10 to 2.56 compared with CountGD [20]. Note that the errors on CountBench [5] are much smaller than those on CARPK [4], mainly because of lower object densities, *i.e.*, 6.04 vs. 66.25 per image on average, and more similar viewpoints to FSC-147, as CARPK images are captured in a bird's-eye view.

4.3.5 Ablation Study of Major Components

Table 4.5 summarizes the ablation of major components on the FSC-147 dataset [3] under the FSC setting. GDino [18] is adapted as the baseline by replacing provided text exemplars with textual descriptions generated by LLaVA. We then gradually add GICE, replace the feature enhancer of GDino by CSR, and replace the cross-modality decoder by SSGA for ablation. All

the compared methods utilize four sets of inputs as CSG. Compared with the baseline model, all three proposed modules individually achieve significant performance gains over all four metrics, demonstrating their effectiveness. The combinations of two modules further improve all four evaluation metrics. Lastly, the utilization of all three proposed components achieves the lowest error rates. The ablation results demonstrate the contribution of each component to the final performance.

4.3.6 Ablation of Number of Visual Exemplars

For few-shot counting, the number of visual exemplars plays a key role in the counting accuracy. We investigate the impact of the number of visual exemplars on the FSC-147 dataset. The results are summarized in Table 4.4. We can see that increasing the number of visual exemplars consistently reduces all errors on both the validation and test sets, as more exemplars provide richer visual cues to guide more accurate counting. Also note that when the shot number is 0, it is equivalent to zero-shot counting, as reported in Table 1 of the manuscript.

Table 4.4: Ablation study of the number of visual exemplars on the FSC-147 dataset [3].

Shot Num.	ϵ_M^T	ϵ_R^T	ϵ_M^V	ϵ_R^V
0	8.52	83.26	9.40	49.08
1	7.17	62.97	8.84	41.21
2	6.31	49.80	8.32	39.74
3	5.23	27.95	7.79	34.53

4.3.7 Failure Case Analysis

Fig. 4.4 shows five representative failure cases in which CountGD [20], GeCo [27], and CSG struggle to yield accurate counts. For example, in the first case, severe self-similarity leads to over-counting repetitive parts of a single chair; In the third case, apples located at a distance or partially occluded are frequently overlooked, leading to systematic underestimation. In the last case, the bookshelf example highlights difficulties in segmenting and enumerating densely packed and overlapping instances. These cases expose key limitations of current class-agnostic counting frameworks, particularly in dealing with self-similarity, occlusion, low saliency, and high-density objects.

4.3.8 Model Complexity Analysis

Table 4.6 summarizes training time, inference time, model size, and GFLOPs for the top-three performing methods, CountGD [20] and GeCo [27], and CSG. CSG maintains competitive inference time and computational efficiency, balancing between accuracy and efficiency.

Table 4.5: Ablation of major components on FSC-147 [3].

$\mathcal{F}^{\text{GICE}}$	$\mathcal{F}^{\text{CSR}}$	$\mathcal{F}^{\text{SGGA}}$	$\epsilon_{\text{M}}^{\text{T}}$	$\epsilon_{\text{R}}^{\text{T}}$	$\epsilon_{\text{M}}^{\text{V}}$	$\epsilon_{\text{R}}^{\text{V}}$
✗	✗	✗	10.77	99.51	9.32	51.72
✓	✗	✗	8.06	90.14	8.72	43.95
✗	✓	✗	7.49	77.87	8.10	44.96
✗	✗	✓	7.84	86.38	8.35	41.59
✓	✗	✓	6.75	67.92	7.83	38.86
✓	✓	✗	5.99	52.38	7.93	37.19
✗	✓	✓	5.82	36.06	7.71	34.86
✓	✓	✓	5.23	27.95	7.26	28.77

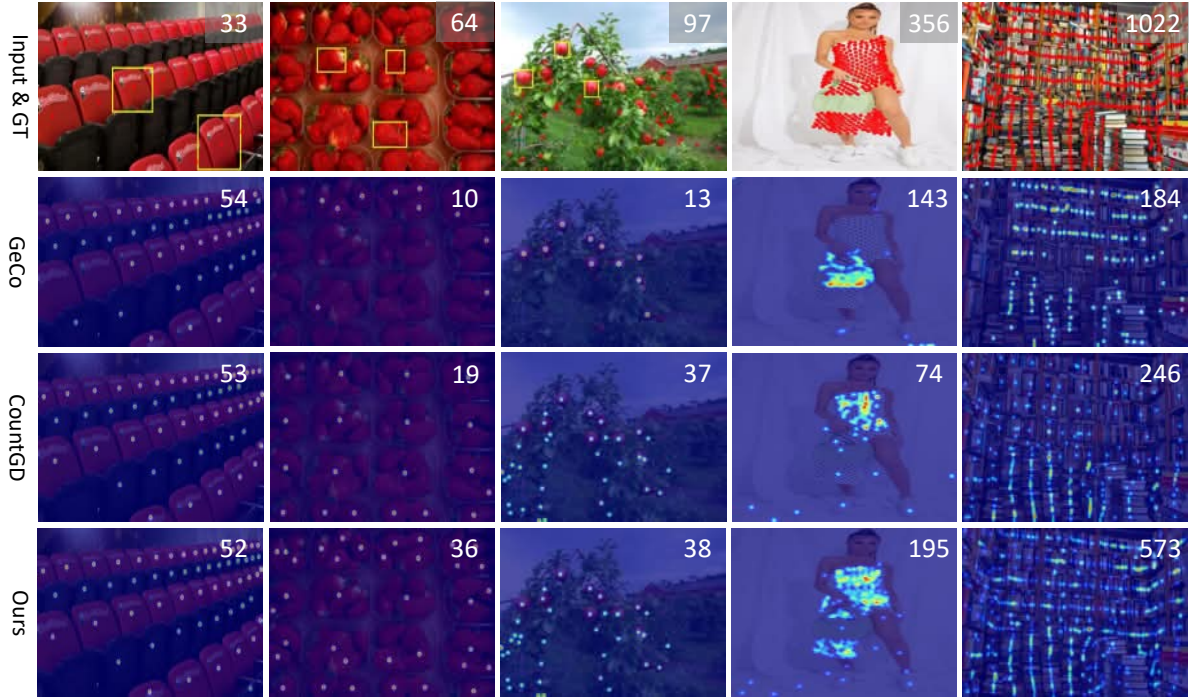


Figure 4.4: Failure cases for FSC on the FSC-147 dataset [3].

Table 4.6: Comparison of model complexity.

Method	Training	Inference	#Params.	GFLOPs
GeCo [27]	200h	0.97s	641M	1075
CountGD [20]	24h	0.32s	204M	437
Proposed CSG	40h	0.36s	301M	680

4.3.9 More Qualitative results

More comparison results on the FSC-147 dataset, CARPK dataset, and CountBench dataset are shown in Fig. 4.5, Fig. 4.6, and Fig. 4.7, respectively. The proposed CSG demonstrates superior counting performance on all three datasets.

Qualitative Comparisons on FSC-147 [3].

In Fig. 4.5, we present qualitative comparisons of the proposed CSG framework on both the test and validation sets of FSC-147 [3], covering a range of representative scenarios. The illustrated categories demonstrate a consistent trend: as object count increases, all evaluated methods tend to exhibit higher counting errors, primarily due to increased scene complexity and occlusion. Despite these challenges, our CSG model consistently delivers superior performance, effectively mitigating occlusion and density-related ambiguities and maintaining lower error rates even under high-density conditions.

Qualitative Comparisons on CARPK [4].

We present a qualitative comparison between the proposed CSG framework and the state-of-the-art CountGD [20] under the FSC setting on the CARPK dataset [4], as illustrated in Fig. 4.6. The results demonstrate that CSG consistently yields more accurate and robust counting outcomes across both low- and high-density scenarios. In particular, regardless of object density or exemplar variation, CSG reliably outperforms CountGD, highlighting its superior generalization capability and enhanced adaptability to varying visual complexities.

Qualitative Comparisons on CountBench [5].

In Fig. 4.7, we present detailed visualization results of the proposed CSG on the CountBench [5] dataset. As illustrated, the model consistently achieves high localization accuracy by effectively leveraging both the textual exemplar and the query text, resulting in semantically-aware and precise predictions.

4.4 Chapter Summary

The proposed CSG well tackles the challenges of few-shot counting by mapping an input image with visual exemplars and two generated text descriptions into countable objects. The GICE blocks and CSR module capture the intra-domain and cross-domain interactions between the query and exemplars, respectively, enabling efficient alignment and integration of heterogeneous features. Significant object candidates and their group common characteristics are extracted in SGGA and exploited to highlight candidates using group attention. Extensive experiments on the FSC-147, CARPK, and CountBench datasets under four CAC settings demonstrate that CSG consistently and significantly outperforms state-of-the-art models in both intra-domain and cross-domain evaluations. Future work includes counting similar objects but of different classes in an image.

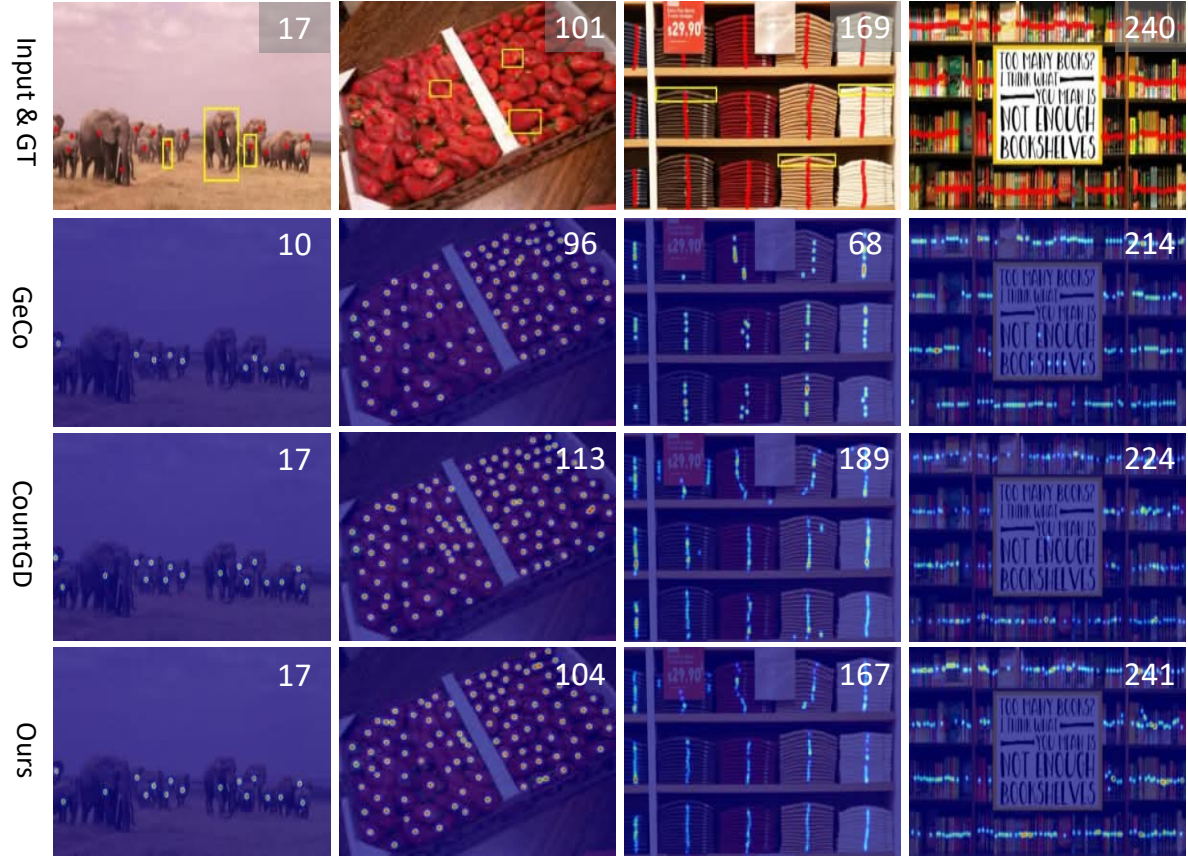


Figure 4.5: Additional qualitative comparisons on FSC-147 dataset [3].

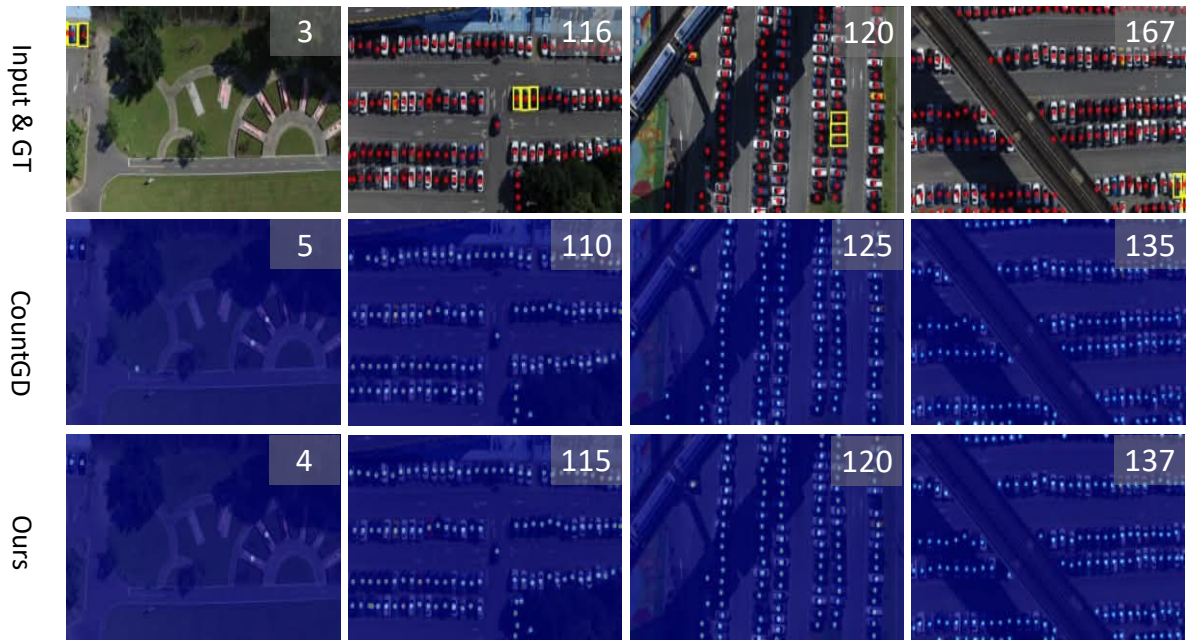


Figure 4.6: Qualitative comparisons on CARPK dataset [4].

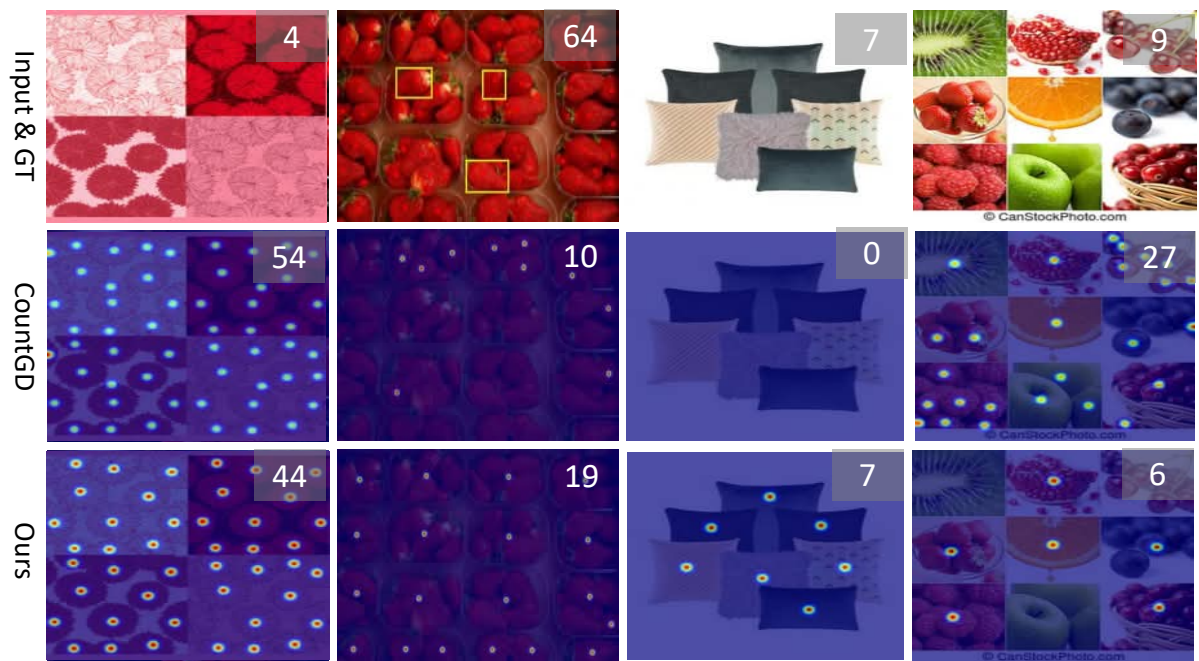


Figure 4.7: Qualitative comparison on CountBench dataset [5].

Chapter 5

Summary and Reflections

This chapter provides a comprehensive summary of the project and presents key reflections. Section 5.1 outlines the overall strategy and practical approach adopted for project management, detailing each developmental phase and its corresponding objectives. Section 5.2 offers critical reflections on various aspects of the project, including time management, risk mitigation, failure handling, and the experience gained through conducting formal research. Sections 5.3 and 5.4 conclude the chapter by highlighting the main achievements, identifying current system limitations, and proposing potential directions for future work.

5.1 Project management

The Gantt chart in Fig. 5.1 presents a 47-week timeline extending from 27 May to 21 April. Each task bar is proportionally scaled based on its scheduled start date and duration. To align with the targeted conference deadline, key development phases and both paper submissions were strategically concentrated during the summer break.

Project Preparation, Literature Review, and Proof-of-Concept (May – Early July)

The project commenced at the start of the summer break, with all preparatory tasks, *i.e.*, the project plan and proposal, completed within the first week. This was followed by a two-week literature review focusing on the application of the recently proposed methods to core computer vision tasks and their potential adaptation to CAC. Over ten baseline models across segmentation and detection were implemented and evaluated; however, their performance fell short of expectations, resulting in a two-week delay. Consequently, the initial proof-of-concept, initially scheduled for early July, was completed by mid-July.

Design, Experimentation, and Theoretical Validation (Mid-July – Early August)

A two-week period was devoted to model design and experimental evaluation. During this phase, a theoretical framework was formalized, and three novel modules were developed through iterative consultation with the supervisor. Each module underwent empirical validation and demonstrated consistent improvements over baseline architectures.

Conference Preparation and Manuscript Drafting (August – Mid-September)

This stage focused on the ICASSP submission. Initial efforts involved aligning with the conference’s formatting and stylistic standards. A preliminary manuscript draft was completed in

Project Plan

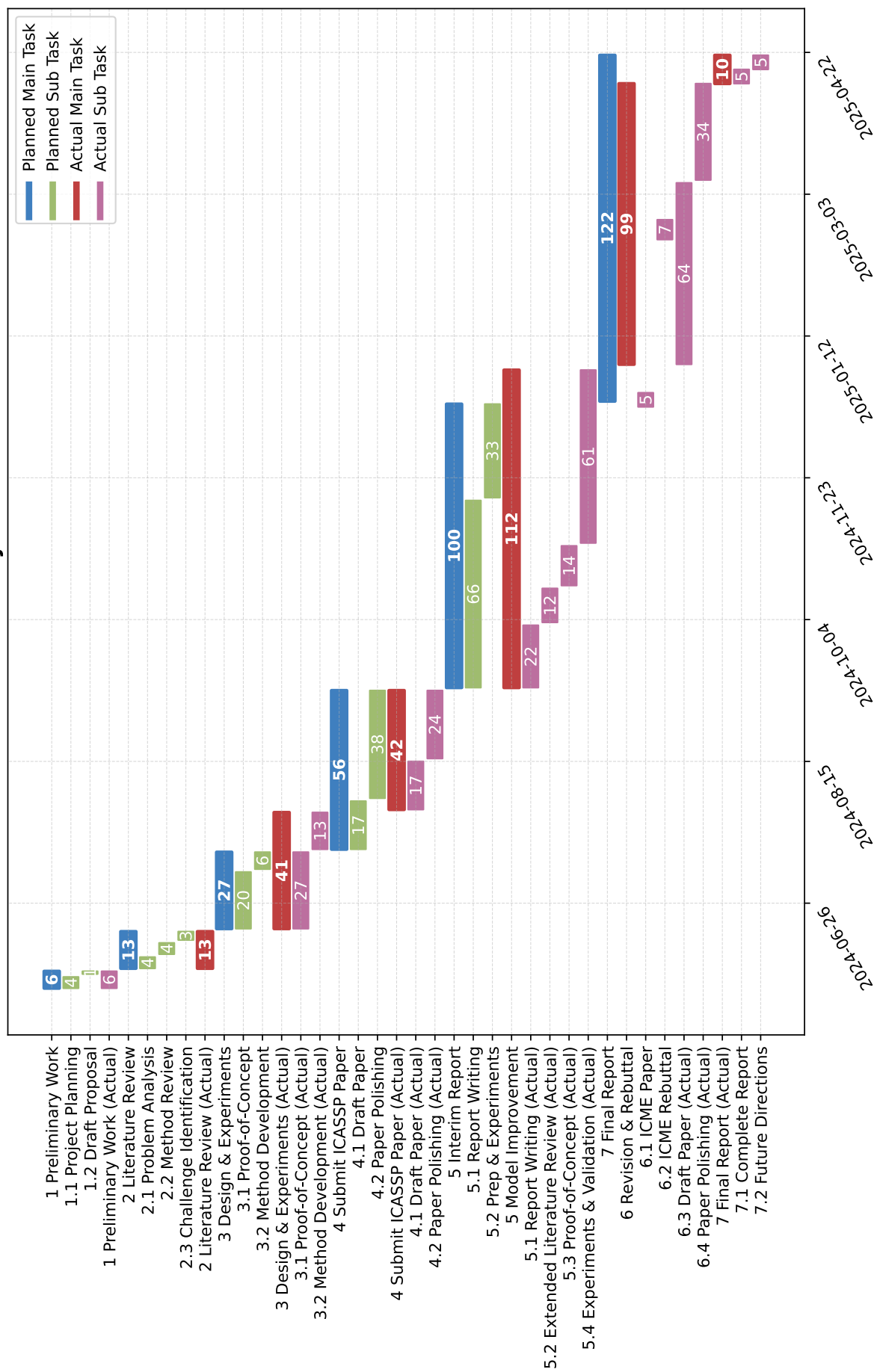


Figure 5.1: Overview of Project Timeline.

mid-August. Subsequent iterations, informed by feedback from both the supervisor and peers, involved additional experiments and extensive revisions, culminating in a finalized submission in early September.

Extension, Refinement, and Subsequent Submissions (Post-September – December)

Building upon insights gained during the ICASSP phase, a new framework targeting the FSC task was conceptualized. This involved a targeted literature review and the development of a refined proof-of-concept, which was completed by late October. Comprehensive experiments were then conducted, and results were finalized in December. This work was expanded into a full study intended for submission to a second conference. Concurrently, the ICASSP rebuttal was conducted in November. Following the notification of rejection in mid-December, a resubmission to ICME was completed, accompanied by the drafting of the mid-term project report.

Final Revisions and Thesis Submission (January – April)

Between January and mid-February, final experiments for the new paper were completed, and manuscript revisions were undertaken. The ICME rebuttal was addressed in late February. From early March through 11 April, ongoing discussions and verification rounds with the supervisor helped finalize the submission. The final week (11–21 April) was dedicated to compiling and polishing the thesis report, with an emphasis on academic rigour and timely delivery.

The timeline exemplifies a balance between structured planning and adaptive project management, reflecting the iterative cycle of experimentation, theoretical refinement, and academic writing essential for producing conference-grade research.

The task list is below:

1. Conduct preliminary preparations, including planning and administrative setup.
2. Perform an in-depth literature review to establish the research foundation.
3. Design and implement experiments for the first research contribution.
4. Draft and submit a conference paper based on the first contribution.
5. Design and conduct experiments for the second research contribution.
6. Prepare and submit the rebuttal for the first paper submission.
7. Revise and resubmit the first paper following rejection.
8. Compile and submit the interim progress report.
9. Prepare a rebuttal for the resubmitted version of the first paper.
10. Draft and submit a conference paper for the second research contribution.
11. Finalize and submit the dissertation, and prepare a demonstration video.

5.2 Reflections

This section offers a reflective evaluation of the project from three perspectives. Sections 5.2.1 and 5.2.2 explore how experiences from previous projects, particularly in time and risk management, were applied, while also highlighting the unique challenges faced in this project and the approaches used to overcome them. Section 5.2.3 presents the key insights gained from engaging in formal academic research.

5.2.1 Time Management

Positive Experience This project represented our first engagement in a research endeavor aimed at conference submission. With limited prior exposure to deep learning, we encountered a steep learning curve that demanded rapid assimilation of domain knowledge and rigorous time management. One of the earliest challenges was underestimating the time required for proof-of-concept development. Inexperience led to insufficient allocation for selecting and validating baseline models, resulting in delays and underscoring the necessity of more structured planning in the initial phase. The tight timeline imposed by the conference deadline emphasized the importance of buffer periods to accommodate potential architectural revisions and unexpected setbacks. Several rounds of model restructuring in the early phase significantly disrupted the original schedule.

In contrast, the development of the second framework unfolded with greater efficiency. Although not originally planned, it benefited from a clearer problem formulation, improved familiarity with the domain, and an already established experimental infrastructure. These factors enabled rapid prototyping and early validation, culminating in a second conference submission. This experience illustrated the compounding advantages of iterative research experience, not only in implementation but also in planning and coordination. Compared to the first framework, the second required fewer debugging iterations and less trial-and-error, highlighting the value of reusable components and methodological refinement.

Despite the increased workload, integrating a second submission successfully highlighted the importance of flexible scheduling and early investment in infrastructure. The lessons learned from the first iteration translated into significantly improved execution in the second. Overall, this process greatly strengthened our time management skills. We came to recognize that well-defined objectives, a detailed execution roadmap, and proactive risk planning are indispensable. The Gantt chart, in particular, proved to be an effective tool for tracking milestones and maintaining clarity throughout the project lifecycle.

Negative Experience This project marked, to the best of our knowledge, the first instance in which an undergraduate student independently led two concurrent research efforts aimed at high-tier conferences, *i.e.*, one submitted to a top CCF-B conference and the other to a prestige CCF-A conference. These projects encompassed the full research pipeline, from problem formulation and model development to experimental validation and manuscript preparation. In hindsight, we underestimated the technical depth, iterative rigor, and coordination required for success, particularly at the CCF-A level, where expectations demand not only technical maturity but also rapid iteration and collaborative precision.

For the first project, we overestimated the quality of the manuscript and failed to prepare a robust fallback plan in case of rejection. As a result, the resubmission process collided with final examinations in the first semester, causing severe time pressure and disrupting both academic and research workflows.

Regarding the second project, we now recognize that we overcommitted. Driven by the ambition to achieve a high-impact contribution, we set goals that were overly ambitious relative to our current skill set. This misalignment exposed deficiencies in foundational knowledge, research planning, and task delegation. We had, in effect, reached a “local maxima” often observed in early-stage researchers, *i.e.*, pursuing an elite target without a sufficiently mature technical base or support infrastructure. Several project management issues followed. Most notably, we failed to allocate buffer time during the manuscript phase to absorb feedback cycles and inevitable delays. The absence of a contingency timeline resulted in acute time pressure close to the submission deadline. Moreover, our lack of proactive coordination with PhD-level collaborators reduced their involvement, placing disproportionate responsibility on our supervisor. This highlighted a clear shortcoming in leadership and communication. These challenges were further compounded by overlapping academic responsibilities, part-time commitments, and graduate school applications. Managing these competing priorities required frequent reprioritization and intense bursts of focus, particularly during manuscript submission and rebuttal stages. While the process was both intellectually and emotionally demanding, it ultimately proved to be deeply instructive. The tight schedule also left me with only one week to finalize the project report and another to prepare the demonstration video.

This process emphasized the importance of realistic time estimation, early-stage risk mitigation, and structured communication within research teams. Most importantly, it taught me that effective project leadership is not solely defined by technical contributions—it also requires foresight, accountability, and the capacity to adapt under pressure. These insights will guide our approach to future research planning and team collaboration as we progress toward our academic and professional aspirations.

5.2.2 Risk Management

Positive Experience

This research project involved several potential risks, the most significant being discrepancies between empirical outcomes and theoretical expectations.

To proactively address such challenges, we conducted early-stage risk assessments and implemented preventive strategies. Drawing on insights from the Machine Learning module, we restructured the training pipeline by enhancing parameter logging and refining hyperparameter tuning mechanisms. These adjustments substantially improved the stability and efficiency of model training in later stages.

Moreover, we validated the effectiveness of newly introduced components through an extensive review of related literature, ensuring that architectural modifications were grounded in established research. Routine code inspections and peer reviews also played a crucial role in promptly identifying issues and maintaining the reliability of the experimental workflow.

Negative Experience

Despite these precautionary efforts, several setbacks revealed deficiencies in our risk management process. A notable instance involved a latent bug in the evaluation code, which inadvertently inflated performance metrics. This led to an overestimation of the model’s capabilities and an erroneous conclusion that the proof-of-concept phase had been completed.

As a result, we prematurely transitioned to subsequent research stages. Although the issue was eventually detected and corrected, this incident underscores the necessity of rigorous validation protocols before interpreting experimental results. Fortunately, the adjustment was made without significantly impacting the overall project schedule.

5.2.3 Lessons Learned from Conducting Research

This research-driven project adhered to formal academic standards, and the supervisor’s consistent guidance facilitated smooth and continuous progress.

An early lesson involved the selection and validation of baseline models. When entering a new domain, it is more appropriate to adopt state-of-the-art baselines specific to that field rather than transferring models from unrelated tasks. Once a deeper understanding of the domain is acquired, cross-domain adaptation can be reconsidered as a more informed choice.

During the writing phase, we learned the importance of maintaining frequent communication with our supervisor, especially when shaping the narrative and contributions of the paper. Supervisors often bring valuable experience and stronger storytelling skills, which are essential for effectively presenting the novelty and significance of the work. Trusting and leveraging this expertise proved to be highly beneficial.

In the rebuttal and resubmission stages, we developed a mindset of anticipating and accepting potential setbacks. Rather than being discouraged by initial rejections, we learned to treat them as part of the iterative research process and remain committed to continuous improvement.

Lastly, we realized that effective communication plays a pivotal role in collaborative research. Coordinating with co-authors and managing shared responsibilities requires proactive engagement and clear delegation, which are as crucial as the technical content itself.

5.3 Conclusion and future work

5.3.1 Contributions of GCA-SUNet in ZSC

The GCA-SUNet represents a significant step forward in reducing annotation dependency by eliminating exemplar inputs. This framework leverages the robust feature extraction and integrates three novel gated modules that serve dual purposes. First, they suppress irrelevant features or background noise via a gate mechanism, thereby emphasizing the regions corresponding to objects of interest. Second, by directly mapping the input image to a density map, the model is capable of capturing intrinsic visual cues such as latent object-level semantics, spa-

tial self-similarity, and context-aware dependencies. Empirical evidence on challenging benchmarks confirms that GCA-SUNet not only reduces the reliance on manual annotations but also generalizes effectively in complex, real-world scenarios.

5.3.2 Contributions of CSG framework in FSC

The Cross-modal Semantic-guided Group-attended (CSG) framework addresses the inherent limitations in the FSC task by refining the interplay between visual and generated textual information. Specifically, to achieve fine-grained alignment between the visual and linguistic domains, two gated blocks and a semantic refinement module are introduced. These blocks refine the query features by highlighting tokens that are highly correlated with the exemplar features. Moreover, through group attention, the framework effectively aggregates common semantic properties, thereby enhancing object reasoning. This results in superior generalization, particularly for novel object categories and under low-annotation conditions.

In summary, the research contributions address two central challenges in CAC vision tasks through innovative deep learning strategies. The GCA-SUNet offers a robust zero-shot counting solution by autonomously learning and exploiting intrinsic visual cues, thus eliminating the need for human-annotated exemplars. In parallel, the CSG framework advances few-shot counting by harnessing cross-modal interactions between automatically generated textual descriptions and visual features, leading to improved alignment, object reasoning, and generalization. Collectively, these contributions reduce annotation overhead and enhance the scalability and robustness of counting systems, paving the way for more flexible and generalizable approaches in open-world visual environments.

5.3.3 Limitation and Future Work

Despite recent advances, as illustrated in Fig. 5.2, employing CSG for demonstration purposes, the additional failure cases on FSC-147 [3] expose several critical limitations of the current approaches. 1) In scenes with densely packed objects, the model struggles to delineate individual instances, causing systematic under-counting due to severe occlusion and overlap. 2) We observe self-multiple counting, where the predicted count is erroneously inflated by structurally similar or repetitive patterns occurring within a single object or across adjacent objects. These false positives frequently arise in images dominated by repetitive textures, symmetric structures, or recurring motifs, where visual similarity is misinterpreted as object multiplicity.

To remedy these shortcomings, future work will pursue more adaptive and efficient counting strategies. Specifically, we will refine the feature-extraction pipeline to better preserve the fine-grained details required to distinguish tightly clustered instances, and incorporate stronger discriminative modules to suppress false positives induced by visual similarity. The overarching goal is to guarantee a faithful one-to-one correspondence between model predictions and real-world object instances without compromising computational efficiency or scalability.

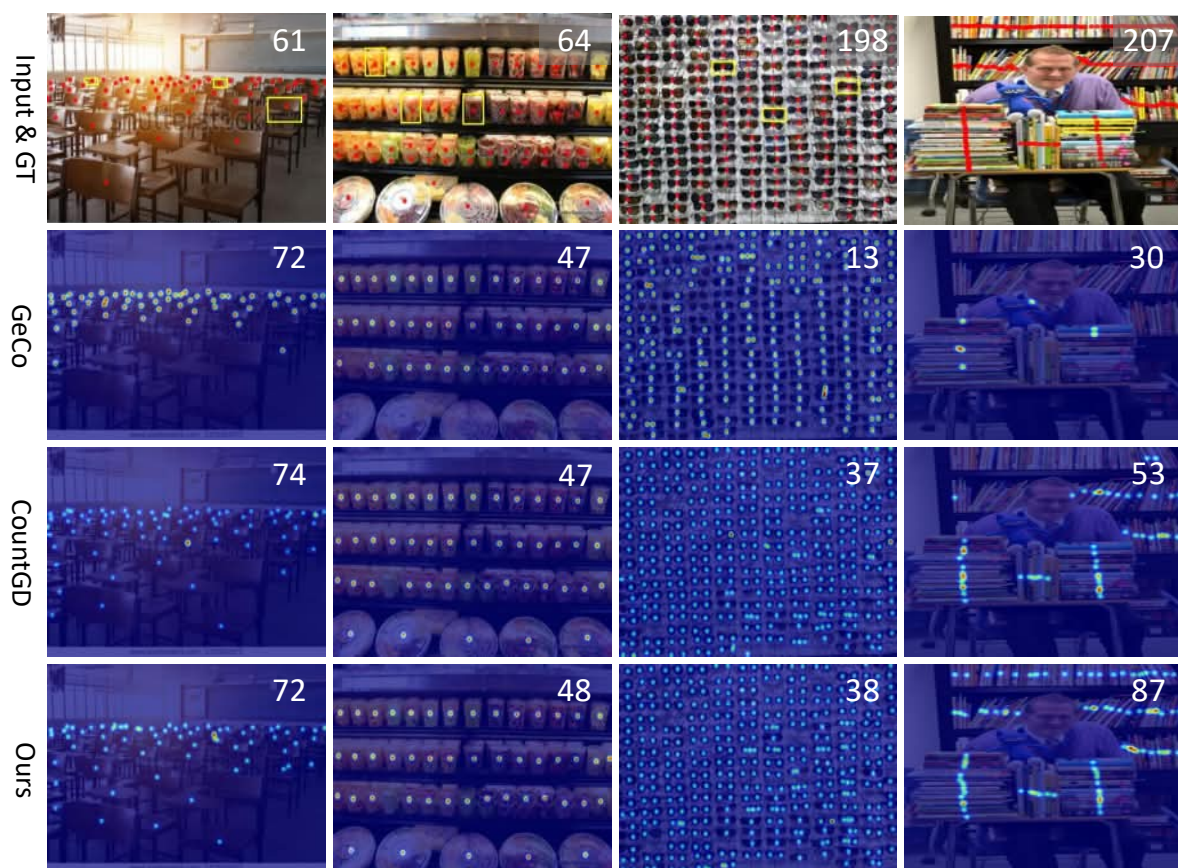


Figure 5.2: Additional failure cases on the FSC-147 dataset [3].

References

- [1] Liu Chang, Zhong Yujie, Zisserman Andrew, and Xie Weidi. CounTR: Transformer-based Generalised Visual Counting. In *Brit. Mach. Vis. Conf.*, 2022.
- [2] Michael Hobley and Victor Prisacariu. Learning to Count Anything: Reference-less Class-agnostic Counting with Weak Supervision. In *IEEE Conf. Comput. Vis. Pattern Recog.*, 2023.
- [3] Viresh Ranjan, Udbhav Sharma, Thu Nguyen, and Minh Hoai. Learning to count everything. In *IEEE Conf. Comput. Vis. Pattern Recog.*, pages 3394–3403, 2021.
- [4] Meng-Ru Hsieh, Yen-Liang Lin, and Winston H Hsu. Drone-based object counting by spatially regularized regional proposal network. In *Int. Conf. Comput. Vis.*, pages 4145–4153, 2017.
- [5] Roni Paiss, Ariel Ephrat, Omer Tov, Shiran Zada, Inbar Mosseri, Michal Irani, and Tali Dekel. Teaching clip to count to ten. In *Int. Conf. Comput. Vis.*, pages 3170–3180, 2023.
- [6] Meng-Ru Hsieh, Yen-Liang Lin, and Winston H Hsu. Drone-based object counting by spatially regularized regional proposal network. In *Int. Conf. Comput. Vis.*, pages 4145–4153, 2017.
- [7] Mingliang Dai, Zhizhong Huang, Jiaqi Gao, Hongming Shan, and Junping Zhang. Cross-head supervision for crowd counting with noisy annotations. In *IEEE Int. Conf. Acoust. Speech Signal Process.*, pages 1–5, 2023.
- [8] Ersin Kilic and Serkan Ozturk. An accurate car counting in aerial images based on convolutional neural networks. *J. Ambient Intell. Humanized Comput.*, pages 1–10, 2023.
- [9] Weidi Xie, J Alison Noble, and Andrew Zisserman. Microscopy cell counting and detection with fully convolutional regression networks. *Comput. Methods Biomech. Biomed. Eng. Imag. Vis.*, 6:283–292, 2018.
- [10] Sravya Vardhani Shivapuja, Mansi Pradeep Khamkar, Divij Bajaj, Ganesh Ramakrishnan, and Ravi Kiran Sarvadevabhatla. Wisdom of (binned) crowds: A bayesian stratification paradigm for crowd counting. In *ACM Int. Conf. Multimed.*, pages 3574–3582, 2021.
- [11] Yuting Liu, Zheng Wang, Miaoqing Shi, Shin’ichi Satoh, Qijun Zhao, and Hongyu Yang. Towards unsupervised crowd counting via regression-detection bi-knowledge transfer. In *ACM Int. Conf. Multimed.*, pages 129–137, 2020.
- [12] Erika Lu, Weidi Xie, and Andrew Zisserman. Class-agnostic counting. In *Proc. Asian Conf. Comput. Vis.*, pages 669–684. Springer, 2019.
- [13] Michael Hobley and Victor Prisacariu. Abc easy as 123: A blind counter for exemplar-free multi-class class-agnostic counting. In *Eur. Conf. Comput. Vis.*, pages 304–319. Springer, 2024.

- [14] Jer Pelhan, Vitjan Zavrtanik, Matej Kristan, et al. DAVE-A Detect-and-Verify Paradigm for Low-Shot Counting. In *IEEE Conf. Comput. Vis. Pattern Recog.*, pages 23293–23302, 2024.
- [15] Viresh Ranjan and Minh Hoai Nguyen. Exemplar free class agnostic counting. In *Proc. Asian Conf. Comput. Vis.*, pages 3121–3137, 2022.
- [16] Nikola Djukic, Alan Lukezic, Vitjan Zavrtanik, and Matej Kristan. A low-shot object counting network with iterative prototype adaptation. In *IEEE Conf. Comput. Vis. Pattern Recog.*, pages 18872–18881, 2023.
- [17] Zhicheng Wang, Liwen Xiao, Zhiguo Cao, and Hao Lu. Vision transformer off-the-shelf: A surprising baseline for few-shot class-agnostic counting. In *Proc. AAAI Conf. Artif. Intell.*, volume 38, pages 5832–5840, 2024.
- [18] Shilong Liu, Zhaoyang Zeng, Tianhe Ren, Feng Li, Hao Zhang, Jie Yang, Qing Jiang, Chunyuan Li, Jianwei Yang, Hang Su, et al. Grounding dino: Marrying dino with grounded pre-training for open-set object detection. In *Eur. Conf. Comput. Vis.*, pages 38–55, 2025.
- [19] Siyang Dai, Jun Liu, and Ngai-Man Cheung. Referring expression counting. In *IEEE Conf. Comput. Vis. Pattern Recog.*, pages 16985–16995, 2024.
- [20] Niki Amini-Naieni, Tengda Han, and Andrew Zisserman. Countgd: Multi-modal open-world counting. *Advances in Neural Information Processing Systems*, 37:48810–48837, 2024.
- [21] Hu Cao, Yueyue Wang, Joy Chen, Dongsheng Jiang, Xiaopeng Zhang, Qi Tian, and Manning Wang. Swin-Unet: Unet-like Pure Transformer for Medical Image Segmentation. In *Eur. Conf. Comput. Vis. Worksh.*, 2022.
- [22] Chaitanya Desai, Deva Ramanan, and Charless C Fowlkes. Discriminative models for multi-class object layout. *International journal of computer vision*, 95:1–12, 2011.
- [23] Wang Li, Li Yongbo, and Xue Xiangyang. Coda: Counting objects via scale-aware adversarial density adaption. In *Int. Conf. Multimedia and Expo*, pages 193–198. IEEE, 2019.
- [24] Zhilin Qiu, Lingbo Liu, Guanbin Li, Qing Wang, Nong Xiao, and Liang Lin. Crowd counting via multi-view scale aggregation networks. In *Int. Conf. Multimedia and Expo*, pages 1498–1503, 2019.
- [25] Yuzhe Wu, Yipeng Xu, Tianyu Xu, Jialu Zhang, Jianfeng Ren, and Xudong Jiang. Gca-sun: A gated context-aware swin-unet for exemplar-free counting. *arXiv preprint arXiv:2409.12249*, 2024.
- [26] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. *Proc. Int. Conf. Learn. Represent.*, 2021.

- [27] Jer Pelhan, Alan Lukezic, Vitjan Zavrtanik, and Matej Kristan. A novel unified architecture for low-shot counting by detection and segmentation. *Adv. Neural Inform. Process. Syst.*, 37:66260–66282, 2024.
- [28] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C Berg, Wan-Yen Lo, et al. Segment anything. In *Int. Conf. Comput. Vis.*, pages 4015–4026, 2023.
- [29] Jingyi Xu, Hieu Le, Vu Nguyen, Viresh Ranjan, and Dimitris Samaras. Zero-shot object counting. In *IEEE Conf. Comput. Vis. Pattern Recog.*, pages 15548–15557, 2023.
- [30] Ruixiang Jiang, Lingbo Liu, and Changwen Chen. Clip-count: Towards text-guided zero-shot object counting. In *ACM Int. Conf. Multimed.*, pages 4535–4545, 2023.
- [31] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *Int. Conf. Mach. Learn.*, pages 8748–8763, 2021.
- [32] Seunggu Kang, WonJun Moon, Euiyeon Kim, and Jae-Pil Heo. Vlcounter: Text-aware visual representation for zero-shot object counting. In *Proc. AAAI Conf. Artif. Intell.*, volume 38, pages 2714–2722, 2024.
- [33] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*, pages 4171–4186, 2019.
- [34] Olga Barinova, Victor Lempitsky, and Pushmeet Kohli. On detection of multiple object instances using hough transforms. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 34(9):1773–1784, 2012.
- [35] Min Shi, Hao Lu, Chen Feng, Chengxin Liu, and Zhiguo Cao. Represent, compare, and learn: A similarity-aware framework for class-agnostic counting. In *IEEE Conf. Comput. Vis. Pattern Recog.*, pages 9529–9538, 2022.
- [36] Yuanwu Xu, Feifan Song, and Haofeng Zhang. Learning spatial similarity distribution for few-shot object counting. In *IJCAI*, pages 1507–1515, 2024.
- [37] Wentao He, Jianfeng Ren, Ruibin Bai, and Xudong Jiang. Two-stage rule-induction visual reasoning on RPMs with an application to video prediction. *Pattern Recog.*, 160:111151, 2025.
- [38] Jialu Zhang, Xiaoying Yang, Wentao He, Jianfeng Ren, Qian Zhang, Yitian Zhao, Ruibin Bai, Xiangjian He, and Jiang Liu. Scale optimization using evolutionary reinforcement learning for object detection on drone imagery. *Proc. AAAI Conf. Artif. Intell.*, 38(1):410–418, 2024.
- [39] Enrico Bellocchio, Thomas A Ciarfuglia, Gabriele Costante, and Paolo Valigi. Weakly supervised fruit counting for yield estimation using spatial consistency. *IEEE Robot. Autom. Lett.*, 4:2348–2355, 2019.

- [40] Jayme Garcia Arnal Barbedo, Luciano Vieira Koenigkan, Patricia Menezes Santos, and Andrea Roberto Bueno Ribeiro. Counting cattle in UAV images—dealing with clustered animals and animal/background contrast changes. *Sensors*, 20:2126, 2020.
- [41] Seunggu Kang, WonJun Moon, Euiyeon Kim, and Jae-Pil Heo. Vlcounter: Text-aware visual representation for zero-shot object counting. In *Proc. AAAI Conf. Artif. Intell.*, volume 38, pages 2714–2722, 2024.
- [42] Hüseyin Gökhan Akçay, Bekir Kabasakal, Duygugül Aksu, Nusret Demir, Melih Öz, and Ali Erdoğan. Automated bird counting with deep learning for regional bird distribution mapping. *Animals*, 10:1207, 2020.
- [43] Vitjan Zavrtanik, Martin Vodopivec, and Matej Kristan. A segmentation-based approach for polyp counting in the wild. *Eng. Appl. Artif. Intell.*, 88:103399, 2020.
- [44] Zhikang Liu, Yiming Zhou, Yuansheng Xu, and Zilei Wang. Simplenet: A simple network for image anomaly detection and localization. In *IEEE Conf. Comput. Vis. Pattern Recog.*, pages 20402–20411, 2023.
- [45] Xingke Song, Xiaoying Yang, Chenglin Yao, Jianfeng Ren, Ruibin Bai, Xin Chen, and Xudong Jiang. ERL-MPP: Evolutionary reinforcement learning with multi-head puzzle perception for solving large-scale jigsaw puzzles of eroded gaps. *Proc. AAAI Conf. Artif. Intell.*, 2025.
- [46] Chenglin Yao, Jianfeng Ren, Ruibin Bai, Heshan Du, Jiang Liu, and Xudong Jiang. Progressively-orthogonally-mapped EfficientNet for action recognition on time-range-doppler signature. *Expert Syst. Appl.*, 255:124824, 2024.
- [47] Jialu Zhang, Xinyi Wang, Chenglin Yao, Jianfeng Ren, and Xudong Jiang. Visual-linguistic cross-domain feature learning with group attention and gamma-correct gated fusion for extracting commonsense knowledge. *ACM Int. Conf. Multimed.*, page 4650–4659, 2024.
- [48] Chengtai Li, Yee Yang Tan, Yuting He, Jianfeng Ren, Ruibin Bai, Yitian Zhao, Heng Yu, and Xudong Jiang. DARR: A dual-branch arithmetic regression reasoning framework for solving machine number reasoning. *Proc. AAAI Conf. Artif. Intell.*, 2025.
- [49] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *Int. Conf. Comput. Vis.*, pages 10012–10022, 2021.
- [50] Ji Zhang, Jian-Jun Qiao, Xiao Wu, and Wei Li. Vehicle counting network with attention-based mask refinement and spatial-awareness block loss. In *ACM Int. Conf. Multimed.*, pages 2889–2898, 2021.
- [51] Hui Lin, Zhiheng Ma, Xiaopeng Hong, Yaowei Wang, and Zhou Su. Semi-supervised crowd counting via density agency. In *ACM Int. Conf. Multimed.*, pages 1416–1426, 2022.
- [52] Lingbo Liu, Jiaqi Chen, Hefeng Wu, Tianshui Chen, Guanbin Li, and Liang Lin. Efficient crowd counting via structured knowledge transfer. In *ACM Int. Conf. Multimed.*, pages 2645–2654, 2020.

- [53] Walid Balid, Hasan Tafish, and Hazem H Refai. Intelligent vehicle counting and classification sensor for real-time traffic surveillance. *IEEE Transactions on Intelligent Transportation Systems*, 19(6):1784–1794, 2017.
- [54] Guangshuai Gao, Qingjie Liu, and Yunhong Wang. Counting from sky: A large-scale data set for remote sensing object counting and a benchmark method. *IEEE Transactions on geoscience and remote sensing*, 59(5):3642–3655, 2020.
- [55] Zach Eaton-Rosen, Thomas Varsavsky, Sebastien Ourselin, and M Jorge Cardoso. As easy as 1, 2... 4? uncertainty in counting tasks for medical imaging. In *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Interv.*, pages 356–364. Springer, 2019.
- [56] Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. Improved baselines with visual instruction tuning. In *IEEE Conf. Comput. Vis. Pattern Recog.*, pages 26296–26306, 2024.
- [57] Minjoon Seo, Aniruddha Kembhavi, Ali Farhadi, and Hannaneh Hajishirzi. Bidirectional attention flow for machine comprehension. In *Proc. Int. Conf. Learn. Represent.*, 2017.
- [58] Hao Tan and Mohit Bansal. Lxmert: Learning cross-modality encoder representations from transformers. *arXiv preprint arXiv:1908.07490*, 2019.
- [59] A Vaswani. Attention is all you need. *Advances in Neural Information Processing Systems*, 2017.
- [60] John Arevalo, Thamar Solorio, Manuel Montes-y Gómez, and Fabio A González. Gated multimodal units for information fusion. *arXiv preprint arXiv:1702.01992*, 2017.
- [61] Ethan Perez, Florian Strub, Harm De Vries, Vincent Dumoulin, and Aaron Courville. Film: Visual reasoning with a general conditioning layer. In *Proc. AAAI Conf. Artif. Intell.*, volume 32, pages 3942–3951, 2018.
- [62] Junnan Li, Ramprasaath Selvaraju, Akhilesh Gotmare, Shafiq Joty, Caiming Xiong, and Steven Chu Hong Hoi. Align before fuse: Vision and language representation learning with momentum distillation. *Adv. Neural Inform. Process. Syst.*, 34:9694–9705, 2021.
- [63] Zhuofan Xia, Xuran Pan, Shiji Song, Li Erran Li, and Gao Huang. Vision transformer with deformable attention. In *IEEE Conf. Comput. Vis. Pattern Recog.*, pages 4794–4803, 2022.
- [64] Hao Zhang, Feng Li, Shilong Liu, Lei Zhang, Hang Su, Jun Zhu, Lionel M Ni, and Heung-Yeung Shum. Dino: Detr with improved denoising anchor boxes for end-to-end object detection. *arXiv preprint arXiv:2203.03605*, 2022.
- [65] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. In *Int. Conf. Comput. Vis.*, pages 2980–2988, 2017.
- [66] Niki Amini-Naieni, Kiana Amini-Naieni, Tengda Han, and Andrew Zisserman. Open-world text-specified object counting. In *Brit. Mach. Vis. Conf.*, 2023.
- [67] Yifei Qian, Zhongliang Guo, Bowen Deng, Chun Tong Lei, Shuai Zhao, Chun Pong Lau, Xiaopeng Hong, and Michael P Pound. T2icount: Enhancing cross-modal understanding for zero-shot counting. In *IEEE Conf. Comput. Vis. Pattern Recog.*, 2025.

- [68] Wei Lin, Kunlin Yang, Xinzhu Ma, Junyu Gao, Lingbo Liu, Shinan Liu, Jun Hou, Shuai Yi, and Antoni B Chan. Scale-prior deformable convolution for exemplar-guided class-agnostic counting. In *Brit. Mach. Vis. Conf.*, page 313, 2022.
- [69] Christoph Schuhmann, Richard Vencu, Romain Beaumont, Robert Kaczmarczyk, Clayton Mullis, Aarush Katta, Theo Coombes, Jenia Jitsev, and Aran Komatsuzaki. Laion-400m: Open dataset of clip-filtered 400 million image-text pairs. *arXiv preprint arXiv:2111.02114*, 2021.

Appendix A

More Dataset Details.

We use three benchmark datasets: FSC-147 [3], CARPK [4], and CountBench [5] for the object counting task in this study. Table A.1 provides a summary of the key statistics and characteristics of these datasets for comparison.

Table A.1: Summary of dataset statistics and characteristics for three representative object counting benchmarks: FSC-147 [3], CARPK [4], and CountBench [5].

Dataset	#Imgs.	Image Resolution	Object Density			#Categories	Visual Exemplar	Textual Exemplar
			Min.	Avg.	Max.			
FSC-147 [3]	6135	(384, 384) - (1918, 384)	7	65	3701	147	✓	✓
CARPK(Test Set) [4]	459	(1280, 720)	2	66.25	188	1	✓	✓
CountBench [5]	540	(384, 384) - (1160, 384)	2	6.04	10	277	✗	✓

Table A.2: List of categories in FSC-147 dataset [3], ordered by training, validation, and test subsets from top to bottom.

Training Subset							
alcohol bottle	baguette roll	ball	banana	bead	bee	birthday candle	biscuit
boat	bottle	bowl	box	bread roll	brick	buffalo	bun
calamari ring	can	candle	cap	car	cartridge	cassette	cement bag
cereal	chewing gum piece	chopstick	clam	coffee bean	coin	cotton ball	cow
crane	crayon	croissant	crow	cup	cupcake	cupcake holder	fish
gemstone	go game piece	goat	goldfish snack	goose	ice cream	ice cream cone	instant noodle
jade stone	jeans	kidney bean	kitchen towel	lighter	lipstick	m&m piece	macaron
match	meat skewer	mini blind	mosaic tile	naan bread	nail	nut	onion ring
orange	pearl	pen	pencil	penguin	pepper	person	pigeon
plate	polka dot tile	potato	rice bag	roof tile	screw	shoe	spoon
spring roll	stair	stapler pin	straw	supermarket shelf	swan	tomato	watermelon
window	zebra						
Validation Subset							
ant	bird	book	bottle cap	bullet	camel	chair	chicken wing
donut	donut holder	flamingo	flower	flower pot	grape	horse	kiwi
milk carton	oyster	oyster shell	package of fresh cut fruit	peach	pill	polka dot	prawn cracker
sausage	seagull	shallot	shirt	skateboard	toilet paper roll		
Test Subset							
apple	candy piece	carrom board piece	cashew nut	comic book	crab cake	deer	egg
elephant	finger food	green pea	hot air balloon	keyboard key	lego	marble	marker
nail polish	potato chip	red bean	round dessert	sauce bottle	sea shell	sheep	ski
stamp	sticky note	strawberry	sunglasses	tree log	watch	yellow lego stud	

A.1 FSC-147 Dataset

The FSC-147 dataset [3] was curated specifically for CAC tasks, comprising 6,135 high-quality images across 147 diverse categories, such as kitchen utensils, office stationery, vehicles, and animals. These images were collected through a keyword-based web scraping approach from

sources including Flickr, Google, and Bing. Each image was manually filtered to ensure the presence of at least seven objects with uniform appearance and minimal occlusion. Objects in each image are annotated with a dot at their approximate center, and three randomly selected objects are further marked with axis-aligned bounding boxes as exemplars. The dataset is partitioned into three disjoint subsets: a training set with 3,659 images from 89 categories, a validation set with 1,286 images from 29 categories, and a testing set with 1,190 images from 29 categories. Notably, the object categories in the three subsets do not overlap, thereby ensuring that models are evaluated on entirely unseen object types. Table A.2 lists the categories included in each subset. On average, images have a resolution of 774×938 pixels and contain 56 objects (ranging from 7 to 3,701 objects per image), totaling 343,818 annotations. Figure A.1 shows some representative sample images in the FSC-147 dataset.



Figure A.1: Sample images in the FSC-147 dataset [3].

A.2 CARPK Dataset

The CARPK dataset [4] is a large-scale benchmark specifically curated for car counting in aerial parking lot scenarios. It comprises 1,448 bird’s-eye view images with approximately 90,000 annotated vehicles captured across four distinct parking lots using drone-mounted cameras. Each image is annotated with precise bounding boxes, defined by the top-left and bottom-right coordinates of each visible vehicle, including those at the image boundaries, provided the vehicle is discernible. CARPK offers a unique drone-based perspective. This top-down view introduces varied spatial layouts and occlusion patterns, enhancing the diversity and generalization of training samples for deep learning-based counting models. CARPK is officially divided into 989 and 459 samples for training and testing, respectively [4]. Following [20, 25], we employ its test set for cross-domain evaluation, with vehicles ranging from 2 to 188 per image. Figure A.2 presents representative samples from the CARPK dataset, highlighting its visual diversity and detailed annotations.

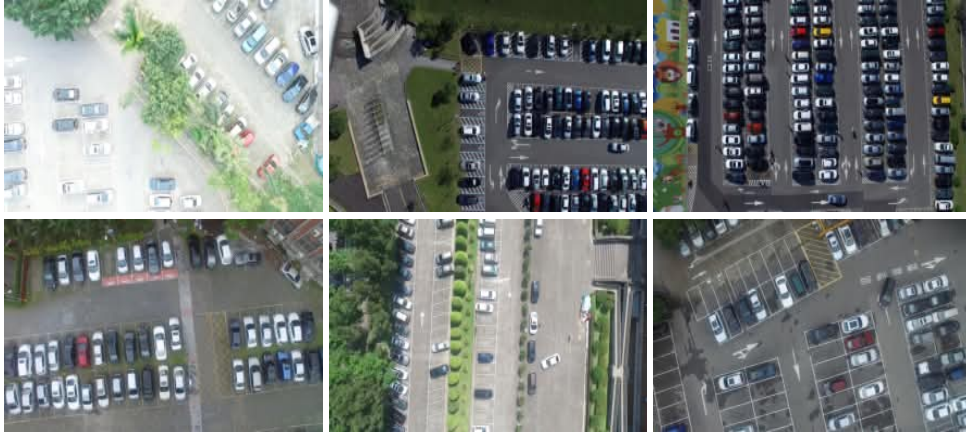


Figure A.2: Sample images from the CARPK dataset [4].

A.3 CountBench Dataset

The CountBench dataset [5] is a newly developed benchmark for object counting, derived from the LAION-400M image-text dataset [69]. It comprises 540 image-caption pairs, with the number of visible objects in each image ranging from two to ten. Each pair includes an image accompanied by a corresponding caption that accurately reflects the count of visible objects, ensuring a balanced distribution of object counts across the dataset. Following [20], the entire dataset serves as a test set to assess the cross-domain generalization capabilities. This dataset is exclusively intended for evaluation purposes and provides a diverse range of visual scenarios, as illustrated in Figure A.3.



Figure A.3: Sample images from the CountBench dataset [5].

Appendix B

Detailed Architecture of CSG.

Detailed architectures of the proposed major components, *i.e.*, GICE, CSR, and SGGA, are presented in Table B.1.

Table B.1: Overall configuration of GICE, CSR, and SGGA modules.

Module	Sub Module	Layer Operations	Parameters	Input	Output
GICE	Token Saliency	MLP	C256C2048C1024	$(B, L, 256)$	$(B, L, 1)$
		AvgPooling Sigmoid Max			
	Contrastive Embedding	MLP	C256C2048C1024	$(B, L^E, 256), (B, L^Q, 1024)$	$(B, L, 1)$
		Dot Product Sigmoid Max			
	Gate	Linear	C256C1024	$(B, L, 256)$ $(B, L, 1024), (B, L, 1)$	
		Repeat			
		Hadamard Product	C1024		$(B, L, 1024)$
		Layer Norm			
		Repeat	C256C256	$(B, L, 1024), (B, L, 1)$	
		Hadamard Product			
		Linear	C256		$(B, L, 256)$
		Layer Norm			
		Addition	C256C768	$(B, L, 256), (B, L, 256)$	
		Linear			
CSR	Semantic Aggregation	Layer Norm	C768		$(B, L, 768)$
		Self-Attention	C768D4H8	$(B, L^V, 768), (B, L^L, 768)$	$(B, 1, 768)$
		CLSPooling			
		Self-Attention	C768D4H8	$(B, L^V, 768), (B, L^L, 768)$	$(B, 1, 768)$
		CLSPooling			
	Semantic Alignment	Concat	C1536C2048C256	$(B, 1, 768), (B, 1, 768)$	$(B, 1, 256)$
		MLP			
		Concat	C512D6H8		$(B, L, 256), (B, L, 256)$
		Self-Attention			
	Bi-Attention	Linear	C512C256		$(B, L, 256)$
SGGA	Significant Candidates Selection	Deformable-Attention	C256D6	$(B, L, 256)$	$(B, L, 256)$
		Dot Product	C256C2048C256	$(B, L^Q, 256), (B, L^S, 256)$	$(B, L, 256)$
		Softmax			
		Dot Product			
		MLP $\times 2$			
	Group Attention	Top-K Selection	K900	$(B, L^Q, 256), (B, L^S, 256)$	$(B, 900, 256)$
		Self-Attention	C256D6H8	$(B, 900, 256)$	
		Cross-Attention	C256D6H8	$(B, 900, 256), (B, L, 256)$	
		Cross-Attention	C256D6H8	$(B, 900, 256), (B, L, 256)$	
		AvgPooling	C256D3H8	$(B, 900, 256)$	$(B, 2, 256)$
		Cross-Attention		$(B, 2, 256), (B, 900, 256)$	$(B, 2, 256)$
		Self-Attention		$(B, 2, 256)$	
		Cross-Attention		$(B, 900, 256), (B, 2, 256)$	$(B, 900, 256)$
	Head	MLP	C256C4	$(B, 900, 256)$	$(B, 900, 4)$
		Dot Product		$(B, 900, 256)$	
		Max			$(B, 900, 1)$

Appendix C

ICME Accepted Paper

GCA-SUNet: A Gated Context-Aware Swin-UNet for Exemplar-Free Counting

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Abstract—Exemplar-Free Counting aims to count objects of interest without intensive annotations of objects or exemplars. To achieve this, we propose a Gated Context-Aware Swin-UNet (GCA-SUNet) to directly map an input image to the density map of countable objects. Specifically, a set of Swin transformers form an encoder to derive a robust feature representation, and a Gated Context-Aware Modulation block is designed to suppress irrelevant objects or background through a gate mechanism and exploit the attentive support of objects of interest through a self-similarity matrix. The gate strategy is also incorporated into the bottleneck network and the decoder of the Swin-UNet to highlight the features most relevant to objects of interest. By explicitly exploiting the attentive support among countable objects and eliminating irrelevant features through the gate mechanisms, the proposed GCA-SUNet focuses on and counts objects of interest without relying on predefined categories or exemplars. Experimental results on the real-world datasets such as FSC-147 and CARPK demonstrate that GCA-SUNet significantly and consistently outperforms state-of-the-art methods. The code is available at <https://github.com/Amordia/GCA-SUNet>.

Index Terms—Object Counting, Exemplar-Free Counting, Gate Mechanism, Self-Similarity Matrix

I. INTRODUCTION

Object counting determines the number of instances of a specific object class in an image, *e.g.*, vehicles [1], [2], crowd [3], and cells [4]. It can be broadly categorized into the following. 1) Class-Specific Counting (CSC), counting specific categories like fruits [5] and animals [6]; 2) Class-Agnostic Counting (CAC), counting objects based on visual exemplars [7]–[9] or text prompts [10], [11]; 3) Exemplar-Free Counting (EFC), counting objects without exemplars, presenting a significant challenge in discerning countable objects and determining their repetitions [9], [12], [13].

Exemplar-Free Counting shows promise for automated systems such as wildlife monitoring [14], healthcare [15], and anomaly detection [16]. Hobley and Prisacariu directly regressed the image-level features learned by attention modules into a density map [13]. CounTR [9] and LOCA [17] are originally designed for CAC tasks, but can be adapted to EFC tasks by using trainable components to simulate exemplars. RepRPN-Counter identifies exemplars from region proposals

by majority voting [12], and DAVE selects valuable objects using a strategy similar to majority vote [18].

Existing models [9], [17], [18] often explicitly require exemplars to count similar objects. EFC methods such as RepRPN-Counter do not require exemplars but generate them through region proposal [12]. Either explicit or implicit exemplars may induce sample bias as exemplars can't cover the sample distribution. To address the challenge, we propose Gated Context-Aware Swin-UNet (GCA-SUNet), which directly maps an input image to the density map of countable objects, without any exemplars. Specifically, the encoder consists of a set of Swin Transformers to extract features, and a set of Gated Context-Aware Modulation (GCAM) blocks to exploit the attentive supports of countable objects. The bottleneck network includes a Gated Enhanced Feature Selector (GEFS) to emphasize the encoded features that are relevant to countable objects. The decoder includes a set of Swin transformers for generating the density map, with the help of Gated Adaptive Fusion Units (GAFUs) to selectively weigh features based on their relevance to countable objects. Finally, a regression head is utilized to derive the density map from the aggregated features.

One key challenge in EFC is to effectively differentiate countable objects from other objects. The GCAM blocks tackle this challenge by first evaluating the feature quality by computing the feature score for each token, and then prioritizing those with informative content. To uncover the informative features, GCAM computes pairwise similarities between tokens through a self-similarity matrix, exploiting the support of repeating objects in the same scene. Lastly, a gate mechanism is incorporated to highlight the most relevant features while suppressing irrelevant ones.

Another challenge is that foreground objects often share similar low-level features with background content. The skip connections directly fuse low-level features in the encoder with high-level semantics in the decoder, potentially impeding counting performance as the background could disturb the foreground objects. To tackle this issue, gate mechanisms are incorporated into both GEFS and GAFU to suppress irrelevant low-level features while preserving as much information on objects of interest as possible. The former selectively enhances the condensed features at the bottleneck, and the latter filters the irrelevant features in the decoder.

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Our contributions can be summarized as follows. 1) The proposed GCA-SUNet achieves exemplar-free counting through a UNet-like architecture that utilizes Swin transformer blocks for feature encoding and decoding, avoiding the sample bias of exemplar-based methods [12]. 2) The proposed GCAM focuses on countable objects by exploiting attentive support of repetitive objects through the self-similarity matrix. 3) The gate mechanism is integrated into various modules, *e.g.*, GCAM, GEFS and GAFU, which suppresses the features of irrelevant objects or background while highlighting the most relevant features to countable objects. 4) Our GCA-SUNet is evaluated on the FSC-147 and CARPK datasets. It outperforms state-of-the-art methods for exemplar-free counting.

II. RELATED WORK

Object counting has evolved through various paradigms, each with obvious tradeoffs on annotation overhead, adaptability, and generalization. Existing methods can be broadly grouped into Class-Specific Counting (CSC), Class-Agnostic Counting (CAC) and Exemplar-Free Counting (EFC).

Class-Specific Counting. Early object counting efforts are tightly coupled with class-specific detectors trained on pre-defined categories [19]–[21]. These frameworks offer accurate localization and counting, but their reliance on class-specific training data hinders their ability to generalize to unseen object classes. As a result, scaling these approaches to more diverse scenarios remains challenging and computationally expensive.

Class-Agnostic Counting. To overcome the limitations of class specificity, CAC methods [7]–[9] have been developed. Instead of modelling each class individually, these methods learn a universal mapping from image features to density maps, guided by visual exemplars [7]–[9], [22] or textual prompts [10], [11]. This strategy alleviates the need for category-specific annotations and enables broader generalization. Prominent examples include Generic Matching Network (GMN) [23], which infers counts through similarity comparisons between query images and exemplar shots, and Fam-Net [7], which refines query features through feature correlation and adaptive parameter updates. CounTR [9] enhances the token of exemplars and the query image using cross-attention mechanics. DAVE [18] utilizes a detector to generate a high-recall object set and excludes outliers by cosine-similarity metrics. Models exploiting textual exemplars [10], [11] often rely on rich language models and vision language pre-training to interpret instructions and align them with visual content. Despite these advancements, the above methods inherently remain dependent on the quality and relevance of the guidance, which might falter if these cues are not informative or biased.

Exemplar-Free Counting. In contrast, Exemplar-Free Counting (EFC) eliminates all external guidance, *i.e.*, no predefined categories, no exemplars, and no textual prompts. Instead, EFC models autonomously discover and emphasize regions containing countable objects solely from visual cues [9], [12], [13]. This reduces annotation burden and prevents biases introduced by external supports, making EFC more suitable for diverse and unpredictable real-world settings. For instance,

some approaches leverage pre-trained vision transformers [12] with weakly-supervised training, or automatically generate region proposals as exemplars [13], facilitating the counting without any provided exemplar.

While prior models have progressively moved from strict CSC tasks to more flexible EFC tasks, the complete absence of guiding signals beyond the visual images remains an ambitious goal. The proposed Gated Context-Aware Swin-UNet (GCA-SUNet) tackles this challenge by directly predicting a density map for countable objects without any exemplar or textual guidance. More importantly, by strategically integrating gate mechanisms within the encoder, bottleneck, and decoder, and leveraging self-similarity to filter out irrelevant backgrounds, the proposed method advances the state-of-the-art in Exemplar-Free Counting significantly. This strategy achieves both robustness and scalability, opening new possibilities for counting arbitrary objects in unconstrained scenarios.

III. PROPOSED METHOD

A. Overview of Proposed Method

The proposed GCA-SUNet is built upon a Swin-UNet architecture [24], with three new types of building blocks, GCAM, GEFS and GAFU to exploit attentive support of countable objects and suppress irrelevant tokens or features, as outlined in Fig. 1. The primary motivation to choose Swin-UNet is that it has demonstrated superior performance in many tasks [25]. It is also well suited for our task in which we need to generate object density maps of the same size as the input image. Specifically, the proposed GCA-SUNet begins with patched image feature $\mathbf{F}$, following by feature encoding,

$$\mathbf{F}_i^E = \mathcal{F}_i^{\text{Down}}(\mathcal{F}_i^{\text{Swin-T}}(\mathcal{F}_i^{\text{GCAM}}(\mathbf{F}_{i-1}^E))), \quad (1)$$

where $\mathcal{F}_i^{\text{Down}}$, $\mathcal{F}_i^{\text{Swin-T}}$, $\mathcal{F}_i^{\text{GCAM}}$ denote down-sampling, GCAM, and Swin-T processing, and $\mathbf{F}_{i-1}^E$ and $\mathbf{F}_i^E$ are the input and output features at the i -th stage, respectively. GCAM enhances the token for countable objects and suppresses others.

At the bottleneck, the features are enhanced through GEFS, *i.e.*, $\mathbf{F}^{\text{BN}} = \mathcal{F}^{\text{GEFS}}(\mathbf{F}_K^E)$, where $\mathcal{F}^{\text{GEFS}}(\cdot)$ denotes the operation of GEFS, and $\mathbf{F}_K^E$ denotes the output features at the K -th stage of the encoder. GEFS selects the features corresponding to the countable object using a gate mechanism.

Subsequently, a set of Swin transformer blocks are utilized as the decoder to derive the density map. Specifically, the features at the j -th stage of the decoder are derived as,

$$\mathbf{F}_j^D = \mathcal{F}_j^{\text{Up}}(\mathcal{F}_j^{\text{Swin-T}}(\mathcal{F}_j^{\text{GAFU}}(\mathbf{F}_{j-1}^D, \mathbf{F}_{K+1-j}^E))), \quad (2)$$

where $\mathcal{F}_j^{\text{Up}}$, $\mathcal{F}_j^{\text{Swin-T}}$, and $\mathcal{F}_j^{\text{GAFU}}$ denote the operation of up-sampling, Swin transformer, and GAFU block, respectively. The GAFU enhances features through a gate mechanism, prioritizing crucial information with a dynamic assigned weight.

Finally, features are processed through a regression head,

$$\mathbf{F}^{\text{head}} = \mathcal{F}^{\text{Head}}(\mathbf{F}_K^D), \quad (3)$$

where $\mathcal{F}^{\text{Head}}$ denotes the regression head consisting of a series of convolutional blocks. The output is a density map that accurately represents the object count.

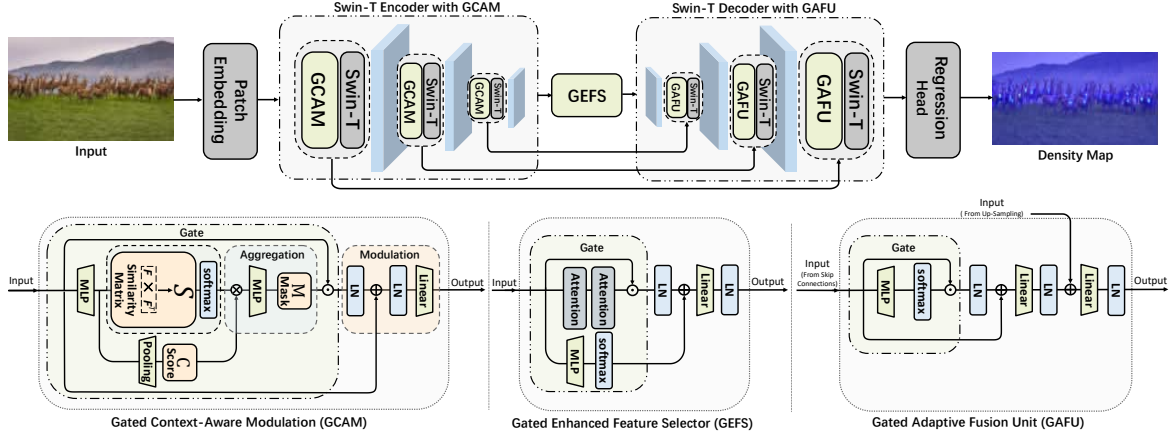


Fig. 1. Overview of proposed GCA-SUNet. It consists of an encoder, bottleneck, and a decoder. The encoder consists of a set of GCAM blocks to highlight the features relevant to countable objects while suppressing others, and a set of Swin transformers to extract features. The GEFS in the bottleneck and the GAFU in the decoder enhance the features of objects of interest. Finally, a regression head generates a density map for estimating the number of objects.

B. Swin-T Encoder with GCAM

A set of SWIN transformer blocks form the encoder to extract features related to countable objects. The GCAM adapts a dynamic token modulation to simultaneously exploit the attentive support of tokens relevant to countable objects and suppress features of irrelevant objects [26]–[28]. This process facilitates self-probing among objects and precise capture of objects of the same category for exemplar-free counting.

Specifically, we first compress token features F_i^E using an MLP, $F_i^{\text{proj}} = \mathcal{F}^{\text{MLP}}(F_i^E)$. To identify the objects of interest, we resort to two key observations: 1) The objects should be salient enough to step out from the background; 2) Similar objects could support each other to boost the saliency. The former is exploited by computing the average feature score C_i for each token through average pooling $\mathcal{F}^{\text{AVG}}$ as,

$$C_i = \mathcal{F}^{\text{AVG}}(F_i^{\text{proj}}). \quad (4)$$

The score reflects the importance of tokens, prioritizing those with rich content. Tokens that frequently appear in similar contexts are more likely to be related to the target object of interest. To identify them, we employ a similarity matrix,

$$S_i = \sigma(F_i^{\text{proj}} \times F_i^{\text{proj}T}), \quad (5)$$

where σ is a softmax function to normalize similarities across rows. S_i captures the semantic similarity of tokens in a spatial context to emphasize tokens that repeatedly share similar features, thereby emphasizing potential countable objects. A mask M_i is derived by aggregating S_i and C_i as,

$$M_i = \sigma(\mathcal{F}^{\text{MLP}}(S_i \times C_i)). \quad (6)$$

C_i encodes the token importance when considering the token alone, while S_i encodes the token importance after interacting with other tokens. The tokens are then filtered by the mask as,

$$F_i^{\text{GCAM}} = \mathcal{F}^{\text{Linear}}(\mathcal{F}^{\text{LN}}(\mathcal{F}^{\text{LN}}(F_i^E \odot M_i) + F_i^{\text{proj}})), \quad (7)$$

where $\odot$, $\mathcal{F}^{\text{LN}}$ and $\mathcal{F}^{\text{Linear}}$ denote element-wise product, layer normalization and linear layer, respectively. The GCAM applies the mask M_i to F_i^E , filtering out less relevant features and reinforcing those critical ones for countable objects.

The proposed GCAM selectively amplifies the importance of tokens related to significant object features through pairwise similarities. It is significantly different from LOCA [17] and DAVE [18] which depends on predefined prototypes to predict object densities. Instead, our GCAM leverages the self-similarity matrix for more dynamic and precise modulation of features. It is also different from RCC [13] that relies on global feature comparisons, and CounTR [9] that utilizes attention-driven similarity matrices. The GCAM emphasizes a clear distinction between relevant and irrelevant tokens.

C. Bottleneck with GEFS

The proposed GEFS selectively filters out features in the bottleneck that are semantically irrelevant to the objects of interest, while allowing critical condensed features to pass. It is implemented by first deriving the local token weights as $W^{\text{GEFS}} = \sigma(\mathcal{F}^{\text{MLP}}(F_K^E))$, and applying them on features as,

$$F_0^D = F_K^E + (W^{\text{GEFS}} \odot (\mathcal{F}^{\text{ATTN}}(\mathcal{F}^{\text{ATTN}}(F_K^E)))). \quad (8)$$

The GEFS is positioned at the bottleneck where features transit from the down-sampling pathway to the up-sampling pathway. As a vital bottleneck, the GEFS compresses and filters essential object-related features, ensuring that only the most relevant information of countable objects is advanced into the up-sampling pathway. Specifically, the attention blocks allow the model to capture a more comprehensive range of relationships and refine its ability to extract high-level semantic representations, resulting in more accurate and semantically enriched feature embeddings. Furthermore, a gate mechanism is incorporated into GEFS to selectively prioritize specific aspects of this condensed representation, effectively filtering out less relevant semantics. This process not only refines features by strengthening inter-dependencies between relevant

TABLE I
COMPARISON WITH OTHER METHODS ON THE FSC-147 DATASET [7], WITH BEST RESULTS HIGHLIGHTED IN BOLD.

Category	Method	Backbone	Resolution	Test Set		Val Set	
				MAE	RMSE	MAE	RMSE
CAC	GMN [23] ACCV'18	ResNet-50	255	37.86	141.39	39.02	106.06
	FamNet [7] CVPR'21	ResNet-50	384	32.27	131.46	32.15	98.75
	LOCA [17] ICCV'23	ResNet-50	512	16.22	103.96	17.43	54.96
	DAVE [18] CVPR'24	ResNet-50	512	15.14	103.49	15.54	52.67
	CounTR [9] BMVC'22	ViT-B & CNN	384	14.71	106.87	18.07	71.84
EFC	RepRPN-C [12] ACCV'22	ResNet-50	[384,1504]	26.66	129.11	29.24	98.11
	RCC [13] CVPRW'23	ViT-B	224	17.12	104.53	17.49	58.81
	Proposed GCA-SUNet	SwinT-B	384	14.08	84.77	17.02	55.37

features, but also lays a solid foundation for comprehensive reconstruction of the up-sampling pathway.

D. Swin-T Decoder with GAFU

The decoder contains a set of Swin-T blocks to articulate the density map, and a set of Gated Adaptive Fusion Units (GAFUs) to integrate low-level features from the encoder with abstract features from the up-sampling pathway. In each GAFU, the token weights are determined by a gate mechanism as, $\mathbf{W}^{\text{GAFU}} = \sigma(\mathcal{F}^{\text{MLP}}(\mathbf{F}_i^{\text{E}}))$, and modulate the features as,

$$\mathbf{F}_i^{\text{G}} = \mathbf{F}_i^{\text{E}} + (\mathbf{W}^{\text{GAFU}} \odot \mathbf{F}_i^{\text{E}}). \quad (9)$$

Subsequently, these features are fused with the decoder features as $\mathbf{F}_j^{\text{GAFU}} = \mathcal{F}^{\text{Linear}}([\mathbf{F}_j^{\text{D}}, \mathbf{F}_i^{\text{G}}])$. By weighing the features during the fusion process, the GAFU effectively concentrates on semantic information pertinent to countable objects, minimizing the interference from irrelevant details.

E. Discussion of Gate Mechanisms

Gate mechanisms are incorporated into the three proposed modules, GCAM, GEFS and GAFU. They share a similar underlying mechanism, both they are tailored to address distinct challenges. 1) Gates in GCAM tackle the challenges of differentiating countable objects from others, by emphasizing the relevant tokens through a similarity-based gate. It effectively probes countable objects within the same category through a similarity matrix. 2) Gates in GEFS or GAFU tackle the challenges of effectively differentiating foreground features from the background context, *i.e.*, to suppress irrelevant low-level features while preserving essential information on objects of interest. By reinforcing relevant inter-dependencies, it prioritizes the semantic information of countable objects, thereby enhancing its ability to distinguish the objects from the background when constructing the density map.

IV. EXPERIMENTAL RESULTS

A. Experimental Settings

Two benchmark datasets are utilized for evaluation. Following [9], [17], [18], we employ the Mean Average Error (MAE) and Root Mean Squared Error (RMSE) as evaluation metrics. **FSC-147** [7] is collected from real world and it consists of 6,135 images of 147 categories with 7 to 3,731 objects per

image, mainly composed of foods, animals, kitchen utensils, and vehicles. It is officially split into 3,659, 1,286 and 1,190 images for training, validation and testing, respectively.

CARPK [19] comprises 1,448 images taken from four parking lots using a bird's-eye view. It is primarily intended for object counting and vehicle localization tasks, and it is officially split into 989 training images and 459 testing images.

The Swin-T blocks are pre-trained on ImageNet-22k [29], and other modules are randomly initialized. AdamW optimizer is employed, with an initial learning rate of 0.003, a decay rate of 0.95 and a batch size of 16. The model is trained with a warm-up period of 50 epochs and 300 epochs in total. The input image size is 384×384 . Data augmentation [9] is employed to facilitate efficient training. All experiments are conducted using two NVIDIA RTX A5000 GPUs.

B. Comparison Results on FSC-147 Dataset

Comparison experiments are conducted on the FSC-147 dataset. The results are summarized in Table I. Following the practice in [9], [13], we report the errors on the test and validation sets. We have the following observations. 1) The proposed GCA-SUNet outperforms all the compared methods regarding test errors, while performing slightly poorer than DAVE [18] regarding validation errors. The superior results demonstrate its effectiveness for EFC tasks. It outperforms not only the dedicated methods for solving EFC tasks such as RepRPN-C [12] and RCC [13], but also state-of-the-art models for CAC tasks. Compared to the second-best method, CounTR [9], the performance gain is 0.63 for of MAE and 16.47 for RMSE. 2) Although the GCA-SUNet performs slightly poorer than DAVE [18] in terms of validation errors, it significantly outperforms DAVE in terms of test errors, *i.e.*, a performance gain of 1.06 for MAE and 18.72 for RMSE. DAVE performs well on the validation set and generalizes poorly on the test set. In contrast, the GCA-SUNet generalizes well on the novel test set with minimal errors.

We visually compare the density maps with CounTR [9] and RCC [13] on the FSC-147 dataset. As shown in Fig. 2, our method can capture fine-grained details of objects. In contrast, CounTR sometimes generates density maps that do not accurately distinguish between individual objects in the



Fig. 2. Visual comparisons of the generated density maps with CounTR [9] and RCC [13] on the FSC-147 dataset.

map, *e.g.*, in the fourth column, CounTR can't identify the far-way small fruit, while our method can.

C. Cross-Domain Evaluation

Following [30], we conduct a cross-domain evaluation, *i.e.*, training the model on the FSC-147 dataset [7] while directly evaluating on the CARPK dataset [19]. The results are summarized in Table II. The results for all the compared methods are obtained from [9], [13], [17] under the same settings. As shown in Table II, our model shows superior cross-domain performance compared with other methods, achieving a performance gain of 0.56 on MAE and 0.61 on RMSE compared to the previous best-performing method CounTR. Compared to the earlier EFC model [13], the gains are even more significant, highlighting GCA-SUNet's superior generalization over the compared methods.

TABLE II
TRAINING ON THE FSC-147 DATASET [7] WHILE TESTING ON THE CARPK DATASET [19].

Category	Method	Test Set	
		MAE	RMSE
CAC	LOCA [17] ICCV'23	16.84	19.72
	CounTR [9] BMVC'22	11.52	14.56
EFC	RCC [13] CVPR'23	21.38	26.61
	Proposed GCA-SUNet	10.96	13.95

D. Ablation Study of Major Modules

We ablate the three major modules of proposed method on the FSC-147 dataset [7]. Swin-UNet [24] serves as the baseline model. The results are summarized in Table III. The GCAM module alone significantly reduces the MAE by 2.32 on the test set and by 1.72 on the validation set compared with the baseline, underscoring its capability to enhance feature selectivity that is essential for complex scenes. Similarly, utilizing GEFS or GAFU individually also greatly reduces errors on both test and validation sets, showcasing the proposed gate mechanism's ability to highlight relevant features while suppressing irrelevant ones. The combination

TABLE III
ABLATION STUDY OF EACH COMPONENT ON THE FSC-147 DATASET [7].

$\mathcal{F}^{\text{GCAM}}$	$\mathcal{F}^{\text{GEFS}}$	$\mathcal{F}^{\text{GAFU}}$	Test Set		Val Set	
			MAE	RMSE	MAE	RMSE
×	×	×	18.86	89.00	22.85	85.16
✓	×	×	16.54	85.20	21.13	81.30
×	✓	×	16.63	80.33	20.13	62.98
×	×	✓	16.87	84.36	21.60	79.37
✓	×	✓	15.27	84.81	20.08	81.58
×	✓	✓	15.07	78.79	18.58	59.17
✓	✓	×	15.21	85.71	17.97	55.67
✓	✓	✓	14.08	84.77	17.02	55.37

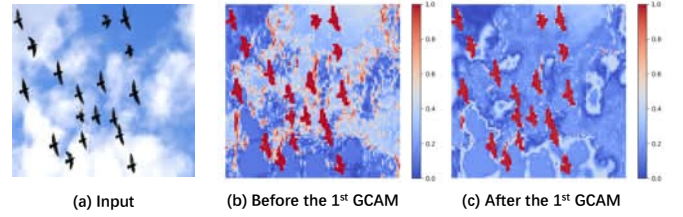


Fig. 3. Feature maps before/after the 1st GCAM in the Swin-T encoder.

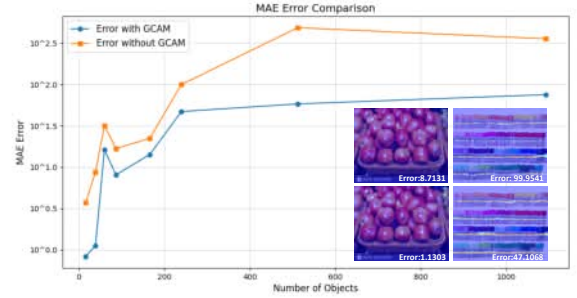


Fig. 4. Evaluation of GCAM in various object density scenarios.

of these modules also achieves significant performance gains. Specifically, the overall framework with all three components achieves the most substantial improvement compared with the baseline, reducing the MAE by 4.78 on the test set and by 5.83 on the validation set. The results show the effectiveness of the synergistic interaction among different modules and emphasize the individual contribution of each major component.

E. Analysis of GCAM

As shown in Table III, GCAM contributes most significantly to the overall performance gain. We further visualize its effects as the density maps in Fig. 3. Following the practice in [13], sub-figure (b) and (c) are obtained by projecting the density maps into two-dimensional spaces. Clearly by applying GCAM, the noisy estimation in the background areas (sky) is significantly reduced, with the foreground objects (birds) becoming more prominent. The results show that the proposed GCAM effectively enhances the representation of foreground tokens while suppressing irrelevant ones in the background.

We further compare the results with/without the GCAM across scenarios of different object densities in Fig. 4. For different object densities, the GCAM could consistently reduce the MAE, signifying its importance. On the other hand, Fig. 4 reveals a clear trend of increasing error as the number of objects grows. Indeed, as the density of objects increases, they become smaller and counting tasks become more challenging. Future work will focus on optimizing the proposed approach to improve its performance in such challenging scenes.

F. Model Complexity Analysis

Tab. IV compares the inference time, model size, FLOPs and training epochs with the previous best method, CounTR [9]. The proposed model requires inference time comparable to CounTR. It significantly outperforms CounTR in terms of accuracy, at a higher but reasonable computational cost.

TABLE IV
COMPARISON OF INFERENCE TIME, MODEL SIZE AND FLOPs, AND TRAINING EPOCHS.

Method	Inf. time (s)	Parameters (M)	FLOPs (G)	Training (epochs)
CounTR [9]	0.08	100	84	1000
Proposed GCA-SUNet	0.11	192	173	300

V. CONCLUSION

The proposed GCA-SUNet effectively tackles the problems of exemplar-free counting by using a Swin-UNet architecture to directly map the input image to the density map of countable objects. The proposed GCAM exploits the attention information among the tokens of repetitive objects through the self-similarity matrix, and suppresses the features of irrelevant objects through a gate mechanism. The gate mechanism is also incorporated into the GEFS module and the GAFU module, which highlight the features most relevant to countable objects while suppressing irrelevant ones. Our experiments on the FSC-147 and CARPK datasets demonstrate that GCA-SUNet outperforms state-of-the-art methods, achieving superior performance in both intra-domain and cross-domain scenarios.

REFERENCES

- [1] W. He, J. Ren, R. Bai, and X. Jiang, "Two-stage rule-induction visual reasoning on RPMs with an application to video prediction," *Pattern Recognit.*, vol. 160, p. 111151, 2025.
- [2] J. Zhang, X. Yang, W. He, J. Ren, Q. Zhang, Y. Zhao, R. Bai, X. He, and J. Liu, "Scale optimization using evolutionary reinforcement learning for object detection on drone imagery," *Proc. AAAI Conf. Artif. Intell.*, vol. 38, no. 1, pp. 410–418, 2024.
- [3] M. Dai, Z. Huang, J. Gao, H. Shan, and J. Zhang, "Cross-head supervision for crowd counting with noisy annotations," in *IEEE Int. Conf. Acoust. Speech Signal Process.*, 2023, pp. 1–5.
- [4] W. Xie, J. A. Noble, and A. Zisserman, "Microscopy cell counting and detection with fully convolutional regression networks," *Comput. Methods Biomech. Biomed. Eng. Imag. Vis.*, vol. 6, pp. 283–292, 2018.
- [5] E. Bellocchio, T. A. Ciarfuglia, G. Costante, and P. Valigi, "Weakly supervised fruit counting for yield estimation using spatial consistency," *IEEE Robot. Autom. Lett.*, vol. 4, pp. 2348–2355, 2019.
- [6] J. G. A. Barbedo, L. V. Koenigkan, P. M. Santos, and A. R. B. Ribeiro, "Counting cattle in UAV images—dealing with clustered animals and animal/background contrast changes," *Sensors*, vol. 20, p. 2126, 2020.
- [7] V. Ranjan, U. Sharma, T. Nguyen, and M. Hoai, "Learning to count everything," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2021, pp. 3394–3403.
- [8] M. Shi, H. Lu, C. Feng, C. Liu, and Z. Cao, "Represent, compare, and learn: A similarity-aware framework for class-agnostic counting," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2022, pp. 9529–9538.
- [9] C. Liu, Y. Zhong, A. Zisserman, and W. Xie, "CounTR: Transformer-based Generalised Visual Counting," in *Brit. Mach. Vis. Conf.*, 2022, p. 370.
- [10] J. Xu, H. Le, V. Nguyen, V. Ranjan, and D. Samaras, "Zero-shot object counting," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2023, pp. 15 548–15 557.
- [11] S. Kang, W. Moon, E. Kim, and J.-P. Heo, "Vlcounter: Text-aware visual representation for zero-shot object counting," in *Proc. AAAI Conf. Artif. Intell.*, vol. 38, 2024, pp. 2714–2722.
- [12] V. Ranjan and M. H. Nguyen, "Exemplar free class agnostic counting," in *Proc. Asian Conf. Comput. Vis.*, 2022, pp. 3121–3137.
- [13] M. Hobley and V. Prisacariu, "Learning to count anything: Reference-less class-agnostic counting with weak supervision," *arXiv preprint arXiv:2205.10203*, 2022.
- [14] H. G. Akçay, B. Kabasakal, D. Aksu, N. Demir, M. Öz, and A. Erdoğan, "Automated bird counting with deep learning for regional bird distribution mapping," *Animals*, vol. 10, p. 1207, 2020.
- [15] V. Zavrtanik, M. Vodopivec, and M. Kristan, "A segmentation-based approach for polyp counting in the wild," *Eng. Appl. Artif. Intell.*, vol. 88, p. 103399, 2020.
- [16] Z. Liu, Y. Zhou, Y. Xu, and Z. Wang, "Simplenet: A simple network for image anomaly detection and localization," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2023, pp. 20 402–20 411.
- [17] N. Djukic, A. Lukezic, V. Zavrtanik, and M. Kristan, "A low-shot object counting network with iterative prototype adaptation," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2023, pp. 18 872–18 881.
- [18] J. Pelhan, V. Zavrtanik, M. Kristan *et al.*, "DAVE-A Detect-and-Verify Paradigm for Low-Shot Counting," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2024, pp. 23 293–23 302.
- [19] M.-R. Hsieh, Y.-L. Lin, and W. H. Hsu, "Drone-based object counting by spatially regularized regional proposal network," in *Int. Conf. Comput. Vis.*, 2017, pp. 4145–4153.
- [20] L. Wang, Y. Li, and X. Xue, "Coda: Counting objects via scale-aware adversarial density adaption," in *Int. Conf. Multimedia and Expo. IEEE*, 2019, pp. 193–198.
- [21] Z. Qiu, L. Liu, G. Li, Q. Wang, N. Xiao, and L. Lin, "Crowd counting via multi-view scale aggregation networks," in *Int. Conf. Multimedia and Expo*, 2019, pp. 1498–1503.
- [22] Y. Zhang, H. Zhang, and Q. Zhao, "Counting and locating anything: Class-agnostic few-shot object counting and localization," in *Int. Conf. Multimedia and Expo*, 2023, pp. 2921–2926.
- [23] E. Lu, W. Xie, and A. Zisserman, "Class-agnostic counting," in *Proc. Asian Conf. Comput. Vis.*, 2019, pp. 669–684.
- [24] H. Cao, Y. Wang, J. Chen, D. Jiang, X. Zhang, Q. Tian, and M. Wang, "Swin-Unet: Unet-like Pure Transformer for Medical Image Segmentation," in *Eur. Conf. Comput. Vis. Worksh.*, 2022, p. 205–218.
- [25] X. Song, X. Yang, C. Yao, J. Ren, R. Bai, X. Chen, and X. Jiang, "ERL-MPP: Evolutionary reinforcement learning with multi-head puzzle perception for solving large-scale jigsaw puzzles of eroded gaps," *Proc. AAAI Conf. Artif. Intell.*, 2025.
- [26] C. Yao, J. Ren, R. Bai, H. Du, J. Liu, and X. Jiang, "Progressively-orthogonally-mapped EfficientNet for action recognition on time-range-doppler signature," *Expert Syst. Appl.*, vol. 255, p. 124824, 2024.
- [27] J. Zhang, X. Wang, C. Yao, J. Ren, and X. Jiang, "Visual-linguistic cross-domain feature learning with group attention and gamma-correct gated fusion for extracting commonsense knowledge," *ACM Int. Conf. Multimedia*, p. 4650–4659, 2024.
- [28] C. Li, Y. Y. Tan, Y. He, J. Ren, R. Bai, Y. Zhao, H. Yu, and X. Jiang, "DARR: A dual-branch arithmetic regression reasoning framework for solving machine number reasoning," *Proc. AAAI Conf. Artif. Intell.*, 2025.
- [29] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, "Swin transformer: Hierarchical vision transformer using shifted windows," in *Int. Conf. Comput. Vis.*, 2021, pp. 10 012–10 022.
- [30] Z. Wang, L. Xiao, Z. Cao, and H. Lu, "Vision transformer off-the-shelf: A surprising baseline for few-shot class-agnostic counting," in *Proc. AAAI Conf. Artif. Intell.*, vol. 38, 2024, pp. 5832–5840.

Appendix D

ACMMM Submitted Paper

CSG: A Cross-Modal Semantic-Guided Group-Attended Framework for Few-Shot Counting

Anonymous Author(s)

ABSTRACT

Class-agnostic counting (CAC) aims to accurately count objects of interest using only a few visual exemplars or text prompts without relying on extensive annotations. In this work, we present a Cross-modal Semantic-guided Group-attended (CSG) framework for CAC tasks. Our method first leverages a large vision-language model (LLaVA) to generate dual text descriptions for both the query image and exemplars and encoded by BERT. Simultaneously, visual features are extracted from the query and exemplars using a Swin Transformer, forming four sets of multimodal inputs. To achieve deep and effective cross-modal interactions, two Gated Intra-modal Contrastive Embedding (GICE) blocks are employed to refine the query features by highlighting tokens that are highly correlated with exemplar features in both the visual and linguistic domains. In addition, a Cross-modal Semantic Refinement (CSR) module is used to extract and refine exemplars' semantics through cross-modal interactions, and a Semantic-Guided Group-Attention (SGGA) module further aggregates shared characteristics among candidate objects via a set of attention mechanisms. Extensive experiments on FSC-147, CARPK, and CountBench datasets demonstrate that CSG significantly outperforms state-of-the-art methods in both accuracy and robustness. Code will be available upon paper acceptance.

CCS CONCEPTS

• Computing methodologies → Computer vision tasks.

KEYWORDS

Few-Shot Counting, Class-Agnostic Counting, Group Attention

1 INTRODUCTION

Counting objects such as vehicles [47], crowd [17, 22], and cells [43] is crucial in many applications, *e.g.*, traffic monitoring [4], remote sensing [10], and biomarker statistics [9]. Early methods count object of specific classes, but can't count objects not covered in training [24, 37]. Recently, Class-agnostic counting has emerged as a promising research, counting objects during inference without retraining the model [34]. They do not require large-scale annotated samples, but only a few visual exemplars [20, 40] and/or text prompts [14, 44] to specify classes. CAC tasks are more challenging but potentially have wider applications than counting specific classes of objects.

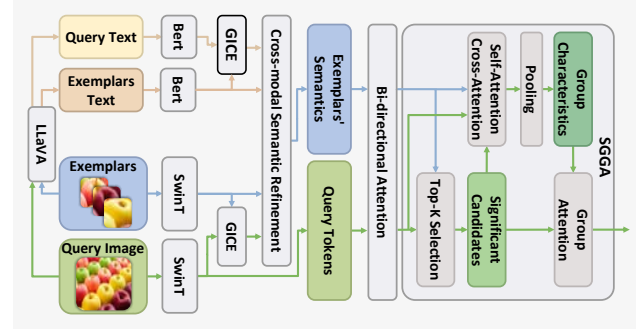


Figure 1: The proposed CSG leverages LLaVA to generate text descriptions of both query image and exemplars, forming four sets of features. The query and exemplars perform intra-domain interactions through GICEs and cross-domain interactions through CSR modules. Significant candidates are extracted from query features, and group common characteristics are extracted in SGGA, which in turn highlight the significant candidates, further enhancing the counting accuracy.

Existing CAC models can be broadly grouped as follows. 1) Few-Shot Counting (FSC) [20], which counts objects using limited visual exemplars extracted from the query image. Recent models focus on estimating object density maps using exemplar patches [40, 45], or detecting candidate objects and selecting relevant ones through exemplar comparison [2, 29]. 2) Prompt-Based Counting (PBC) [13, 44], which utilizes textual prompts to quantify objects, leveraging Vision-language models to establish cross-modal alignment between textual prompts and visual query features, and counting objects without explicit visual exemplars. 3) Zero-Shot Counting (ZSC) [11, 41], which autonomously identifies countable regions by self-probing potential objects through pre-trained vision transformers [11] or synthesizing pseudo-exemplars [33], without explicit external guidance.

Despite these advancements, existing CAC models still face challenges. 1) Visual exemplars often overemphasize visual details but limited exemplars may not provide sufficient semantic guidance [34]. In contrast, textual cues provide a comprehensive understanding of the target object but lack a specific visual context clearly assigned to that object [14]. Only very recently, Amini-Naieni et al. integrated both visual exemplars and text prompts to exploit visual clues and textual semantics of countable objects [2], but they heavily relied on expensive prompt annotations. 2) Four sets of visual and linguistic features can be extracted from the query and exemplars, posing significant challenges for deeply interacting them, while most methods utilize either visual clues only [28], or text prompts only [13]. 3) Lastly, the shared characteristics of countable objects in the query images are crucial clues but only very few methods partially exploit

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them, *e.g.*, Xu et al. aggregated high-quality exemplars from class-relevant patches [44], and Amini-Naieni et al. aggregated a unified semantic of countable objects through the token similarity [2].

To leverage multi-modal features while avoiding extensive annotations of prompts, we propose a **Cross-modal Semantic-guided Group-attended (CSG)** framework for few-shot counting. As shown in Fig. 1, we first leverage LLaVA [21] to automatically generate two distinct descriptions with high-level semantics by using two specifically designed prompts. Trained on a large number of vision-language pairs, LLaVA incorporates extensive knowledge about the target objects that limited exemplars alone cannot provide. These descriptions eliminate the need for labour-intensive annotations while enabling multi-modal object counting. We then utilize BERT [6] to extract two sets of linguistic features from the descriptions, and Swin Transformer [25] to extract two sets of visual features from the query and exemplars, respectively, forming four sets of features.

Then, we propose two modules to deeply interact these four sets of features. 1) **Gated Intra-modal Contrastive Embedding (GICE)**. Two GICE blocks are developed to interact the query and exemplars in the visual and linguistic domain, separately. Specifically, we leverage the contrastive embeddings [23] to derive the similarity matrix between the query and exemplars, highlighting the query tokens that are most similar to the exemplars. Then, salient tokens are emphasized by using saliency scores that are derived by average pooling over the query tokens. These two mechanisms help the model dynamically adapt to multimodal inputs and selectively enhance salient and relevant query tokens. 2) **Cross-modal Semantic Refinement (CSR)**, which extracts and refines the exemplars' semantics through cross-modal interaction among the four sets of features, and mutually enhances the query tokens and exemplars' semantics through bi-directional attention. Following [6], we embed the query's semantics in the [CLS] token by interacting their visual and linguistic features through a self-attention mechanism, and embed the exemplars' semantics in a similar way. The two semantics are then concatenated and compressed by a Multi-Layer Perceptron (MLP), and further interacted with exemplars' visual tokens and linguistic tokens through two self-attention blocks, respectively, to derive the exemplified semantics. Then, visual query tokens and exemplified semantics are mutually enhanced through bi-directional attention [23], achieving both query-to-semantic and semantic-to-query cross-attention enhancements.

Lastly, to exploit the common characteristics of significant objects, we propose a **Semantic-Guided Group-Attention (SGGA)** module. We first perform Top-*K* selection to select the most relevant object candidates from the query, guided by the exemplars' semantics. The selected candidates are further boosted by two cross-attention operations, *i.e.*, the first integrates the visual-linguistic clues from the query image, and the second incorporates visual-linguistic semantics from the exemplars. Then, we obtain the common characteristics of candidates through average pooling, and utilize them to highlight the relevant candidates through a set of attention mechanisms. The proposed SGGA significantly improves the distinctiveness and granularity of candidate features by deeply exploiting the common characteristics embedded in the group of candidates.

Our contributions can be summarized as follows. 1) The proposed CSG leverages LLaVA to generate text descriptions, extracting and exploiting four sets of visual and linguistic features from the query

image and exemplars, avoiding extensive annotations of prompts. 2) The two GICE blocks deeply interact the four distinct sets of features in visual and linguistic domains separately, and the CSR module extracts the visual-linguistic semantics of exemplars and mutually enhances the query tokens and exemplars' semantics through bi-directional attention. 3) The SGGA further highlights countable objects under the guidance of exemplars' semantics, and exploits the group characteristics of objects to further enhance their features, effectively differentiating salient objects from background and other obfuscations. 4) The proposed model is comprehensively evaluated on the FSC-147 [34], CARPK [12] and CountBench datasets [27], outperforming state-of-the-art methods for few-shot counting.

2 RELATED WORK

Class-agnostic counting has evolved through various paradigms. Existing methods can be grouped into Few-Shot Counting [28, 29, 40], Prompt-Based Counting [2, 14] and Zero-Shot Counting [41].

Few-Shot Counting. Class-agnostic counting was first introduced in the context of Few-Shot Counting [26] to address the limitations of class-specific counting. Many methods have been designed to map from image features to density maps, guided by visual exemplars [34]. This exemplar-based strategy reduces the reliance on class-specific annotations, enabling broader generalization across diverse scenarios. For example, FamNet [34] refines query features by leveraging feature correlations and enhances count estimation accuracy through test-time adaptation. CountTR [20] employs cross-attention mechanisms to strengthen interactions between exemplars' features and query features, thereby refining these features. CACViT [40] employs decoupled attention in Vision Transformer [8] to maximize the interactions between the query and exemplars. DAVE [29] integrates a detector to generate a high-recall object set and filters outliers using cosine similarity. Recently, GeCo [28] integrates Segment Anything Model [15] into an encoder-decoder model and dynamically generates candidate regions.

Prompt-Based Counting. PBC aims to minimize the effort of annotating visual exemplars by utilizing text prompts instead, *i.e.*, utilizing the capabilities of large language models and vision-language pre-training to interpret prompts and align them with visual contents. Xu et al. utilized text prompts to construct prototypes, effectively identifying class-relevant patches as visual exemplars [44]. CLIP-Count [13] utilizes the CLIP text encoder [32] to obtain text descriptions, and adopts a contrastive loss and a hierarchical patch-text interaction module to adapt VLMs for visual counting tasks. VL-Counter [14] combines semantic-aware prompt tokens with visual tokens, highlighting target-specific regions to enhance the features for zero-shot counting. CountGD [2] utilizes BERT [6] to extract the text descriptions, and a set of self-attentions and cross-attentions to facilitate the multi-modal interaction between visual tokens and textual tokens. Despite these advancements, their performance highly depends on the quality and informativeness of the text guidance.

Zero-Shot Counting. ZSC eliminates all forms of external guidance such as exemplars or text prompts. It aims to autonomously identify and focus on regions containing countable objects. ZSC significantly reduces annotation overhead and avoids biases induced by external inputs, making it suitable for diverse and unpredictable real-world scenarios. For example, pre-trained vision transformers

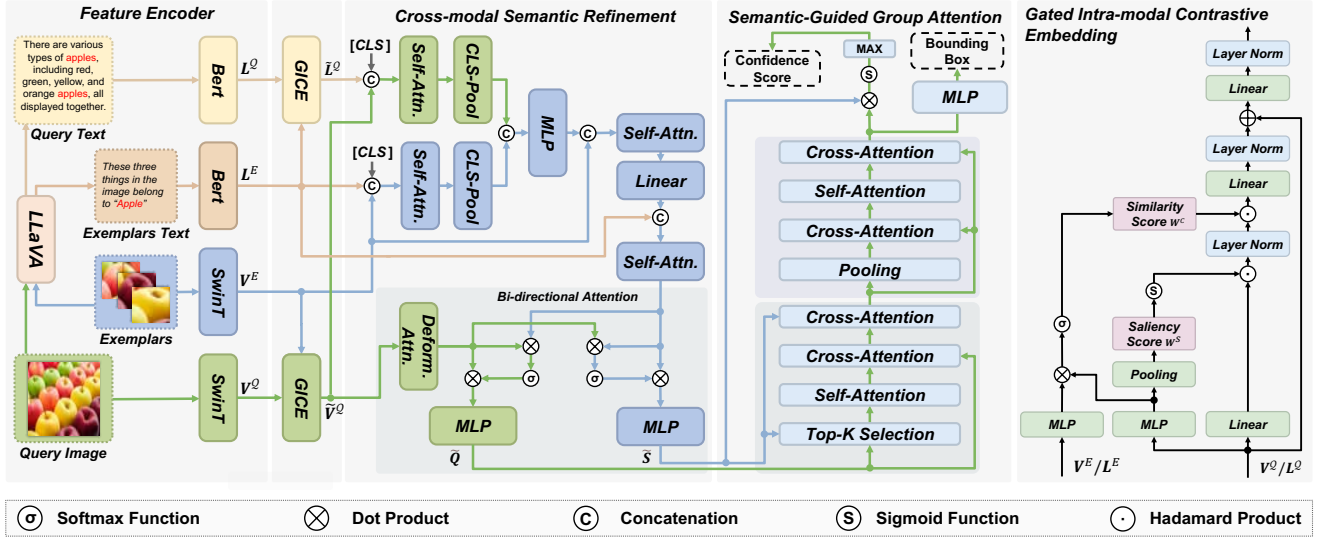


Figure 2: The proposed CSG contains four building blocks. 1) Feature Learning: LLaVA is utilized to generate text descriptions for the query image and exemplars, forming four sets of features by using BERT and SwinT. 2) Gated Intra-modal Contrastive Embedding: The query and exemplars are interacted in visual and linguistic domain separately. 3) Cross-modal Semantic Refinement: Exemplar's semantics are extracted and interact with the query tokens through a bi-directional attention. 4) Semantic-Guided Group Attention: The common features are extracted from object candidates and interact with candidates through group attention.

with weakly-supervised training have been utilized to extract visual patterns without explicit exemplars [11]. Ranjan and Nguyen developed a Repetitive Region Proposal Network (RepRPN) for identifying exemplars from repetitive objects and estimating their approximate count [33]. Wu et al. leveraged a self-similarity matrix and a set of gate mechanisms to selectively enhance the features of countable objects while suppressing irrelevant ones [41]. Indeed, the absence of annotated exemplars presents significant challenges, particularly in reliably discovering and locating countable objects.

3 PROPOSED CSG

3.1 Overview of Proposed Method

To tackle the challenges of few-shot counting, we propose the Cross-modal Semantic-guided Group-attended (CSG) framework. Fig. 1 illustrates the block diagram. Formally, given a query image $I^Q \in \mathbb{R}^{H \times W \times 3}$ and a few visual exemplars $\{I_i^E \in \mathbb{R}^{H^E \times W^E \times 3}\}_{i=1}^K$, the proposed CSG counts the number of objects similar to the exemplars, where H , W , H^E and W^E are the dimensions for the query image and exemplars, respectively, and K is the number of exemplars.

Firstly, a pre-trained vision-language model, LLaVA [21] Φ^{LLaVA} , is utilized to generate text descriptions $t^Q \in \mathbb{R}^{T^Q \times 1}$ and $t^E \in \mathbb{R}^{T^E \times 1}$ for the query and exemplars with two specifically designed prompts respectively, i.e., P^Q = "What kinds of objects are there?" and P^E = "What do these three things in the image belong to?",

$$t^Q = \Phi^{\text{LLaVA}}(I^Q, P^Q), \quad t^E = \Phi^{\text{LLaVA}}\left(\{I_i^E\}_{i=1}^K, P^E\right), \quad (1)$$

where T^Q and T^E denote the lengths of the respective text descriptions. We then utilize a Swin-Transformer Φ^{SwinT} to generate the visual features $V^Q = \Phi^{\text{SwinT}}(I^Q) \in \mathbb{R}^{V^Q \times D}$ and $V^E =$

$\Phi^{\text{SwinT}}(\{I_i^E\}_{i=1}^K) \in \mathbb{R}^{K \times D}$ for the query and exemplars, where D is the channel dimension. Note that one token is generated for each visual exemplar. In parallel, a pre-trained BERT Φ^{BERT} is utilized to obtain the corresponding linguistic embeddings $L^Q = \Phi^{\text{BERT}}(t^Q) \in \mathbb{R}^{L^Q \times D}$ and $L^E = \Phi^{\text{BERT}}(t^E) \in \mathbb{R}^{L^E \times D}$, forming four sets of visual and linguistic features, $\{V^Q, V^E, L^Q, L^E\}$, for object counting.

Secondly, two Gated Intra-modal Contrastive Interaction (GICE) blocks interact the query and exemplars visually and linguistically,

$$\tilde{V}^Q = \Phi^{\text{GICE}}(V^Q, V^E) \in \mathbb{R}^{V^Q \times D}, \quad (2)$$

$$\tilde{L}^Q = \Phi^{\text{GICE}}(L^Q, L^E) \in \mathbb{R}^{L^Q \times D}. \quad (3)$$

Φ^{GICE} highlights the significant query tokens relevant to exemplars.

Thirdly, the Cross-modal Semantic Refinement (CSR) block Φ^{CSR} extracts the exemplars' semantics, progressively refines them by exemplars' visual tokens and linguistic tokens to generate aligned visual-linguistic semantics, and mutually boosts the query tokens through a bi-directional attention [23, 35]. Formally,

$$\{\tilde{Q}, \tilde{S}\} = \Phi^{\text{CSR}}(\tilde{V}^Q, \tilde{L}^Q, V^E, L^E), \quad (4)$$

where $\tilde{Q} \in \mathbb{R}^{V^Q \times D}$ denotes the enhanced query and $\tilde{S} \in \mathbb{R}^{(K+L^E) \times D}$ denotes the exemplar's semantics in the visual-linguistic domain.

Lastly, the proposed Semantic-Guided Group-Attention (SGGA) module Φ^{SGGA} exploits the attentive supports of the group of object candidates. The SGGA first selects the top-K representative object candidates, and then boosts the candidates' features through one self-attention, and two cross-attentions with the query tokens and exemplars' semantics. The common characteristics among these candidates are obtained through an average pooling, and refined through a self-attention to enforce intra-class coherence and two

cross-attentions with candidates' features to highlight relevant features. The refined common characteristics are then applied on candidates' features to derive countable objects. Formally,

$$\{B, \gamma\} = \Phi^{SGGA}(\tilde{Q}, \tilde{S}), \quad (5)$$

where $B \in \mathbb{R}^{Z \times 4}$ denote the set of bounding boxes with associated confidence scores $\gamma \in \mathbb{R}^{Z \times 1}$, and Z is the number of objects.

3.2 Gated Intra-modal Contrastive Embedding

As shown in Fig. 2, the proposed GICE blocks facilitate effective intra-domain interactions between the query and exemplars, one in the visual domain and the other in the linguistic domain. They highlight the query tokens relevant to the exemplars. For illustration, we introduce the GICE block in the visual domain here. The linguistic GICE block follows the same process.

3.2.1 Token Saliency. Salient tokens are emphasized by their saliency. Specifically, we first align the query features V^Q and exemplars' features V^E using an MLP as $F^Q = \mathcal{F}^{\text{MLP}}(V^Q) \in \mathbb{R}^{V^Q \times D}$ and $F^E = \mathcal{F}^{\text{MLP}}(V^E) \in \mathbb{R}^{K \times D}$. The token saliency is derived by an average-pooling function $\mathcal{F}^{\text{AVG}}(\cdot)$ over the query tokens F^Q , followed by a sigmoid activation function $\mathcal{F}^{\text{SIGMOID}}(\cdot)$,

$$\mathbf{w}^S = \mathcal{F}^{\text{SIGMOID}}\left(\mathcal{F}^{\text{AVG}}(F^Q)\right), \quad (6)$$

where $\mathbf{w}^S \in \mathbb{R}^{V^Q \times 1}$ denotes the saliency scores. The average pooling aggregates the query tokens across the channel dimension, summarizing the overall response of each token and indicating their saliency, *i.e.*, tokens that consistently exhibit large responses are more important. We then replicate $\mathbf{w}^S$ D times to form the weighting matrix $W^S \in \mathbb{R}^{V^Q \times D}$, and apply it on F^Q as,

$$F^G = \mathcal{F}^{\text{LN}}\left(W^S \odot \mathcal{F}^{\text{LINEAR}}(F^Q)\right), \quad (7)$$

where F^G , $\mathcal{F}^{\text{LN}}(\cdot)$, $\odot$, and $\mathcal{F}^{\text{LINEAR}}(\cdot)$ denote the resulting salient tokens, layer normalization, Hadamard product and linear transformation, respectively. The linear layer projects F^Q into a new representation space for enhancing the model's dynamics, while the layer normalization mitigates channel-wise scale variations.

Different from uniformly processing all tokens [40], this adaptive weighting strategy enables the model to selectively emphasize informative tokens, thus enhancing the robustness and fine-grained representations of countable objects.

3.2.2 Contrastive Embedding. Many mechanisms have been developed to facilitate interactions between two sets of features, *e.g.*, attention-based fusion [38], cross-attention [39], gated multi-modal unit [3] and feature-wise linear modulation [30]. But these methods fail to establish an effective and tightly coupled feature interaction [16]. In particular, attention-based fusion [38] requires extensive calculations, leading to increased computational overhead. Contrastive embedding [23] has been utilized as a loss for cross-modal matching. We adapt it to emphasize query tokens that are similar to exemplars' tokens with a relatively low computational cost, highlighting strongly correlated query-exemplar token pairs.

Specifically, we obtain the contrastive embedding between the query and exemplars as $E = \sigma(F^Q \times F^{E^T}) \in \mathbb{R}^{V^Q \times K}$, where $\sigma(\cdot)$ denotes the softmax operation. E emphasizes the query tokens that

are similar to the exemplars. Then, we derive the maximum similarity scores for query tokens as $\mathbf{w}^C = \mathcal{F}^{\text{MAX}}(E) \in \mathbb{R}^{V^Q \times 1}$, where $\mathcal{F}^{\text{MAX}}(\cdot)$ denotes max-pooling, highlighting the strongest correspondences between the query's and exemplars' tokens. We then replicate $\mathbf{w}^C$ D times to form the weighting matrix $W^C \in \mathbb{R}^{V^Q \times D}$, and apply it on F^G to produce the enhanced tokens $F^P \in \mathbb{R}^{V^Q \times D}$ as,

$$F^P = \mathcal{F}^{\text{LN}}\left(\mathcal{F}^{\text{LINEAR}}(W^C \odot F^G)\right). \quad (8)$$

Again, the linear layer enhances the model's dynamics, and the layer normalization mitigates scale variations. Compared to the traditional attention mechanism [8], the contrastive embedding requires much less computational power. Additionally, by explicitly modelling the pairwise similarities between query tokens and exemplar tokens, it highlights the query tokens relevant to exemplars. It is also more robust to deal with background noise and other irrelevant objects than cross-attention [39] by filtering out irrelevant or noisy tokens.

Finally, the enhanced tokens are integrated back into the query tokens to produce $\tilde{V}^Q$, the output of GICE block as,

$$\tilde{V}^Q = \mathcal{F}^{\text{LN}}\left(\mathcal{F}^{\text{LINEAR}}(F^P + F^Q)\right). \quad (9)$$

The residual connection preserves the original query tokens.

The proposed GICE integrates two complementary gating mechanisms to enhance intra-domain interactions: 1) Saliency gating assigns token importance based on their informativeness; 2) Contrastive embedding computes query-exemplar similarity in a shared space to emphasize tokens relevant to exemplars. Together, these mechanisms highlight salient and relevant query tokens.

3.3 Cross-Modal Semantic Refinement

The CSR module consists of the follow four steps: 1) It first extracts semantic features from the query and exemplars in the visual-linguistic domain through self-attention. 2) The semantics are further enhanced by visual and linguistic features of exemplars through self-attention. 3) In parallel, visual query tokens are enhanced through a deformable attention [42]. 4) A bi-directional attention [35] is utilized to mutually enhance both the query tokens and exemplars' semantics. As a result, we achieve cross-modal interactions between the query and exemplars in the aligned visual-linguistic domain.

3.3.1 Semantics Extraction. Following [6], we first aggregate two respective semantic tokens, $s^Q \in \mathbb{R}^{1 \times D}$ and $s^E \in \mathbb{R}^{1 \times D}$ for the query and exemplars respectively as,

$$s^Q = \mathcal{F}^{\text{CLS-POOL}}\left(\mathcal{F}^{\text{ATTN}}\left(\{[CLS], \tilde{V}^Q, \tilde{L}^Q\}\right)\right), \quad (10)$$

$$s^E = \mathcal{F}^{\text{CLS-POOL}}\left(\mathcal{F}^{\text{ATTN}}\left(\{[CLS], V^E, L^E\}\right)\right), \quad (11)$$

where $\mathcal{F}^{\text{CLS-POOL}}(\cdot)$ represents the CLS-pooling [6], $\mathcal{F}^{\text{ATTN}}(\cdot)$ represents the self-attention [39], and $[CLS]$ denotes the token encapsulating learnable semantics. s^Q and s^E are then concatenated and passed through an MLP to generate a unified semantic token $s^C = \mathcal{F}^{\text{MLP}}(\{s^Q, s^E\}) \in \mathbb{R}^{1 \times D}$, aggregating the multimodal information into a compact and informative semantic token. It greatly saves the subsequent computations compared to traditional attention mechanism that utilizes the whole set of self-attended tokens [2].

3.3.2 Semantics Enhancement. The semantic token s^C is compact but lacks exemplars' visual and linguistic details. We hence enhance it through two self-attentions with exemplars' visual and linguistic tokens. Firstly, the self-attention followed by a linear projection is applied to s^C and exemplar's visual token V^E to obtain a visually-enhanced semantics $S^V \in \mathbb{R}^{K \times D}$,

$$S^V = \mathcal{F}^{\text{LINEAR}}(\mathcal{F}^{\text{ATTN}}(\{\mathcal{F}^{\text{REP}}(s^C), V^E\})), \quad (12)$$

where $\mathcal{F}^{\text{REP}}(\cdot)$ replicates the semantic token s^C K times, $\mathcal{F}^{\text{ATTN}}(\cdot)$ exploits the interactions between s^C and V^E , and $\mathcal{F}^{\text{LINEAR}}(\cdot)$ compresses the $2K$ tokens back to K tokens. This step integrates high-level semantics s^C with exemplars' fine-grained visual tokens V^E .

Secondly, S^V is further refined by interacting with exemplars' linguistic cues L^E via another self-attention, to derive the linguistically-enhanced semantic tokens $S^A = \mathcal{F}^{\text{ATTN}}(\{S^V, L^E\}) \in \mathbb{R}^{(K+L^E) \times D}$. As a result, the semantic coherence is enhanced with respect to exemplars' visual and linguistic features, ensuring consistent and complementary integration of both visual and linguistic modalities.

Different from cross-attentions [5], our method aligns visual-linguistic semantics within a joint query-exemplar space, with special emphasis on exemplars' tokens. This is particularly critical for FSC tasks, where achieving the semantic consistency often proves more beneficial than directly augmenting query features [13].

3.3.3 Bi-Directional Attention. Bridging the gaps between low-level visual features and high-level semantics remains a fundamental challenge in vision-language understanding. To tackle this challenge, we leverage a Bi-Directional Attention [23, 35] to facilitate mutual refinement of query features and exemplars' semantics. Specifically, visual query tokens V^Q are first refined via a deformable self-attention [42] as $\hat{V}^Q = \mathcal{F}^{\text{DSA}}(V^Q) \in \mathbb{R}^{V^Q \times D}$. As shown in [42], the deformable self-attention could dynamically warp sampling locations based on both image content and spatial structure. Then, a bi-direction attention mechanism [23, 35] is utilized to interact $\hat{V}^Q$ with S^A , capturing cross-modal relevance and generating mutually enhanced query features and exemplars' semantics as,

$$\tilde{Q} = \mathcal{F}^{\text{MLP}}(\sigma(\hat{V}^Q \times S^{A\top}) \times S^A), \quad (13)$$

$$\tilde{S} = \mathcal{F}^{\text{MLP}}(\sigma(S^A \times \hat{V}^{Q\top}) \times \hat{V}^Q). \quad (14)$$

A soft attention map is produced based on the similarity between $\hat{V}^Q$ and S^A , which guides the feature refinement, yielding semantically-enhanced visual query features $\tilde{Q}$ and contextually-informed semantics $\tilde{S}$. Unlike unidirectional attentions, the bi-directional attention enables a symmetric and continuous exchange of complementary information between the two sets of features, leading to more coherent feature integration, more effectively mitigating noise and ambiguity in the inputs, and yielding a richer joint representation.

3.4 Semantic-Guided Group-Attention Module

The SGGa module first extracts a set of object candidates from the query features $\tilde{Q}$ by ranking $\tilde{Q}$ based on their similarity with the semantics $\tilde{S}$, and selects the top- K candidates. Then, a sequence of self-attention and cross-attention are adopted to highlight significant candidates by exploiting their similarities. Lastly, the common characteristics are extracted from these candidates, and progressively interacted with candidates' features through group attention.

Specifically, the significant candidates $C^S \in \mathbb{R}^{Z \times D}$ are generated by selecting K most significant candidates as $C^S = \mathcal{F}^{\text{TOPK}}(\tilde{Q})$, where $\mathcal{F}^{\text{TOPK}}(\cdot)$ denotes the Top- K Selection [46]. Subsequently, C^S is refined by a self-attention and two cross-attentions with the query tokens $\tilde{Q}$ and the semantic guidance $\tilde{S}$ as,

$$C = \mathcal{F}^{\text{CA}}\left(\mathcal{F}^{\text{CA}}\left(\mathcal{F}^{\text{ATTN}}(C^S), \tilde{Q}\right), \tilde{S}\right), \quad (15)$$

where $\mathcal{F}^{\text{CA}}(\cdot)$ represents the cross-attention, and $C \in \mathbb{R}^{Z \times D}$ denotes the refined candidates' features. The first self-attention aggregates the context among the candidates to emphasize the common characteristics, while the cross-attention incorporates complementary cues from both the query features and semantic guidance, further refining the candidates' features. Compared to directly utilizing convolutional matching between features for candidate generation [29], our design leverages semantic similarity to choose highly confident object candidates and further refines them through cross-attentions, resulting in more precise and effective candidates' representations.

The absence of sufficient exemplars often results in biased and ambiguous semantic representations. To tackle this issue, we exploit the common characteristics of candidates through group attention, i.e., we first extract the common characteristics of candidates' features C through average-pooling $\mathcal{F}^{\text{AVG}}(\cdot)$ to generate group features, and then utilize the group features to refine the candidates' features through a sequence of cross-attentions and self-attentions,

$$\tilde{C} = \mathcal{F}^{\text{CA}}(C, \mathcal{F}^{\text{ATTN}}(\mathcal{F}^{\text{CA}}(\mathcal{F}^{\text{AVG}}(C), C))), \quad (16)$$

where the first cross-attention integrates the contexts from candidates' features C into the group features, thus enriching their semantics. The self-attention is applied to further boost the description power, resulting in more coherent and robust group features. Subsequently, the candidates' features are enhanced via the cross-attention with the group features. The entire process enables contextualized refinement of C through the interaction with the group features.

Finally, the enhanced candidates' features $\tilde{C} \in \mathbb{R}^{Z \times D}$ are utilized to generate the bounding box $B \in \mathbb{R}^{Z \times 4}$ via an MLP head as $B = \mathcal{F}^{\text{MLP}}(\tilde{C})$. The confidence scores $\gamma \in \mathbb{R}^{Z \times 1}$ are computed as $\gamma = \mathcal{F}^{\text{MAX}}(\mathcal{F}^{\text{SIGMOID}}(\tilde{C} \times \tilde{S}^T))$, i.e., candidates that are most similar to exemplars' semantics are counted as objects.

The proposed SGGa leverages semantic similarity to extract significant candidates, and then refine them through a set of attentions. Then, it extracts the group features and progressively refines them by interacting with significant candidates, effectively enhancing the semantic coherence among objects and mitigating ambiguity.

3.5 Loss Function

The loss comprises two terms: a localization loss and a classification loss, jointly encouraging spatial precision and robust categorization,

$$\mathcal{L} = \lambda_{\text{LOC}} \|c^P - c^T\|_1 + \lambda_{\text{FL}} \mathcal{F}^{\text{FL}}(\gamma, \gamma^T), \quad (17)$$

where c^P denotes the predicted object centres, derived from the predicted bounding box B . c^T denotes the ground-truth object centres. γ^T denotes the one-hot encoding of ground-truth labels. $\mathcal{F}^{\text{FL}}(\cdot, \cdot)$ denotes the focal loss [18]. λ_{LOC} and λ_{FL} balance the two terms. The localization loss ensures that the objects are located accurately while the focal loss ensures that the objects are classified accurately.

4 EXPERIMENTAL RESULTS

4.1 Experimental Settings

Dataset Description. Experiments are conducted on three datasets. 1) **FSC-147** [34] covers 147 diverse object categories, with significant variations in object density, ranging from 7 objects per image to 3,731 objects per image. The dataset consists of 6,135 images, officially split into 3,659 for training, 1,286 for validation, and 1,190 for testing [34]. The counting categories of the three subsets are mutually exclusive. 2) **CARPK** [12] counts vehicles in aerial imagery. It contains 1,448 bird's-eye view images collected from four parking lots, posing unique challenges of significant scale variations and frequent occlusions. It is officially divided into 989 and 459 samples for training and testing, respectively [12]. Following [2, 41], we employ its test set for cross-domain evaluation, with vehicles ranging from 2 to 188 per image. 3) **CountBench** [27] comprises 540 images, varying from 2 to 10 objects per image. Every sample includes a text description explicitly indicating the object count, facilitating multi-modal research. Following [2], the entire dataset serves as a test set to assess the cross-domain generalization capabilities.

Evaluation Metrics. Following [2, 7, 20, 34], two common evaluation metrics are employed, *i.e.*, Mean Absolute Error (MAE) $\epsilon_M = \frac{1}{N} \sum_{i=1}^N |C_i^P - C_i^G|$, which measures the average absolute difference between the predicted object count C_i^P and the ground truth count C_i^G ; and Root Mean Squared Error (RMSE) $\epsilon_R = \sqrt{\frac{1}{N} \sum_{i=1}^N (C_i^P - C_i^G)^2}$, which computes the square root of the mean of the squared differences between predictions and ground-truth.

Compared Methods. The compared methods can be categorized into four groups. 1) *Zero-Shot Counting* methods, counting objects solely from the query images without any exemplars or prompts, *e.g.*, RCC [11], and GCA-SUNet [41]. 2) *Prompt-Based Counting* methods, which specify the counted objects using text exemplars, *e.g.*, CounTX [1], CLIP-Count [13], VLCounter [14], T2ICount [31], and GroudingREC [5]. 3) *Few-Shot Counting* methods, which utilize visual exemplars for object counting, *e.g.*, BMNet+ [36], SPDCN [19], CounTR [20], LOCA [7], SSD [45], CACViT [40], DAVE [29] and GeCo [28]. 4) *Hybrid* methods, utilizing both visual and textual exemplars to specify classes, *e.g.*, CountGD [2].

Implementation Details. The Adam optimizer with a cosine annealing scheduler is used with an initial learning rate of 1×10^{-5} and a weight decay of 10^{-4} . The hyper-parameters λ_{LOC} and λ_{FL} are empirically set to 1 and 5, respectively. Following [2], we utilize data augmentation to improve generalization, test-time normalization to avoid counting one object multiple times, and adaptive crop to handle small objects and high density scenarios.

4.2 Comparison Results on FSC-147 Dataset

4.2.1 Comparisons to State-of-the-Art Models. We conduct four groups of comparison experiments on the FSC-147 dataset, as summarized in Table 1. Following [2, 34], we report both test errors and validation errors. The results of all the compared methods except GDino [23] are obtained from their original paper.

Few-Shot Counting. We primarily conduct comparison experiments under the FSC setting [34], where three visual exemplars are provided but without text exemplars. We have the following

Table 1: Comparison with state-of-the-art methods on the FSC-147 dataset [34]. V and T represent visual exemplars and text prompts. ϵ_M^T , ϵ_R^T , ϵ_M^V and ϵ_R^V denote MAE and RMSE for test and validation data, respectively. Best results are in bold.

	Method	ϵ_M^T	ϵ_R^T	ϵ_M^V	ϵ_R^V
ZSC	RCC CVPRW'22 [11]	17.12	104.53	17.49	58.81
	LOCA ICCV'23 [7]	16.22	103.96	17.43	54.96
	DAVE CVPR'24 [29]	15.14	103.49	15.54	52.67
	CounTR BMVC'22 [20]	14.71	106.87	18.07	71.84
	GCA-SUNet arxiv [41]	14.08	84.77	17.02	55.37
	GeCo NIPS'24 [28]	13.30	108.72	14.81	64.95
	Proposed CSG	8.52	83.26	9.40	49.08
PBC	Patch-Selection CVPR'23 [44]	22.09	115.17	26.93	88.63
	CounTX BMVC'23 [1]	15.88	106.29	17.10	65.61
	CLIP-Count MM'23 [13]	17.78	106.62	18.79	61.18
	VLCounter AAAI'24 [14]	17.05	106.16	18.06	65.13
	CountGD NIPS'24 [2]	12.98	98.35	12.14	47.51
	T2ICount CVPR'25 [31]	11.76	97.86	13.78	58.78
	GDino ECCV'24 [23]	10.82	104.00	10.32	55.54
	GroundingREC CVPR'24 [5]	10.12	107.19	10.06	58.62
	Proposed CSG	8.08	80.53	8.65	41.25
Hybr.	CountGD NIPS'24 [2]	5.74	24.09	7.10	26.08
	Proposed CSG	4.69	15.23	6.90	28.24
FSC	BMNet+ CVPR'22 [36]	14.62	91.83	15.74	58.53
	SPDCN BMVC'22 [19]	13.51	96.80	14.59	49.97
	CounTR BMVC'22 [20]	11.95	91.23	13.13	49.83
	LOCA ICCV'23 [7]	10.79	56.97	10.24	32.56
	GDino ECCV'24 [23]	10.77	99.51	9.32	51.72
	SSD IJCAI'24 [45]	9.58	64.13	9.73	29.72
	CACViT AAAI'24 [40]	9.13	48.96	10.63	37.95
	DAVE CVPR'24 [29]	8.66	32.36	8.91	28.08
	CountGD NIPS'24 [2]	8.31	91.05	7.46	29.54
	GeCo NIPS'24 [28]	7.91	54.28	9.52	43.00
	Proposed CSG	5.23	27.95	7.26	28.77

observations. 1) The proposed CSG significantly outperforms all the compared methods in terms of both MAE and RMSE on the test set, showing its effectiveness in FSC scenarios. Compared to the second-best method, GeCo [28], it significantly reduces the MAE from 7.91 to 5.23, and reduces the RMSE from 54.28 to 27.95, highlighting its effectiveness in the FSC setting. 2) CountGD [2] produces small RMSE on validation sets, but it tends to overfit to the validation data and produce larger errors on novel test data. In contrast, CSG demonstrates superior generalization performance, which not only achieves excellent performance on the validation set, but also generalizes well on novel test data. 3) CountGD [2] utilizes both text and visual exemplars, but it simply concatenates visual and textual features for multi-modal interactions. Compared with CountGD [2], our model significantly reduces MAE by 3.08 and RMSE by 63.10 on test data, demonstrating the effectiveness of our comprehensive set of cross-modal interactions.

Zero-Shot Counting. Besides FSC, we adapt our method for zero-shot counting, where the visual exemplars are replaced by learnable

tokens as in [20], enabling the model to operate without explicitly providing visual exemplars. We can observe the followings from Table 1. 1) The proposed CSG demonstrates superior performance compared to other ZSC methods, outperforming the second-best method, GeCo [28], with significant reduction of 4.53 in MAE and 19.57 in RMSE. 2) CSG significantly outperforms dedicated ZSC methods such as RCC [11] and GCA-SUNet [41], as it leverages LLaVA to generate text prompts that carry abstract semantics of objects. In addition, the group common characteristics of these object candidates are exploited by SGGA, mitigating the challenges of lack of visual exemplars or text prompts in the ZSC setting. 3) CSG demonstrates excellent generalization ability, outperforming all the compared methods in terms of both validation errors and test errors.

Prompt-Based Counting. For PBC, besides adopting learnable tokens to represent visual exemplars as in ZSC, the provided text exemplars are integrated with the generated text descriptions to provide additional clues. From Table 1, we can observe the following. 1) CSG consistently outperforms all the compared methods on both validation and test sets. Compared to the second-best method, GroundingREC [5], CSG largely reduces the test MAE from 10.12 to 8.08 and the test RMSE from 107.19 to 80.53. The substantial gains are mainly attributed to the customized modality interaction and explicit semantic guidance in the CSR module. 2) Similar performance gains are also confirmed on validation set, demonstrating CSG's robustness and ability to mitigate potential overfitting issues.

Hybrid Counting. Only CountGD [2] explicitly makes use of visual and text exemplars for object counting. The proposed CSG achieves substantial improvements over CountGD [2], reducing the MAE from 5.74 to 4.69 and the RMSE from 24.09 to 15.23 on the test set. The performance gains primarily stem from effective intra-domain interaction in GICE blocks, cross-domain interactions in CSR, and group interaction in SGGA.

Discussions on FSC-147 Dataset. Though originally designed for the FSC task, CSG consistently outperforms all competing approaches on the test set across all settings, demonstrating its task adaptability and superior counting capability. Moreover, since the counting categories are disjoint across the training, validation, and test sets, the outstanding performance on the test set verifies its excellent generalization to unseen object categories. Lastly, the performance gain from ZSC to FSC is more substantial than that from ZSC to PBC, indicating that the integration of visual exemplars provides a more significant performance boost, highlighting the importance of exemplar-based guidance for accurate counting.

4.2.2 Density Analysis. The object density ranges from 7 to 3,731 per image on the FSC-147 dataset. Following [40], we divide the 1,190 test images into two groups: low-density group of 605 images with each containing no more than 37 objects, and high-density group of 585 images, each with 38 or more objects. Table 2 summarizes the results under the FSC setting. CSG outperforms all other methods in both low-density and high-density groups. Particularly in high-density group, compared to the second-best method, GeCo [28], CSG reduces the MSE by 4.91 and RMSE by 42.57, demonstrating its effectiveness in handling high-density images.

4.2.3 Visual Comparisons. CountGD [2], GeCo [28] and CSG are visually compared under the FSC setting. Fig. 3 shows five real-world counting scenarios: elephants in wild, chairs in canteen,

Table 2: Comparison of counting objects on images of different object densities on the FSC-147 dataset [34].

Method	ϵ_M^{Low}	ϵ_R^{Low}	ϵ_M^{High}	ϵ_R^{High}
BMNet+ CVPR'22 [36]	6.23	29.96	23.21	127.62
SPDCN BMVC'22 [19]	5.06	31.53	22.25	133.59
CountTR BMVC'22 [20]	5.44	36.96	18.56	124.11
CACViT AAAI'24 [40]	3.14	6.26	15.33	69.54
CountGD NIPS'24 [2]	1.76	4.97	14.32	120.09
GeCo NIPS'24 [28]	1.97	4.34	13.86	81.59
Proposed CSG	1.30	3.33	8.95	39.02

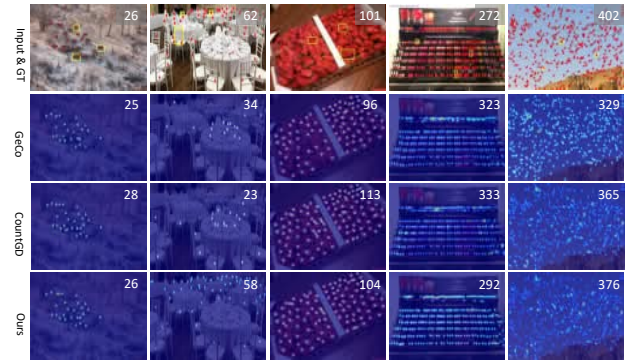


Figure 3: Visual comparisons on the FSC-147 dataset [34].

strawberries in basket, nail polishes on goods shelf and flamingos in the sky. CSG performs well in all five scenarios. As the object density increases, all three methods show a decline in accuracy, while CSG experiences the smallest performance degradation. For example, for counting chairs in canteen, both CountGD [2] and GeCo [28] tend to mistakenly localize tableware instead of chair, whereas CSG accurately delineates and counts chairs.

4.3 Cross-Domain Evaluation

Following [2], we conduct a cross-domain evaluation by training a model on the FSC-147 dataset [34] and evaluating it directly on the CARPK [12] and CountBench [27] datasets without fine-tuning. Following [2], experiments are conducted under ZSC, PBC, and FSC settings for CARPK [12] and under PBC for CountBench [27]. The results of compared methods are taken from their original papers.

CARPK Dataset. As shown in Table 3, CSG consistently and significantly outperforms all the compared methods across all three evaluation settings on CARPK [12]. Compared to GCA-SUNet [41] dedicated for ZSC, CSG significantly reduces MAE from 10.96 to 4.31 and RMSE from 13.95 to 6.28. CSG also surpasses the previous best approach, CountGD [2], in both PBC and FSC, reducing MAE from 3.83 to 3.45 for PBC and from 3.68 to 3.07 for FSC. Notably, only CSG adapts to all settings and it performs best, further highlighting its superior adaptability and generalization capabilities.

CountBench Dataset. The proposed CSG exhibits robust generalization capabilities on CountBench [27], reducing MAE from 0.86 to 0.77 and RMSE from 3.10 to 2.56 compared with CountGD [2].

Table 3: Comparison results of cross-domain evaluation on CARPK dataset [12] and CountBench [27].

	Method	ϵ_M^T	ϵ_R^T
CARPK	ZSC	RCC CVPRW'22 [11]	21.38
		GCA-SUNet arxiv [41]	10.96
		Proposed CSG	4.31
			6.28
	PBC	CLIP-Count MM'23 [13]	11.96
		CountGD NIPS'24 [2]	3.83
		Proposed CSG	3.45
	FSC	LOCA ICCV'23 [7]	9.97
		CACViT AAAI'24 [40]	8.30
		CountGD NIPS'24 [2]	3.68
		Proposed CSG	3.07
CountB	PBC	CountTX BMVC'23 [1]	6.64
		CountGD NIPS'24 [2]	0.86
		Proposed CSG	0.77

Note that the errors on CountBench [27] are much smaller than that on CARPK [12], mainly because of lower object densities, *i.e.*, 6.04 vs. 66.25 per image on average, and more similar viewpoints to FSC-147, as CARPK images are captured in a bird's-eye view.

4.4 Ablation Study of Major Components

Table. 4 summarizes the ablation of major components on the FSC-147 dataset [34] under the FSC setting. GDino [23] is adapted as the baseline, by replacing provided text exemplars with textual descriptions generated by LLaVA. We then gradually add GICE, replace the feature enhancer of GDino by CSR and replace the cross-modality decoder by SSGA for ablation. All the compared methods utilize four sets of inputs as CSG. Compared with the baseline model, all three proposed modules individually achieve significantly performance gains over all four metrics, demonstrating their effectiveness. The combinations of two modules further improve all four evaluation metrics. Lastly, the utilization of all three proposed components achieves the lowest error rates. The ablation results clearly demonstrate the contribution of each component to the final performance.

4.5 Failure Case Analysis

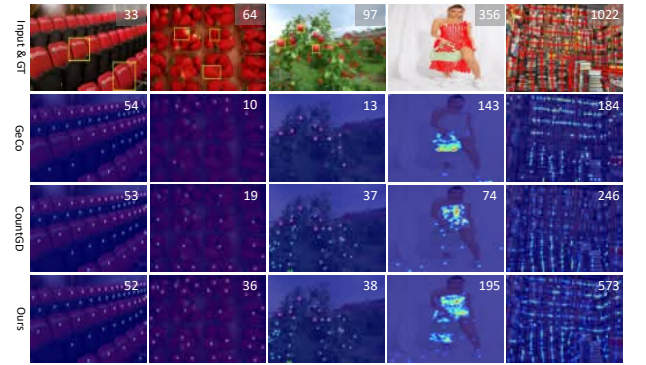
Fig. 4 shows five representative failure cases in which CountGD [2], GeCo [28] and CSG struggle to yield accurate counts. For example, in the first case, severe self-similarity leads to overcounting repetitive parts of a single chair; In the third case, apples located at a distance or partially occluded are frequently overlooked, leading to systematic underestimation. In last case, the bookshelf example highlights difficulties in segmenting and enumerating densely packed and overlapping instances. These cases expose key limitations of current class-agnostic counting frameworks, particularly in dealing with self-similarity, occlusion, low saliency, and high-density objects.

4.6 Model Complexity Analysis

Table 5 summarizes training time, inference time, model size, and GFLOPs for the top-three performing methods, CountGD [2] and GeCo [28], and CSG. CSG maintains competitive inference time and

Table 4: Ablation of major components on FSC-147 [34].

$\mathcal{F}^{\text{GICE}}$	$\mathcal{F}^{\text{CSR}}$	$\mathcal{F}^{\text{SSGA}}$	ϵ_M^T	ϵ_R^T	ϵ_M^V	ϵ_R^V
✗	✗	✗	10.77	99.51	9.32	51.72
✓	✗	✗	8.06	90.14	8.72	43.95
✗	✓	✗	7.49	77.87	8.10	44.96
✗	✗	✓	7.84	86.38	8.35	41.59
✓	✗	✓	6.75	67.92	7.83	38.86
✓	✓	✗	5.99	52.38	7.93	37.19
✗	✓	✓	5.82	36.06	7.71	34.86
✓	✓	✓	5.23	27.95	7.26	28.77

**Figure 4: Failure cases for FSC on the FSC-147 dataset [34].****Table 5: Comparison of model complexity.**

Method	Training	Inference	#Params.	GFLOPs
GeCo [28]	200h	0.97s	641M	1075
CountGD [2]	24h	0.32s	204M	437
Proposed CSG	40h	0.36s	301M	680

computational efficiency, balancing between accuracy and efficiency.

5 CONCLUSION

The proposed CSG well tackles the challenges of few-shot counting by mapping an input image with visual exemplars and two generated text descriptions into countable objects. The GICE blocks and CSR module capture the intra-domain and cross-domain interactions between the query and exemplars respectively, enabling efficient alignment and integration of heterogeneous features. Significant objects candidates and their group common characteristics are extracted in SSGA and exploited to highlight candidates using group attention. Extensive experiments on the FSC-147, CARPK, and CountBench datasets under four CAC settings demonstrate that CSG consistently and significantly outperforms state-of-the-art models in both intra-domain and cross-domain evaluations. Future work includes counting similar objects but of different classes in an image.

REFERENCES

- [1] Niki Amini-Naieni, Kiana Amini-Naieni, Tengda Han, and Andrew Zisserman. 2023. Open-world Text-specified Object Counting. In *Brit. Mach. Vis. Conf.*
- [2] Niki Amini-Naieni, Tengda Han, and Andrew Zisserman. 2024. Countgd: Multimodal open-world counting. *Advances in Neural Information Processing Systems* 37 (2024), 48810–48837.
- [3] John Arevalo, Tamar Solorio, Manuel Montes-y Gómez, and Fabio A González. 2017. Gated multimodal units for information fusion. *arXiv preprint arXiv:1702.01992* (2017).
- [4] Walid Balid, Hasan Tafish, and Hazem H Refai. 2017. Intelligent vehicle counting and classification sensor for real-time traffic surveillance. *IEEE Transactions on Intelligent Transportation Systems* 19, 6 (2017), 1784–1794.
- [5] Siyang Dai, Jun Liu, and Ngai-Man Cheung. 2024. Referring Expression Counting. In *IEEE Conf. Comput. Vis. Pattern Recog.* 16985–16995.
- [6] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*. 4171–4186.
- [7] Nikola Djukic, Alan Lukezic, Vitjan Zavrtanik, and Matej Kristan. 2023. A low-shot object counting network with iterative prototype adaptation. In *IEEE Conf. Comput. Vis. Pattern Recog.* 18872–18881.
- [8] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xi-aohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. 2021. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. *Proc. Int. Conf. Learn. Represent.* (2021).
- [9] Zach Eaton-Rosen, Thomas Varsavsky, Sebastien Ourselin, and M Jorge Cardoso. 2019. As easy as 1, 2... 4? uncertainty in counting tasks for medical imaging. In *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Interv.* Springer, 356–364.
- [10] Guangshuai Gao, Qingjie Liu, and Yunhong Wang. 2020. Counting from sky: A large-scale data set for remote sensing object counting and a benchmark method. *IEEE Transactions on geoscience and remote sensing* 59, 5 (2020), 3642–3655.
- [11] Michael Hobley and Victor Prisacariu. 2022. Learning to count anything: Reference-less class-agnostic counting with weak supervision. *arXiv preprint arXiv:2205.10203* (2022).
- [12] Meng-Ru Hsieh, Yen-Liang Lin, and Winston H Hsu. 2017. Drone-based object counting by spatially regularized regional proposal network. In *Int. Conf. Comput. Vis.* 4145–4153.
- [13] Ruixiang Jiang, Lingbo Liu, and Changwen Chen. 2023. Clip-count: Towards text-guided zero-shot object counting. In *ACM Int. Conf. Multimed.* 4535–4545.
- [14] Seunggu Kang, WonJun Moon, Euiyeon Kim, and Jae-Pil Heo. 2024. VLCounter: Text-Aware Visual Representation for Zero-Shot Object Counting. In *Proc. AAAI Conf. Artif. Intell.*, Vol. 38, 2714–2722.
- [15] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C Berg, Wan-Yen Lo, et al. 2023. Segment anything. In *Int. Conf. Comput. Vis.* 4015–4026.
- [16] Junnan Li, Ramprasaath Selvaraju, Akhilesh Gotmare, Shafiq Joty, Caiming Xiong, and Steven Chu Hong Hoi. 2021. Align before fuse: Vision and language representation learning with momentum distillation. *Adv. Neural Inform. Process. Syst.* 34 (2021), 9694–9705.
- [17] Hui Lin, Zhiheng Ma, Xiaopeng Hong, Yaowei Wang, and Zhou Su. 2022. Semi-supervised crowd counting via density agency. In *ACM Int. Conf. Multimed.* 1416–1426.
- [18] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. 2017. Focal loss for dense object detection. In *Int. Conf. Comput. Vis.* 2980–2988.
- [19] Wei Lin, Kunlin Yang, Xinzhu Ma, Junyu Gao, Lingbo Liu, Shinan Liu, Jun Hou, Shuai Yi, and Antoni B Chan. 2022. Scale-Prior Deformable Convolution for Exemplar-Guided Class-Agnostic Counting.. In *Brit. Mach. Vis. Conf.* 313.
- [20] Chang Liu, Yujie Zhong, Andrew Zisserman, and Weidi Xie. 2022. CounTR: Transformer-based Generalised Visual Counting. In *Brit. Mach. Vis. Conf.*
- [21] Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. 2024. Improved baselines with visual instruction tuning. In *IEEE Conf. Comput. Vis. Pattern Recog.* 26296–26306.
- [22] Lingbo Liu, Jiaqi Chen, Hefeng Wu, Tianshui Chen, Guanbin Li, and Liang Lin. 2020. Efficient crowd counting via structured knowledge transfer. In *ACM Int. Conf. Multimed.* 2645–2654.
- [23] Shilong Liu, Zhaoyang Zeng, Tianhe Ren, Feng Li, Hao Zhang, Jie Yang, Qing Jiang, Chunyuan Li, Jianwei Yang, Hang Su, et al. 2025. Grounding dino: Marrying dino with grounded pre-training for open-set object detection. In *Eur. Conf. Comput. Vis.* 38–55.
- [24] Yuting Liu, Zheng Wang, Miaoqing Shi, Shin'ichi Satoh, Qijun Zhao, and Hongyu Yang. 2020. Towards unsupervised crowd counting via regression-detection bi-knowledge transfer. In *ACM Int. Conf. Multimed.* 129–137.
- [25] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. 2021. Swin transformer: Hierarchical vision transformer using shifted windows. In *Int. Conf. Comput. Vis.* 10012–10022.
- [26] Erika Lu, Weidi Xie, and Andrew Zisserman. 2019. Class-agnostic counting. In *Proc. Asian Conf. Comput. Vis.* 669–684.
- [27] Roni Paiss, Ariel Ephrat, Omer Tov, Shiran Zada, Inbar Mosseri, Michal Irani, and Tali Dekel. 2023. Teaching clip to count to ten. In *Int. Conf. Comput. Vis.* 3170–3180.
- [28] Jer Pelhan, Alan Lukezic, Vitjan Zavrtanik, and Matej Kristan. 2024. A Novel Unified Architecture for Low-Shot Counting by Detection and Segmentation. *Adv. Neural Inform. Process. Syst.* 37 (2024), 66260–66282.
- [29] Jer Pelhan, Vitjan Zavrtanik, Matej Kristan, et al. 2024. DAVE-A Detect-and-Verify Paradigm for Low-Shot Counting. In *IEEE Conf. Comput. Vis. Pattern Recog.* 23293–23302.
- [30] Ethan Perez, Florian Strub, Harm De Vries, Vincent Dumoulin, and Aaron Courville. 2018. Film: Visual reasoning with a general conditioning layer. In *Proc. AAAI Conf. Artif. Intell.*, Vol. 32, 3942–3951.
- [31] Yifei Qian, Zhongliang Guo, Bowen Deng, Chun Tong Lei, Shuai Zhao, Chun Pong Lau, Xiaopeng Hong, and Michael P Pound. 2025. T2ICount: Enhancing Cross-modal Understanding for Zero-Shot Counting. In *IEEE Conf. Comput. Vis. Pattern Recog.*
- [32] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. 2021. Learning transferable visual models from natural language supervision. In *Int. Conf. Mach. Learn.* 8748–8763.
- [33] Viresh Ranjan and Minh Hoai Nguyen. 2022. Exemplar free class agnostic counting. In *Proc. Asian Conf. Comput. Vis.* 3121–3137.
- [34] Viresh Ranjan, Udbhav Sharma, Thu Nguyen, and Minh Hoai. 2021. Learning to count everything. In *IEEE Conf. Comput. Vis. Pattern Recog.* 3394–3403.
- [35] Minjoon Seo, Aniruddha Kembhavi, Ali Farhadi, and Hannaneh Hajishirzi. 2017. Bidirectional Attention Flow for Machine Comprehension. In *Proc. Int. Conf. Learn. Represent.*
- [36] Min Shi, Hao Lu, Chen Feng, Chengxin Liu, and Zhiguo Cao. 2022. Represent, compare, and learn: A similarity-aware framework for class-agnostic counting. In *IEEE Conf. Comput. Vis. Pattern Recog.* 9529–9538.
- [37] Sravya Vardhani Shivapuja, Mansi Pradeep Khamkar, Divij Bajaj, Ganesh Ramakrishnan, and Ravi Kiran Sarvadevabhatla. 2021. Wisdom of (binned) crowds: A bayesian stratification paradigm for crowd counting. In *ACM Int. Conf. Multimed.* 3574–3582.
- [38] Hao Tan and Mohit Bansal. 2019. Lxmert: Learning cross-modality encoder representations from transformers. *arXiv preprint arXiv:1908.07490* (2019).
- [39] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Lukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. *Adv. Neural Inform. Process. Syst.* 30 (2017).
- [40] Zhicheng Wang, Liwen Xiao, Zhiguo Cao, and Hao Lu. 2024. Vision transformer off-the-shelf: A surprising baseline for few-shot class-agnostic counting. In *Proc. AAAI Conf. Artif. Intell.*, Vol. 38, 5832–5840.
- [41] Yuzhe Wu, Yipeng Xu, Tianyu Xu, Jialu Zhang, Jianfeng Ren, and Xudong Jiang. 2024. GCA-SUN: A Gated Context-Aware Swin-UNet for Exemplar-Free Counting. *arXiv preprint arXiv:2409.12249* (2024).
- [42] Zhuofan Xia, Xuran Pan, Shiji Song, Li Erran Li, and Gao Huang. 2022. Vision transformer with deformable attention. In *IEEE Conf. Comput. Vis. Pattern Recog.* 4794–4803.
- [43] Weidi Xie, J Alison Noble, and Andrew Zisserman. 2018. Microscopy cell counting and detection with fully convolutional regression networks. *Comput. Methods Biomech. Biomed. Eng. Imag. Vis.* 6 (2018), 283–292.
- [44] Jingyi Xu, Hieu Le, Vu Nguyen, Viresh Ranjan, and Dimitris Samaras. 2023. Zero-shot object counting. In *IEEE Conf. Comput. Vis. Pattern Recog.* 15548–15557.
- [45] Yuanwu Xu, Feifan Song, and Haofeng Zhang. 2024. Learning Spatial Similarity Distribution for Few-shot Object Counting. In *IJCAI.* 1507–1515.
- [46] Hao Zhang, Feng Li, Shilong Liu, Lei Zhang, Hang Su, Jun Zhu, Lionel M Ni, and Heung-Yeung Shum. 2022. Dino: Detr with improved denoising anchor boxes for end-to-end object detection. *arXiv preprint arXiv:2203.03605* (2022).
- [47] Ji Zhang, Jian-Jun Qiao, Xiao Wu, and Wei Li. 2021. Vehicle counting network with attention-based mask refinement and spatial-awareness block loss. In *ACM Int. Conf. Multimed.* 2889–2898.